MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, HIGH SPEED, CMOS, DECODERS, MONOLITHIC SILICON, POSITIVE LOGIC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for monolithic silicon, high speed, CMOS logic microcircuits. Two product assurance classes and a choice of case outlines and lead finishes are provided and are reflected in the complete part number.

1.2 Part number. The part number shall be in accordance with MIL-M-38510, and as specified herein.

1.2.1 Device types. The device types shall be as follows:

Device type	Circuit
01	BCD-to-decimal decoder
02	Single 3- to 8-line decoder
03	Dual 2- to 4-line decoder
04	4- to 16-line decoder/demultiplexer
52	Single 3- to 8-line decoder (TTL compatible)
53	Dual 2- to 4- line decoder (TTL compatible)

1.2.2 Device class. The device class shall be the product assurance level as defined in MIL-M-38510.

1.2.3 Case outlines. The case outlines shall be designated as follows:

Outline letter	Case outline (see MIL-M-38510, appendix C)
E	D-2 (16-lead, 1/4" x 7/8") dual-in-line package
F	F-5 (16-lead, 1/4" x 3/8") flat package
K	F-6 (24-lead, 3/8" x 5/8") flat package
L	D-9 (24-lead, 1/4" x 1 1/4") dual-in-line package
2	C-2 (20-terminal, .350" x .350") square chip carrier package
3	C-4 (28-terminal, .450" x .450") square chip carrier package

1.3 Absolute maximum ratings.

Supply voltage (V_{CC})	-0.5 V dc to +7.0 V dc
DC input voltage (V_{IN})	-0.5 V dc to V_{CC} +0.5 V dc
DC output voltage (V_{OUT})	-0.5 V dc to V_{CC} +0.5 V dc
Clamp diode current (I_{IK}, I_{OK})	±20 mA
DC output current per pin (I_{OUT})	±25 mA
DC V_{CC} or GND current per pin (I_{OC})	±50 mA
Storage temperature range (T_{STG})	-65°C to +150°C
Maximum power dissipation (P_D)	300 mW
Lead temperature (soldering, 10 seconds)	+300°C
Thermal resistance, junction-to-case (θ_{JC}):	
Cases E, F, K, and L	(See MIL-M-38510, appendix C)
Case 2	60°C/W
Case 3	(TBD)
Junction temperature (T_J)	+175°C

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Rome Air Development Center (RBE-2), Griffiss AFB, NY 13441, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1.4 Recommended operating conditions.

Device types 01, 02, 03, and 04:

Supply voltage (V_{CC})	2 V dc to 6 V dc
Input low (V_{IL}) maximum voltage	0.3 V at V_{CC} = 2 V 0.9 V at V_{CC} = 4.5 V 1.2 V at V_{CC} = 6 V
Input high (V_{IH}) minimum voltage	1.5 V at V_{CC} = 2 V 3.15 V at V_{CC} = 4.5 V 4.2 V at V_{CC} = 6 V
Input rise and fall times (t_r, t_f) maximum:	
V_{CC} = 2 V	1,000 ns
V_{CC} = 4.5 V	500 ns
V_{CC} = 6 V	400 ns

Device types 52 and 53:

Supply voltage (V_{CC})	4.5 V dc to 5.5 V dc
Input low (V_{IL}) maximum voltage	0.8 V at V_{CC} = 4.5 through 5.5 V
Input high (V_{IH}) minimum voltage	2.0 V at V_{CC} = 4.5 through 5.5 V
Input rise and fall times (t_r, t_f) maximum:	
V_{CC} = 4.5 V and 5.5 V	500 ns

All devices:

Output voltage	0 V dc to V_{CC}.
Case operating temperature (T_C)	-55°C to +125°C

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification and standard. The following specification and standard form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-M-38510 - Microcircuits, General Specification for.

STANDARD

MILITARY

MIL-STD-883 - Test Methods and Procedures for Microelectronics.

(Copies of the specification and standard required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Detail specification. The individual item requirements shall be in accordance with MIL-M-38510, and as specified herein.

3.2 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-M-38510 and herein.

3.2.1 Terminal connections. The terminal connections shall be as specified on figure 1.

3.2.2 Logic diagrams. The logic diagrams shall be as specified on figure 2.

3.2.3 Truth tables. The truth tables shall be as specified on figure 3.

3.2.4 Schematic circuits. The schematic circuits shall be submitted to the preparing activity prior to inclusion of a manufacturer's device in this specification and shall be submitted to the qualifying activity as a prerequisite for qualification. All qualified manufacturers' schematics shall be maintained and available upon request.

3.2.5 Case outlines. The case outlines shall be as specified in 1.2.3.

3.3 Lead material and finish. The lead material and finish shall be in accordance with MIL-M-38510 and 6.4 herein.

3.4 Electrical performance characteristics. Unless otherwise specified, the electrical performance characteristics are as specified in table I, and apply over the full recommended case operating temperature range.

3.5 Electrical test requirements. The electrical test requirements for each device class shall be the subgroups specified in table II. The electrical tests for each subgroup are described in table III.

3.6 Marking. Marking shall be in accordance with MIL-M-38510. At the option of the manufacturer, marking of the country of origin may be omitted from the body of the microcircuit, but shall be retained on the initial container.

3.6.1 Total dose radiation hardness identifier. Total dose radiation hardness identifier shall be in accordance with MIL-M-38510 and 4.5.4 herein.

3.6.2 Serialization. All class S devices shall be serialized in accordance with MIL-M-38510.

3.6.3 Correctness of indexing and marking. All devices shall be subjected to the final electrical tests specified in table II after part number marking to verify that they are correctly indexed and identified by part number. Optionally, an approved electrical test may be devised especially for this requirement.

3.7 Microcircuit group assignment. The devices covered by this specification shall be in microcircuit group number 39 (see MIL-M-38510, appendix E).

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-M-38510 and methods 5005 and 5007, as applicable, of MIL-STD-883, except as modified herein.

4.2 Screening. Screening shall be in accordance with method 5004 of MIL-STD-883, and shall be conducted on all devices prior to qualification and quality conformance inspection. The following additional criteria shall apply:

a. Delete the sequence specified in 3.1.9 through 3.1.13 of method 5004 and substitute lines 1 through 7 of table II herein.

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b. Burn-in (method 1015 of MIL-STD-883).

(1) Static tests (test condition A) using circuit shown on figure 4, or equivalent. Ambient temperature (T_A) shall be +125°C minimum. Test duration for each static test shall be 24 hours minimum for class S devices and in accordance with table I of method 1015 for class B devices.

(2) Dynamic test (test condition D) using circuit shown on figure 5, or equivalent. Ambient temperature shall be +125°C minimum. Test duration shall be in accordance with table I of method 1015.

c. Interim and final electrical parameters shall be as specified in table II herein.

d. For class S devices, post dynamic burn-in, or class B devices, post static burn-in, electrical parameter measurements may, at the manufacturer's option, be performed separately or included in the final electrical parameter requirements.

4.2.1 Percent defective allowable (PDA).

a. The PDA for class S devices shall be 5 percent for static burn-in and 5 percent for dynamic burn-in, based on the exact number of devices submitted to each separate burn-in.

b. Static burn-in I and II failures shall be cumulative for determining the PDA.

c. The PDA for class B devices shall be in accordance with MIL-M-38510 for static burn-in. Dynamic burn-in is not required.

d. Those devices whose measured characteristics, after burn-in, exceed the specified delta (Δ) limits or electrical parameter limits specified in table III, subgroup 1, are defective and shall be removed from the lot. The verified failures divided by the total number of devices in the lot initially submitted to burn-in shall be used to determine the percent defective for the lot and the lot shall be accepted or rejected based on the specified PDA.

4.3 Qualification inspection. Qualification inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, D, and E inspections (see 4.4.1 through 4.4.5).

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-M-38510 and as specified herein. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, D, and E inspections (see 4.4.1 through 4.4.5).

4.4.1 Group A inspection. Group A inspection shall be in accordance with table I of method 5005 of MIL-STD-883 and as follows:

a. Tests shall be performed in accordance with table II herein.

b. Subgroups 5 and 6 of table I of method 5005 of MIL-STD-883 shall be omitted.

c. Subgroup 4 (C_i measurement) shall be measured only for initial qualification and after process or design changes which may affect input capacitance. Capacitance shall be measured between the designated terminal and V_{SS} at a frequency of 1 MHz.

d. Subgroups 9 and 11 shall be measured only for initial qualification and after process or design changes which may affect dynamic performance.

Text continues on page 54

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TABLE I. Electrical performance characteristics.

Test	Symbol	Conditions 1/ T_C = +125°C unless otherwise specified	Device type 2/	V_{CC}	Limits Min	Limits Max	Unit
High level output voltage	V_{OH1} 3/	V_{IH} = 1.5 V; V_{IL} = 0.3 V; I_{OH} = -20 µA	01,02, 03,04	2.0 V	1.95		V
	V_{OH2} 3/	V_{IH} = 3.15 V; V_{IL} = 0.9 V; I_{OH} = -20 µA	01,02, 03,04	4.5 V	4.45		V
	V_{OH3}	V_{IH} = 4.2 V; V_{IL} = 1.2 V; I_{OH} = -20 µA	01,02, 03,04	6.0 V	5.95		V
	V_{OH4} 3/	V_{IH} = 3.15 V; V_{IL} = 0.9 V; I_{OH} = -4.0 mA	01,02, 03,04	4.5 V	3.7		V
	V_{OH5}	V_{IH} = 4.2 V; V_{IL} = 1.2 V; I_{OH} = -5.2 mA	01,02, 03,04	6.0 V	5.2		V
	V_{OH6}	V_{IH} = 2.0 V; V_{IL} = 0.5 V; I_{OH} = -20 mA	52,53	4.5 V	4.40		V
	V_{OH7}	V_{IH} = 2.0 V; V_{IL} = 0.8 V; I_{OH} = -4.0 mA	52,53	5.5 V	3.70		V
Low level output voltage	V_{OL1} 3/	V_{IL} = 0.3 V; V_{IH} = 1.5 V; I_{OL} = 20 µA	01,02, 03,04	2.0 V		0.05	V
	V_{OL2} 3/	V_{IL} = 0.9 V; V_{IH} = 3.15 V; I_{OL} = 20 µA	01,02, 03,04	4.5 V		0.05	V
	V_{OL3}	V_{IL} = 1.2 V; V_{IH} = 4.2 V; I_{OL} = 20 µA	01,02, 03,04	6.0 V		0.05	V
	V_{OL4} 3/	V_{IL} = 0.9 V; V_{IH} = 3.15 V; I_{OL} = 4.0 mA	01,02, 03,04	4.5 V		0.4	V
	V_{OL5}	V_{IL} = 1.2 V; V_{IH} = 4.2 V I_{OL} = 5.2 mA	01,02 03,04	6.0 V		0.4	V

See footnotes at end of table.

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TABLE I. Electrical performance characteristics - Continued.

Test	Symbol	Conditions 1/ T_C = +125°C unless otherwise specified	Device type 2/	V_{CC}	Limits Min	Limits Max	Unit
Low level output voltage	V_{OL6}	V_{IL} = 0.8 V; V_{IH} = 2.0 V; I_{OL} = 20 mA	52,53	4.5 V		0.05	V
	V_{OL7}	V_{IL} = 0.8 V; V_{IH} = 2.0 V; I_{OH} = 4.0 mA	52,53	5.5 V		0.4	V
Positive input clamp voltage	V_{IC+}	V_{CC} = GND I_{IN} = 1 mA T_C = +25°C	All			1.5	V
Negative input clamp voltage	V_{IC-}	V_{CC} = Open I_{IN} = -1 mA T_C = +25°C	All			-1.5	V
Input current low	I_{IL}	V_{IN} = GND	01,02, 03,04	6.0 V		-0.1	µA
Input current high	I_{IH}	V_{IN} = V_{CC}	01,02, 03,04	6.0 V		0.1	µA
Input current low	I_{IL}	V_{IN} = GND	52,53	5.5 V		-0.1	µA
Input current high	I_{IH}	V_{IN} = V_{CC}	52,53	5.5 V		0.1	µA
Short-circuit output current	I_{OS1} 3/	T_C = -55°C to +125°C V_O = GND V_I = GND	01,02, 03,04	2.0 V	-2	-50	mA
	I_{OS2} 3/			4.5 V	-15	-150	
	I_{OS3} 3/			6.0 V	-25	-180	
	I_{OS4}			4.0 V	-10	-120	
	I_{OS2} 3/		52,53	4.5 V	-15	-150	
	I_{OS5} 3/			5.5 V	-25	-180	
	I_{OS4}			4.0 V	-10	-120	

See footnotes at the end of table.

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TABLE I. Electrical performance characteristics - Continued.

Test	Symbol	Conditions 1/ T_C = +125°C unless otherwise specified	Device type 2/	V_{CC}	Limits Min	Limits Max	Unit
Supply current quiescent	I_{CC}	V_I = GND or V_{CC}	All	6.0 V		20	µA
Additional supply current quiescent per input pin (one unit load)	$I_{CC\Delta}$ 4/	V_{IL} = 0.8 V V_{IH} = 2.4 V Test pin at V_{IN} = 2.4 V Other pins at 0.8 V or 5.5 V; I_O = 0 V	52,53	5.5 V		1.5	mA
Input capacitance	C_I	T_C = +25°C	All			10	pF
Power dissipation capacitance	C_{PD} 2/ 3/	T_C = +25°C	01			65	pF
			02			85	
			03			75	
			04			90	
			52			85	
			53			75	
Propagation delay times 5/ 6/ Input to output	t_{PHL} t_{PLH}	C_L = 50 pF ±10%	01	4.5 V	4	35	ns
	t_{PLH}		02	4.5 V	4	47	
	t_{PHL}				5		
	t_{PLH} t_{PHL} 4 levels		03	4.5 V	6	51	
	t_{PLH} t_{PHL} 5 levels				8		
	t_{PLH} t_{PHL}		04	4.5 V	6	44	
	t_{PLH}		52	4.5 V	5	47	
	t_{PHL}				6		
	t_{PHL} 4 levels		53	4.5 V	5	41	
	t_{PLH} 5 levels				7		

See footnotes at end of table.

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TABLE I. Electrical performance characteristics - Continued.

Test	Symbol	Conditions 1/ T_C = +125°C unless otherwise specified	Device type 2/	V_{CC}	Limits Min	Limits Max	Unit
Propagation delay times 5/ 6/ Enable to output	t_{PHL} t_{PLH}	C_L = 50 pF ±10%	02	4.5 V	4	41	ns
			03	4.5 V	6	41	
			04	4.5 V	6	42	
			52	4.5 V	4	47	
			53	4.5 V	5	47	
Transition delay times 5/ 6/	t_{THL} t_{TLH}		All	4.5 V	3	20	

1/ Complete terminal conditions and limits shall be as specified in table III.

2/ Power dissipation capacitance (C_{PD}) per decoder/demultiplexer.

3/ Guaranteed but not tested.

4/ Total supply current = I_{CC} + $I_{CC\Delta}$.

5/ Tested at V_{CC} = 4.5 V at +125°C for sample testing and V_{CC} = 4.5 V at +25°C for screening. Guaranteed at other V_{CC} voltages and temperatures, see table IA and exception in 4.4.1d.

6/ For propagation and transition delay times at V_{CC} = 2.0 V, increase limit by a factor of 5. For propagation and transition delay times at V_{CC} = 6.0 V, decrease limit by a factor of .85.

TABLE IA. Calculated dynamic figures at -55 and +25 ambient temperatures (°C).

V_{CC} 1/	T_A = (°C) 2/ +125	T_A = (°C) 2/ -55 and +25
2.0 V	5	5 x 0.75
4.5 V	1	0.75
6.0 V	0.85	0.85 x 0.75

1/ The 2.0 V and 6.0 V numbers are derived from their 4.5 V integer value. Rounding off according 5/4.

2/ Normalized numbers (+125°C equals 1).

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TABLE II. Burn-in and electrical test requirements.

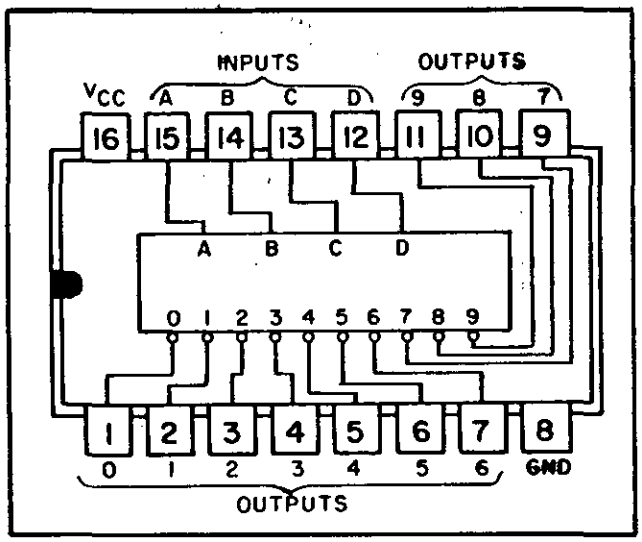
Line no.	Applicable tests and MIL-STD-883 test method	Class S device 1/				Class B device 1/			
		Ref. par.	Table III subgroups 2/ 3/	Table IV delta limits 4/	Test circuit figure	Ref. par.	Table III subgroups 2/ 3/	Table IV delta limits 4/	Test circuit figure
1	Interim electrical parameters (method 5004)		1				1		
2	Static burn-in I (method 1015)	4.2b 4.5.2	Required		3		Not required		
3	Same as line 1		1	Δ					
4	Static burn-in II (method 1015)	4.2b 4.5.2	Required		3	4.2b 4.5.2	Required 5/		3
5	Same as line 1	4.2d	1*	Δ		4.2d	1*	Δ	
6	Dynamic burn-in (method 1015)	4.2b 4.5.2	Required		1/		Not required		
7	Same as line 1	4.2d	1	Δ					
8	Final electrical parameters (method 5004)		1*, 2, 3, 7, 8, 9				1*, 2, 7, 9 5/		
9	Group A test requirements (method 5005)	4.4.1	1, 2, 3, 4, 7, 8, 9, 10, 11			4.4.1	1, 2, 3, 4, 7, 8, 9, 10, 11		
10	Group B end-point electrical parameters (method 5005)	4.4.2	†1, 2, 3, 7, 8, 9, 10, 11	Δ			†1		
11	Group C end-point electrical parameters (method 5005)					4.4.3	1, 2	Δ	
12	Group D end-point electrical parameters (method 5005)	4.4.4	1, 2, 3			4.4.4	1, 2		

1/ Blank spaces indicate tests are not applicable.
2/ * indicates PDA applies to subgroup 1 (see 4.2.1).
3/ † indicates also applies to electrostatic discharge sensitivity tests.
4/ Δ indicates delta limit shall be required only on table III, subgroup 1, where specified, and the delta values shall be computed with reference to the previous interim electrical parameters (line 1).
5/ The device manufacturer may at his option either complete subgroup 1 electrical parameter measurements, including delta measurements, within 96 hours after burn-in completion (removal of bias); or may complete subgroup 1 electrical measurements without delta measurements within 24 hours after burn-in completion (removal of bias).

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Device type 01

Cases E and F



Device type 01

Case 2

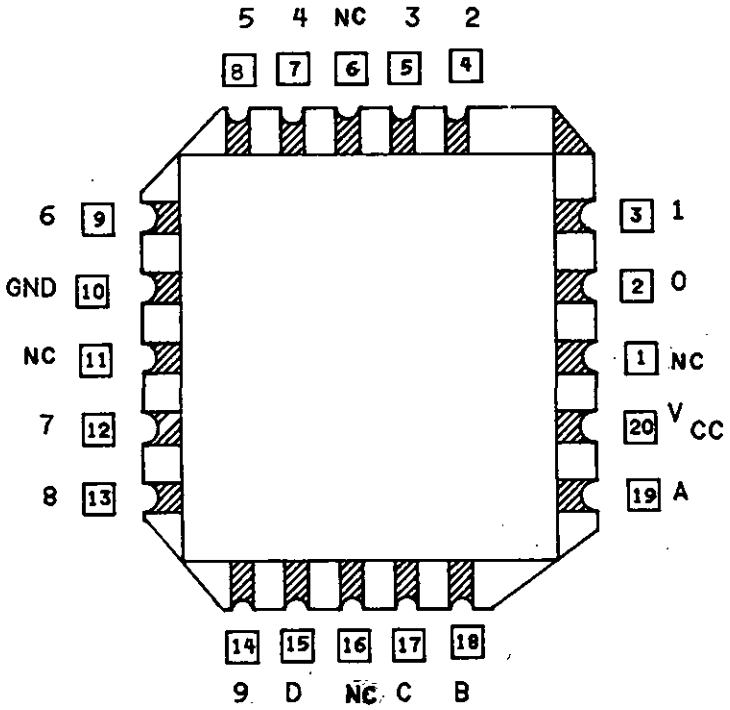
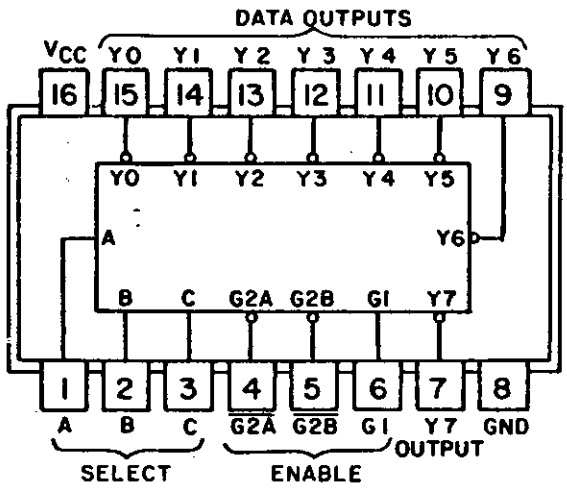


FIGURE 1. Terminal connections (top view).

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Device types 02 and 52

Cases E and F



Device types 02 and 52

Case 2

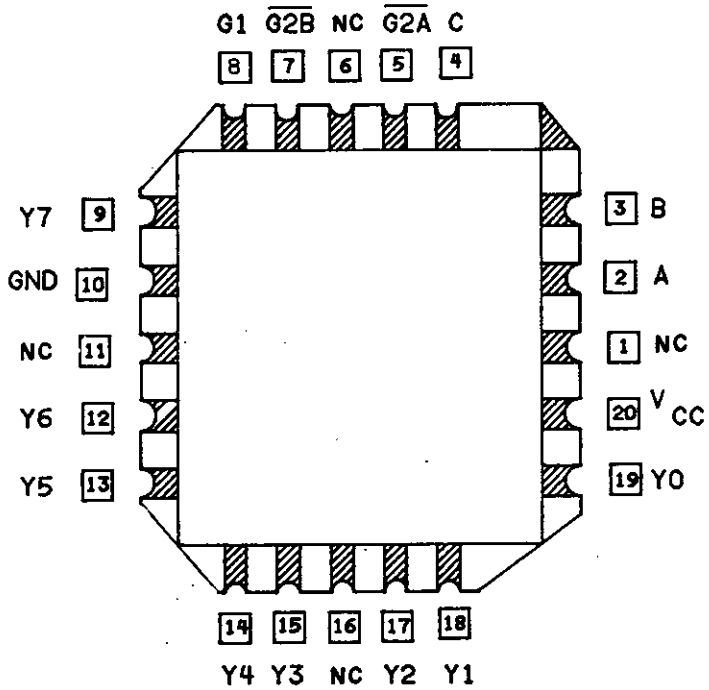
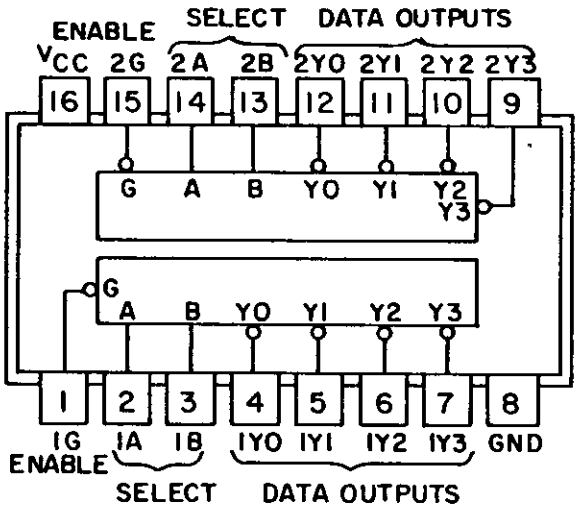


FIGURE 1. Terminal connections (top view) - Continued.

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Device types 03 and 53

Cases E and F



Device types 03 and 53

Case 2

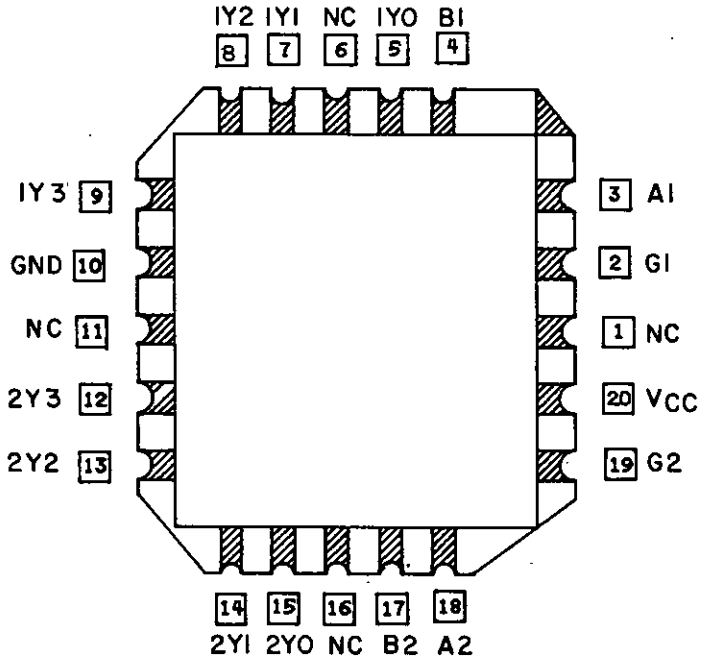
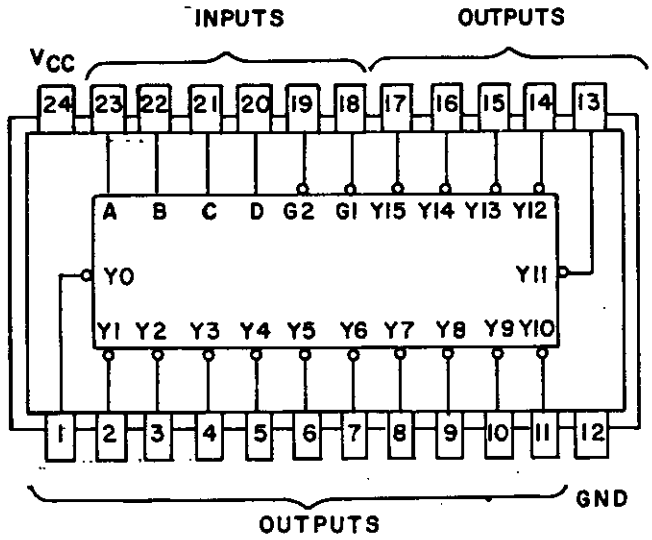


FIGURE 1. Terminal connections (top view) - Continued.

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Device type 04

Cases K and L



Device type 04

Case 3

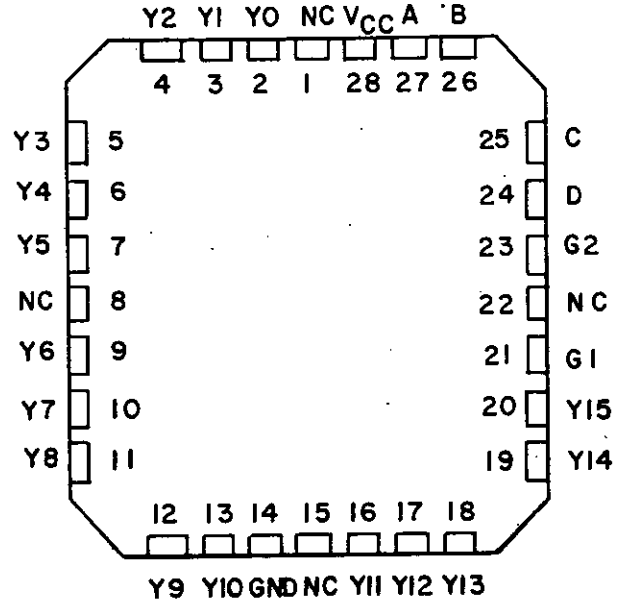


FIGURE 1. Terminal connections (top view) - Continued.

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Device type 01

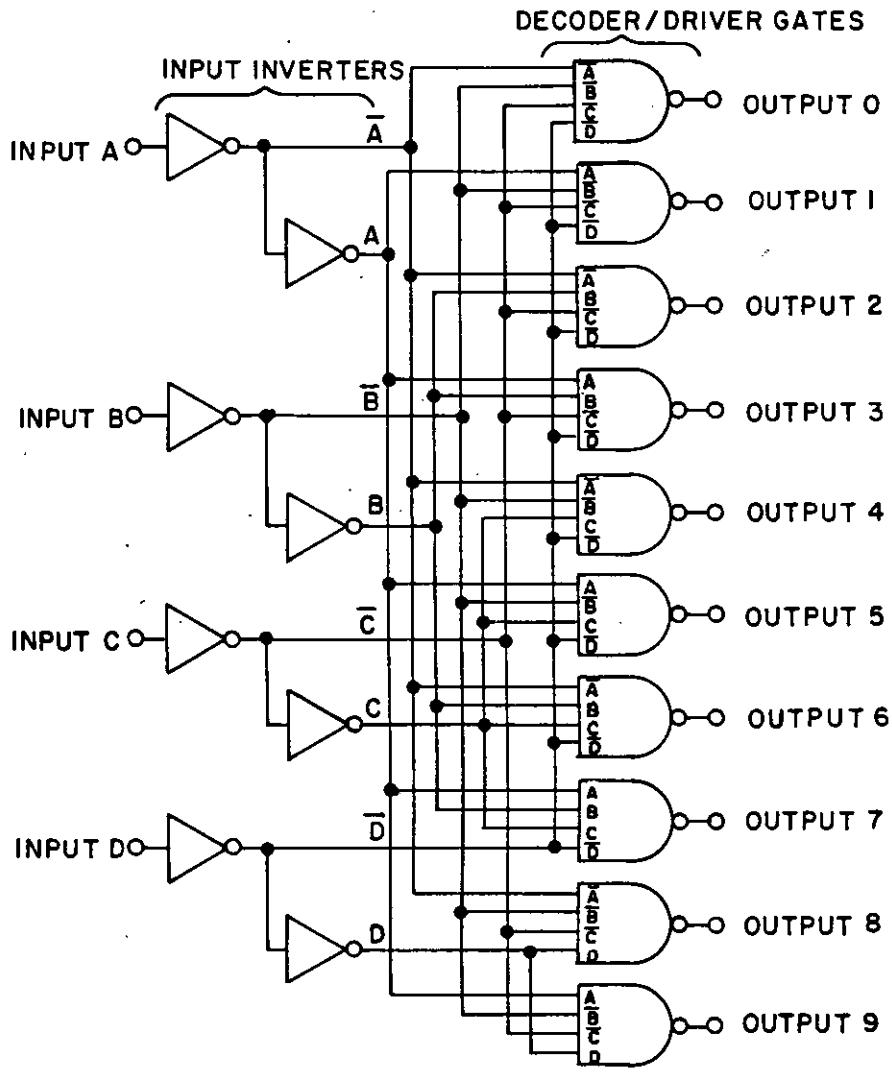


FIGURE 2. Logic diagrams.

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Device types 02 and 52

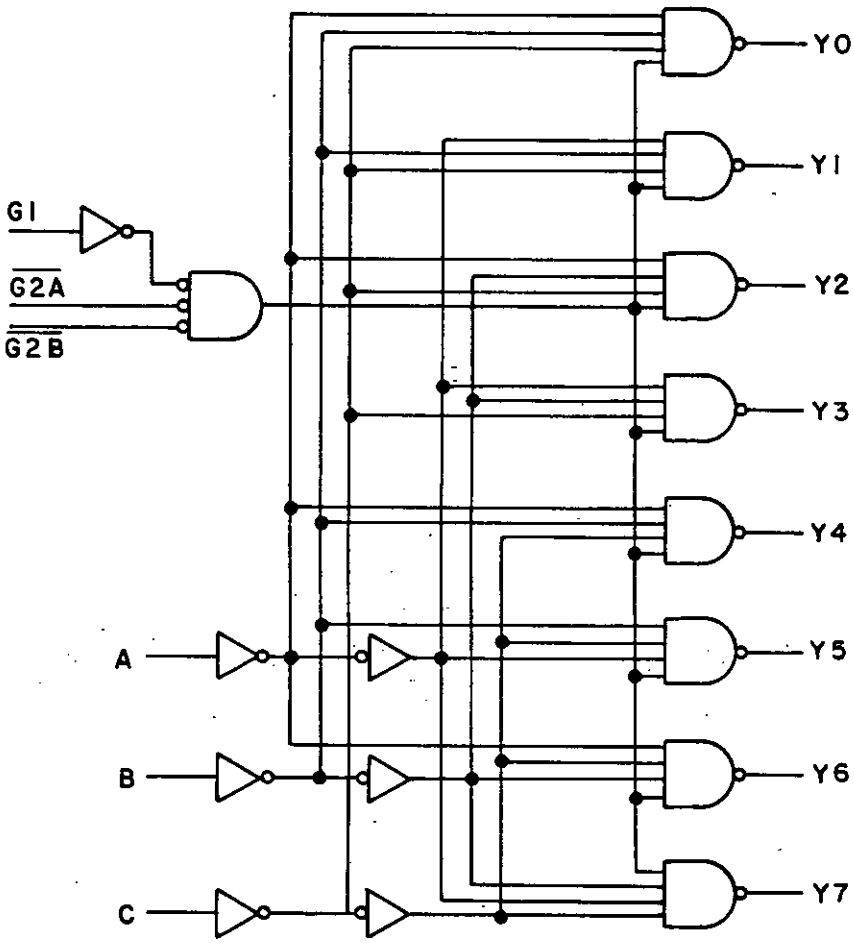


FIGURE 2. Logic diagrams - Continued.

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Device types 03 and 53

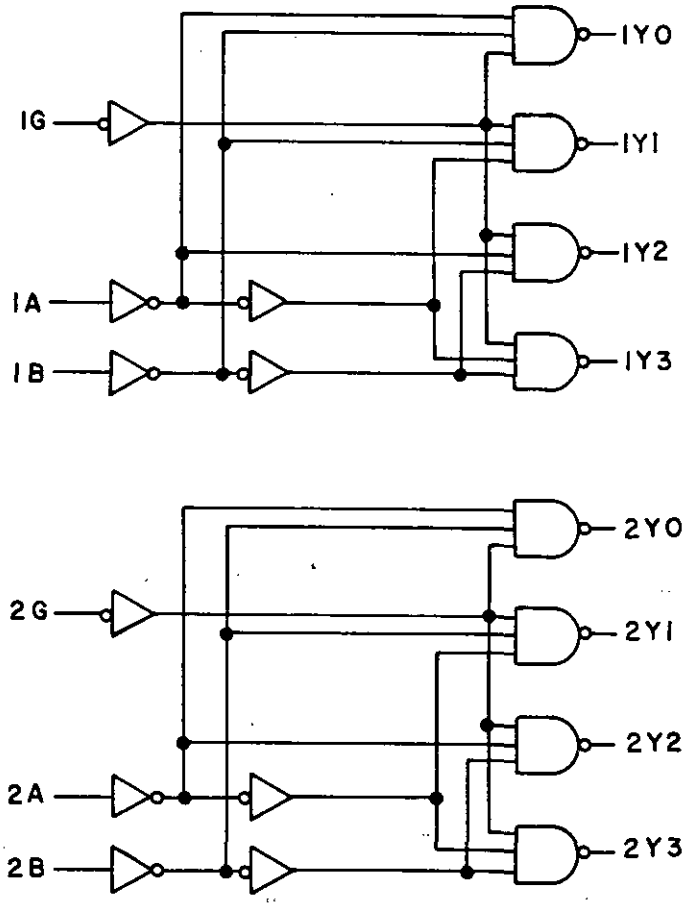


FIGURE 2. Logic diagrams - Continued.

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Device type 04

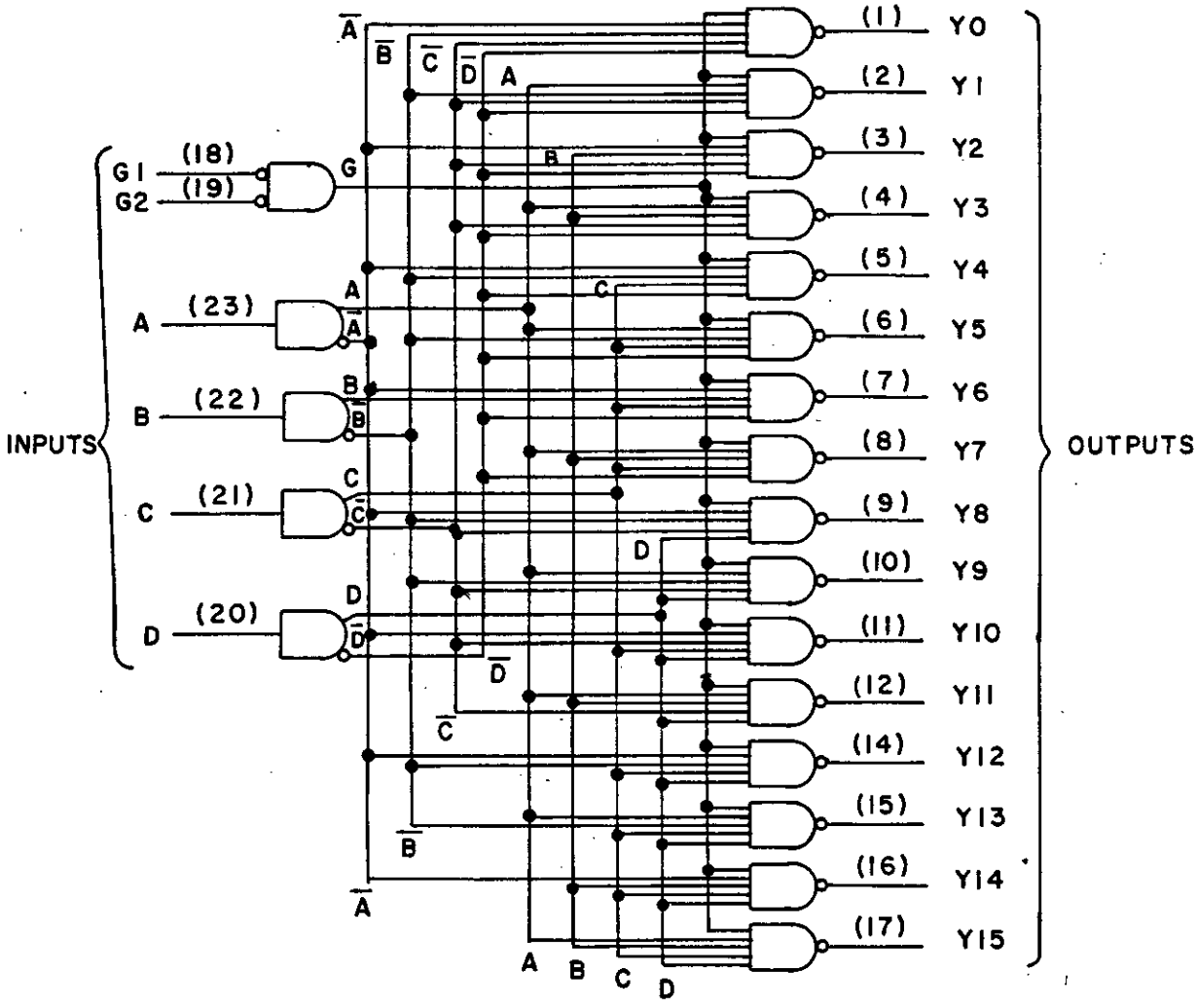


FIGURE 2. Logic diagrams - Continued.

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Device type 01

Inputs				Outputs									
D	C	B	A	0	1	2	3	4	5	6	7	8	9
L	L	L	L	L	H	H	H	H	H	H	H	H	H
L	L	L	H	H	L	H	H	H	H	H	H	H	H
L	L	H	L	H	H	L	H	H	H	H	H	H	H
L	L	H	H	H	H	H	L	H	H	H	H	H	H
L	H	L	L	H	H	H	H	L	H	H	H	H	H
L	H	L	H	H	H	H	H	H	L	H	H	H	H
L	H	H	L	H	H	H	H	H	H	L	H	H	H
L	H	H	H	H	H	H	H	H	H	H	L	H	H
H	L	L	L	H	H	H	H	H	H	H	H	L	H
H	L	L	H	H	H	H	H	H	H	H	H	H	L
H	L	H	L	H	H	H	H	H	H	H	H	H	H
H	L	H	H	H	H	H	H	H	H	H	H	H	H
H	H	L	L	H	H	H	H	H	H	H	H	H	H
H	H	L	H	H	H	H	H	H	H	H	H	H	H
H	H	H	L	H	H	H	H	H	H	H	H	H	H
H	H	H	H	H	H	H	H	H	H	H	H	H	H

FIGURE 3. Truth tables.

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Device types 02 and 52

Inputs					Outputs							
Enable		Select										
G1	$\overline{G2}$*	C	B	A	Y0	Y1	Y2	Y3	Y4	Y5	Y6	Y7
X	H	X	X	X	H	H	H	H	H	H	H	H
L	X	X	X	X	H	H	H	H	H	H	H	H
H	L	L	L	L	L	H	H	H	H	H	H	H
H	L	L	L	H	H	L	H	H	H	H	H	H
H	L	L	H	L	H	H	L	H	H	H	H	H
H	L	L	H	H	H	H	H	L	H	H	H	H
H	L	H	L	L	H	H	H	H	L	H	H	H
H	L	H	L	H	H	H	H	H	H	L	H	H
H	L	H	H	L	H	H	H	H	H	H	L	H
H	L	H	H	H	H	H	H	H	H	H	H	L

*$\overline{G2} = \overline{G2A} + \overline{G2B}$
H = high level, L = low level, X = irrelevant.

Device types 03 and 53

Inputs			Outputs			
Enable	Select					
G	B	A	Y0	Y1	Y2	Y3
H	X	X	H	H	H	H
L	L	L	L	H	H	H
L	L	H	H	L	H	H
L	H	L	H	H	L	H
L	H	H	H	H	H	L

H = high level, L = low level, X = irrelevant

FIGURE 3. Truth tables - Continued.

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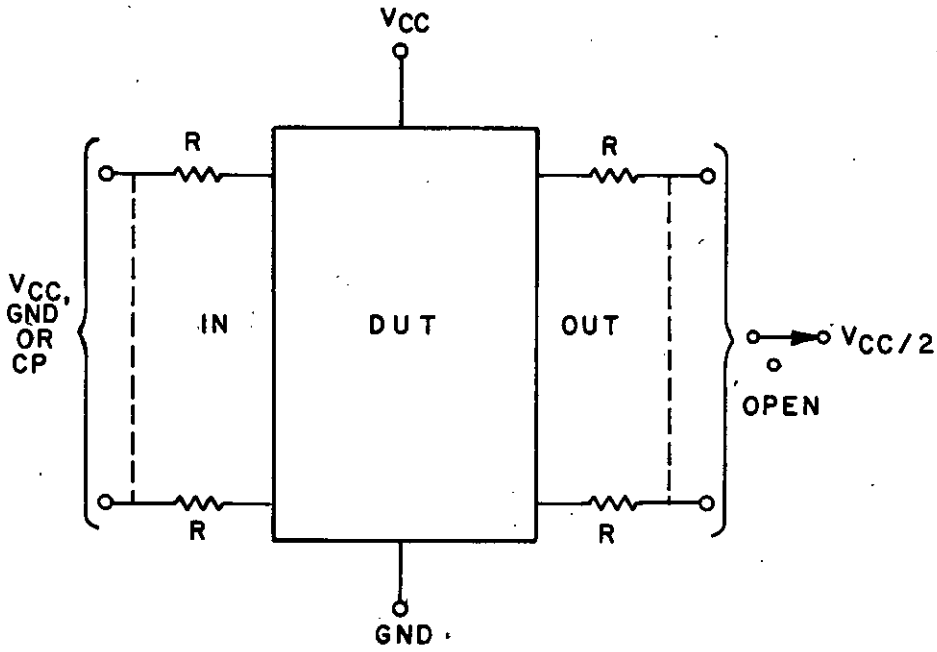
Device type 04

Inputs						Outputs														
G1	G2	D	C	B	A	Y0	Y1	Y2	Y3	Y4	Y5	Y6	Y7	Y8	Y9	Y10	Y11	Y12	Y13	Y14	Y15
L	L	L	L	L	L	L	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H
L	L	L	L	L	H	H	L	H	H	H	H	H	H	H	H	H	H	H	H	H	H
L	L	L	L	H	L	H	H	L	H	H	H	H	H	H	H	H	H	H	H	H	H
L	L	L	L	H	H	H	H	H	L	H	H	H	H	H	H	H	H	H	H	H	H
L	L	L	H	L	L	H	H	H	H	L	H	H	H	H	H	H	H	H	H	H	H
L	L	L	H	L	H	H	H	H	H	H	L	H	H	H	H	H	H	H	H	H	H
L	L	L	H	H	L	H	H	H	H	H	H	L	H	H	H	H	H	H	H	H	H
L	L	L	H	H	H	H	H	H	H	H	H	H	L	H	H	H	H	H	H	H	H
L	L	H	L	L	L	H	H	H	H	H	H	H	H	L	H	H	H	H	H	H	H
L	L	H	L	L	H	H	H	H	H	H	H	H	H	H	L	H	H	H	H	H	H
L	L	H	L	H	L	H	H	H	H	H	H	H	H	H	H	L	H	H	H	H	H
L	L	H	L	H	H	H	H	H	H	H	H	H	H	H	H	H	L	H	H	H	H
L	L	H	H	L	L	H	H	H	H	H	H	H	H	H	H	H	H	L	H	H	H
L	L	H	H	L	H	H	H	H	H	H	H	H	H	H	H	H	H	H	L	H	H
L	L	H	H	H	L	H	H	H	H	H	H	H	H	H	H	H	H	H	H	L	H
L	L	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	L
L	H	X	X	X	X	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H
H	L	X	X	X	X	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H
H	H	X	X	X	X	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H

H = High, L = Low, X = Irrelevant.

FIGURE 3. Truth tables - Continued.

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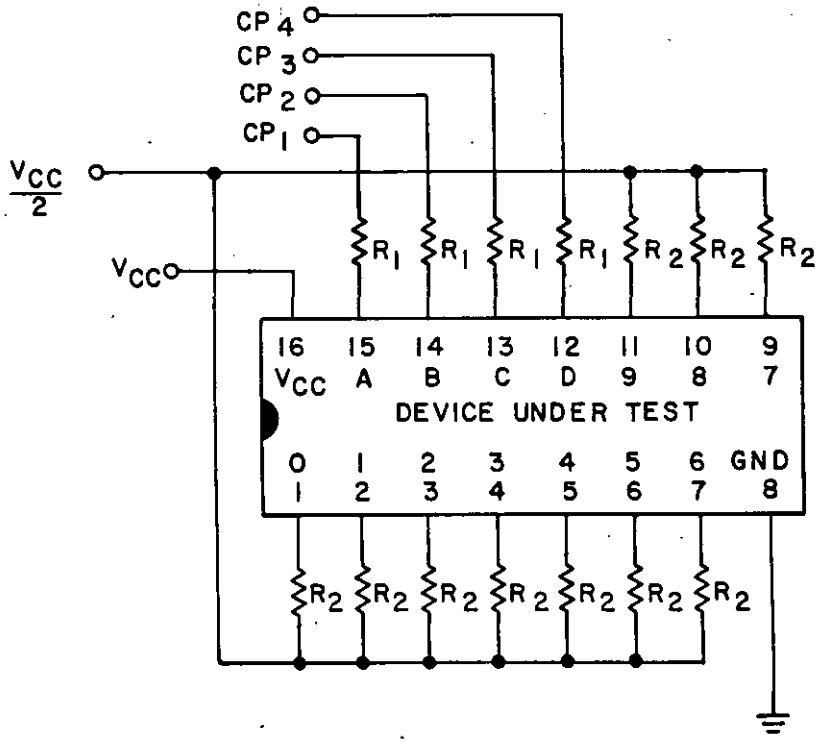


NOTES:
1. For static burn-in I, all inputs shall be connected to GND. Outputs shall be open or connected to $V_{CC/2}$. Resistors are optional on outputs if open. Resistors are required on inputs, and required on outputs connected to $V_{CC/2}$. R = 680Ω to 47 kΩ.
2. For static burn-in II, all inputs shall be connected through the resistors to V_{CC}. Outputs shall be open or connected to $V_{CC/2}$. Resistors are optional on outputs if open. Resistors are required on inputs, and required on outputs connected to $V_{CC/2}$. R = 680Ω to 47 kΩ.
3. $\frac{V_{CC}}{2} = \frac{V_{CC}}{2} \pm 0.5$ V
4. V_{CC} = 6.0 V ±0.5 V, for device types 01 through 04.
   V_{CC} = 5.5 V ±0.5 V, for device types 52 and 53.

FIGURE 4. Static burn-in test circuits.

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Device type 01

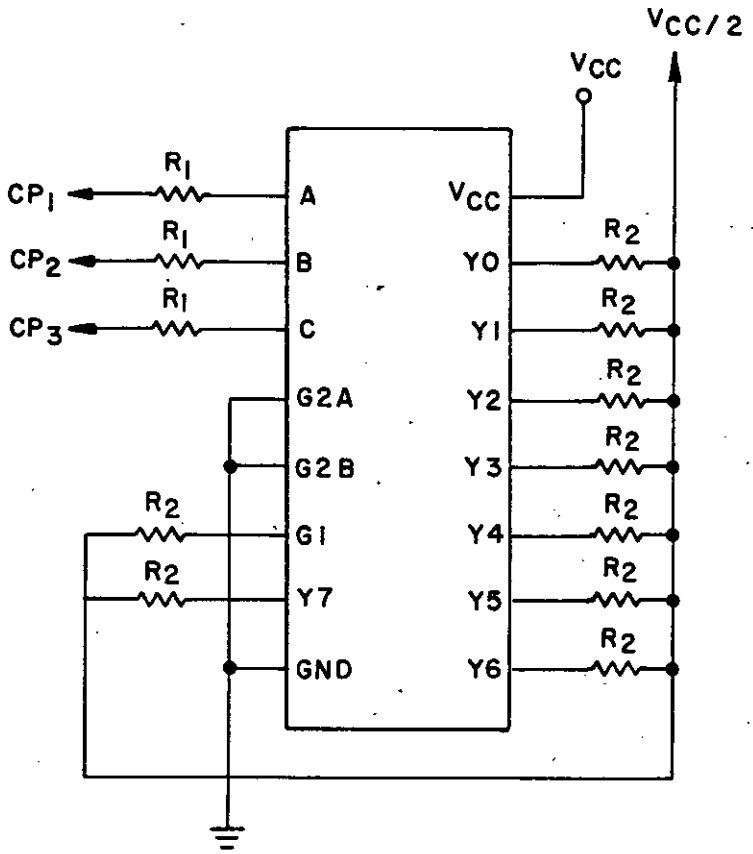


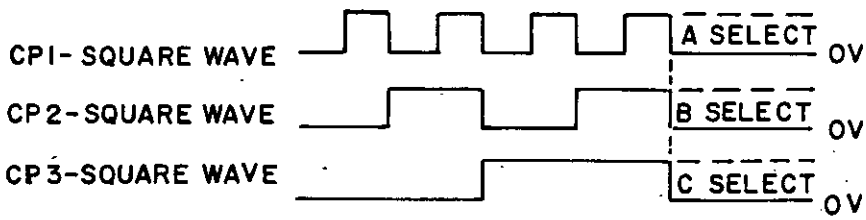
NOTES:
1. Input resistors R_1, = 680Ω to 47 kΩ ±20%.
2. Output resistors R_2 = 1 kΩ ±5%.
3. CP1 = 100 kHz, 50% duty cycle; V_{IH} = 4.5 V to V_{CC}, V_{IL} = 0 ±.5 V, $t_t \leq$ 500 ns
   CP2 = 50 kHz, 50% duty cycle; V_{IH} = 4.5 V to V_{CC}, V_{IL} = 0 ±.5 V, $t_t \leq$ 500 ns
   CP3 = 25 kHz, 50% duty cycle; V_{IH} = 4.5 V to V_{CC}, V_{IL} = 0 ±.5 V, $t_t \leq$ 500 ns
   CP4 = 12.5 kHz, 50% duty cycle; V_{IH} = 4.5 V to V_{CC}, V_{IL} = 0 ±.5 V, $t_t \leq$ 500 ns
4. V_{CC} = 6.0 V ±0.5 V.
5. $V_{\frac{CC}{2}} = V_{\frac{CC}{2}}$ ±0.5 V.

FIGURE 5. Dynamic burn-in and life test circuits.

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Device types 02 and 52



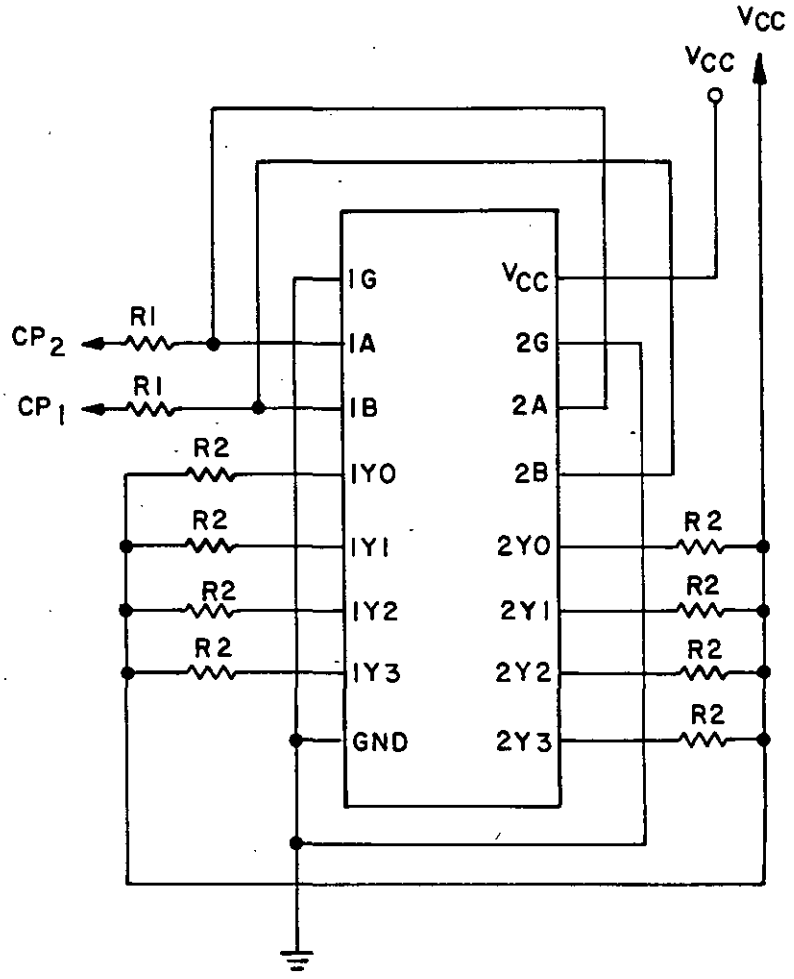


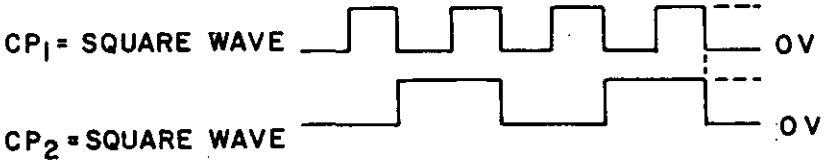
NOTES:
1. R_1 = 680Ω to 47 kΩ; R_2 1kΩ ±5%.
2. V_{CC} = 6.0 V ±0.5 V for device type 02; V_{CC} = 5.5 V ±0.5 V for device type 52.
3. $\frac{V_{CC}}{2} = \frac{V_{CC}}{2}$ ±0.5 V.
4. Duty cycle = 50% ±15%, V_{IH} = 4.5 V min to V_{CC}; V_{IL} = 0.0 V ±0.5 V, $t_t \leq 500$ ns.
5. CP_1, CP_2, CP_3 = 25 kHz to 1 MHz; $CP_3 = \frac{CP_2}{2}$; $CP_2 = \frac{CP_1}{2}$

FIGURE 5. Dynamic burn-in and life test circuits - Continued.

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Device types 03 and 53



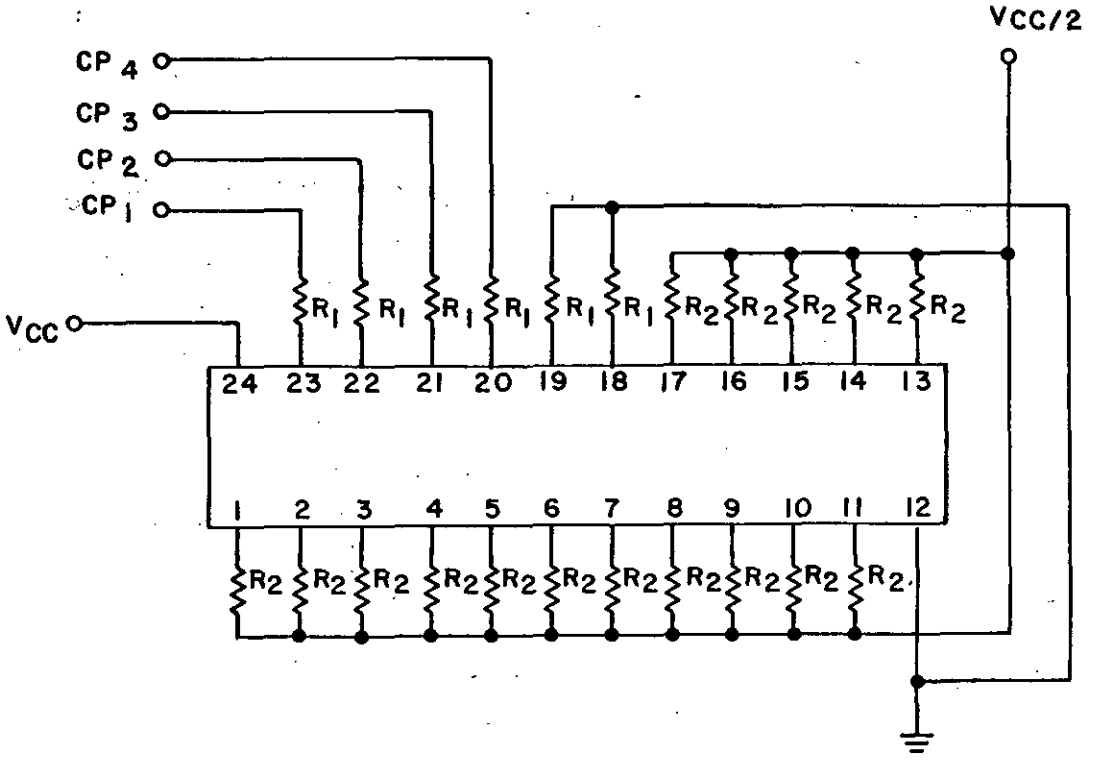


NOTES:
1. Input resistors R_1 = 680Ω to 47 kΩ ±20%, output resistors R_2 = 1 kΩ ±5%.
2. V_{CC} = 6.0 V ±0.5 V for device type 03; V_{CC} = 5.5 V ±0.5 V for device type 53.
3. $\frac{V_{CC}}{2} = \frac{V_{CC}}{2}$ ±0.5 V.
4. Duty cycle = 50% ±15%, V_{IH} = 4.5 V min to V_{CC}; V_{IL} = 0.0 V ±0.5 V, $t_t \leq 500$ ns.
5. CP_1 = 25 kHz to 1 MHz; $CP_2 = \frac{CP_1}{2}$.

FIGURE 5. Dynamic burn-in and life test circuits - Continued.

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Device type 04



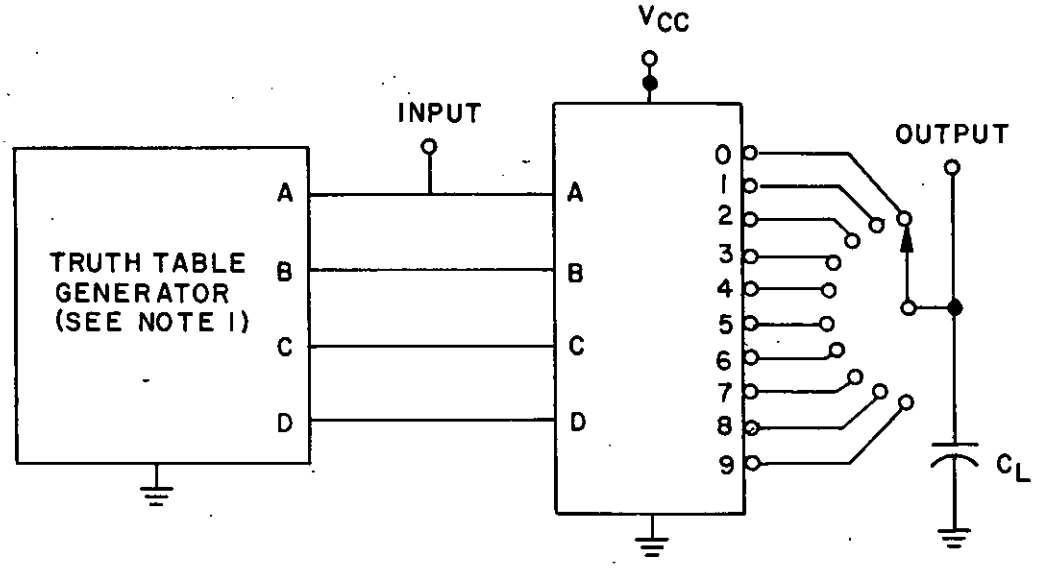
NOTES:
1. Input resistors R_1 = 680Ω to 47 kΩ ±20%, output resistors R_2 = 1 kΩ ±5%.
2. V_{CC} = 6.0 V ±0.5 V.
3. $\frac{V_{CC}}{2} = \frac{V_{CC}}{2}$ ±0.5 V.
4. Duty cycle = 50% ±15%, V_H = 4.5 V minimum to V_{CC}; V_{IL} = 0.0 V ±0.5 V; $t_t \leq$ 500 ns.
5. CP_1, CP_2, CP_3, CP_4 = 25 kHz to 1 MHz; square wave.

$CP_2 = \frac{CP_1}{2}$; $CP_3 = \frac{CP_2}{2}$; $CP_4 = \frac{CP_3}{2}$.

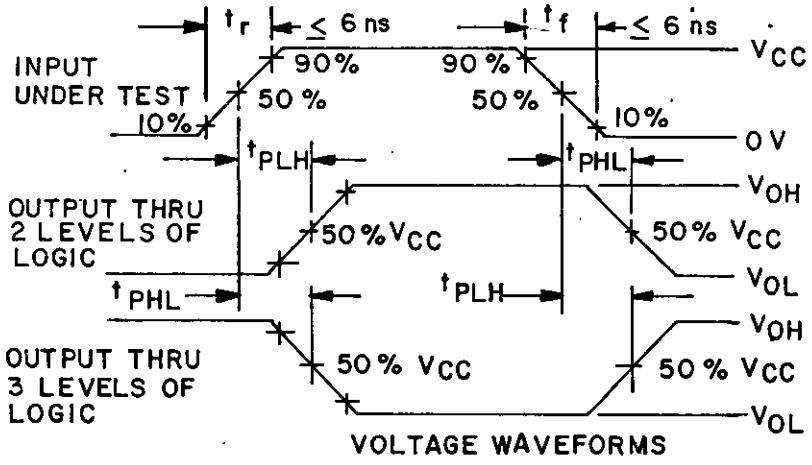
FIGURE 5. Dynamic burn-in and life test circuits - Continued.

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Device type 01



TEST CIRCUIT



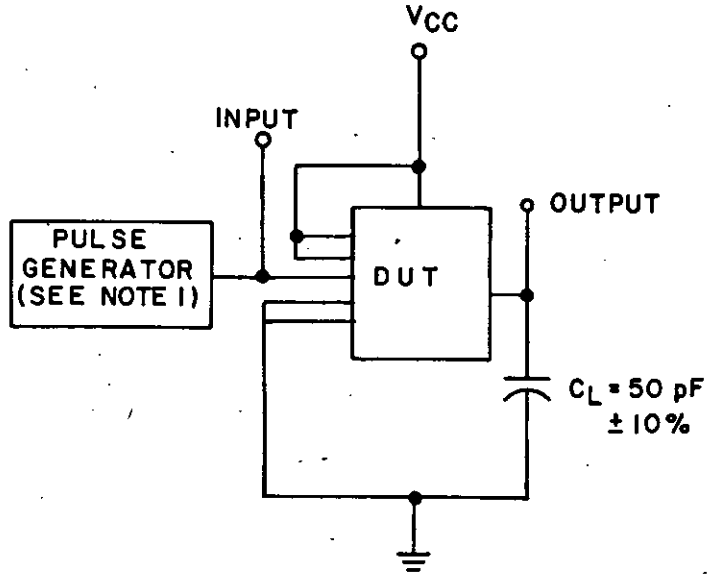
VOLTAGE WAVEFORMS

NOTES:
1. The truth table generator has the following characteristics: PRR ≤ 1 MHz, 50% duty cycle. Transition of the input under test shall occur simultaneously with or following the transition of the other inputs, $t_r = t_f \leq 6$ ns.
2. C_L = 50 pF ± 10%; includes probe and jig capacitance.

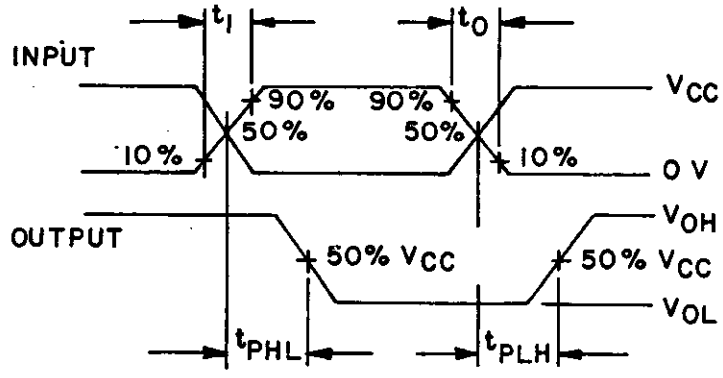
FIGURE 6. Switching time test circuits and waveforms.

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Device types 02, 03, 52, and 53



TEST CIRCUIT

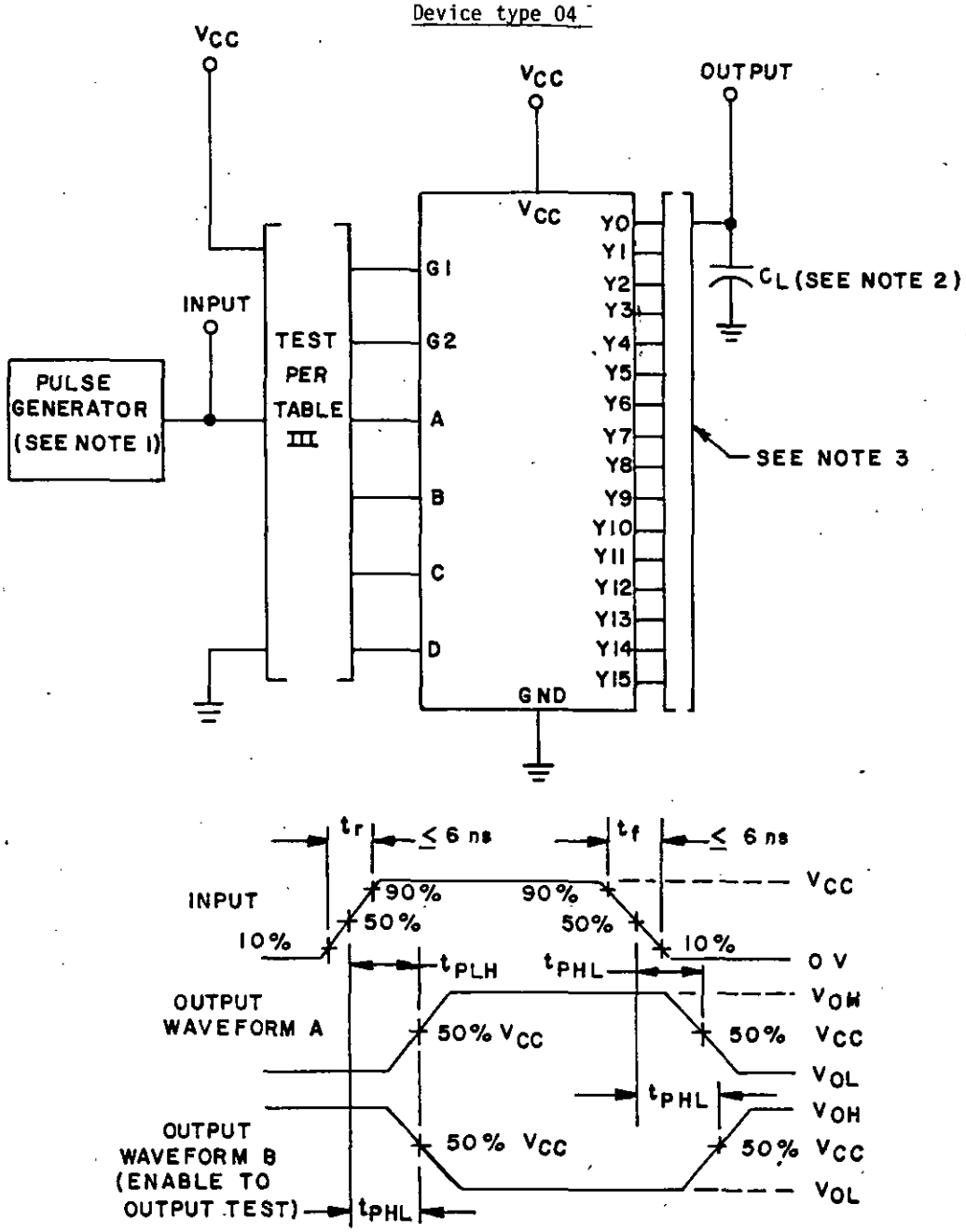


NOTES:
1. Pulse generator has the following characteristics: $t_1 = t_0 \leq 6$ ns; PRR $\leq$ 1 MHz; duty cycle = 50%.
2. C_L = 50 pF ±10% including scope probe, wiring, and stray capacitance, without package in test fixture.
3. Voltage measurements are to be made with respect to network ground terminal.
4. All unused inputs are tied to V_{CC} or ground according to table III.
5. The input signal for HCT device types (52 and 53) will be 0 to 3 volts; however, the 50% V_{CC} measuring point is 1.3 volts for input and output.

FIGURE 6. Switching time test circuits and waveforms - Continued.

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Device type 04



NOTES:
1. The pulse generator has the following characteristics: PRR $\leq$ 1 MHz, duty cycle = 50%, t_r, $t_f \leq$ 6 ns.
2. C_L = 50 pF ±10% minimum and includes probe and jig capacitance.
3. Load circuit is applied separately to each output pin under test. All other outputs may be open or loaded.

FIGURE 6. Switching time test circuits and waveforms - Continued.

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TABLE III. Group A inspection for device type 01.

Symbol	MIL-STD-883 method	Case	Terminal conditions 1/															Measured terminal	Test Limits						Unit
																			Subgroup 1 T_C = +25°C		Subgroup 2 T_C = +125°C		Subgroup 3 T_C = -55°C		
		2	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20							
		E	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16							
		Test no.	0	1	2	3	4	5	6	GND	7	8	9	D	C	B	A	V_{CC}		Min	Max	Min	Max	Min	Max	
$V_{IC(pos)}$		1								1/				1 mA	1 mA			GND	D		1.5					V
		2								"						1 mA		"	C		"					"
		3								"							1 mA	"	B		"					"
		4								"								"	A		"					"
$V_{IC(neg)}$		5								GND				1 mA	-1 mA			1/	D		-1.5					"
		6								"						-1 mA		"	C		"					"
		7								"							-1 mA	"	B		"					"
		8								"								"	A		"					"
I_{CC}	3005	9								"				6.0 V	6.0 V	6.0 V	6.0 V	6.0 V	V_{CC}		0.2		20			µA
V_{OH3}	3006	10	-20 µA							"				4.2 V	4.2 V	4.2 V	4.2 V	"	0	5.95		5.95		5.95		V
		11		-20 µA						"				"	"	"	"	"	1	"		"		"		"
		12			-20 µA					"				"	"	"	"	"	2	"		"		"		"
		13				-20 µA				"				"	"	"	"	"	3	"		"		"		"
		14					-20 µA			"				"	"	"	"	"	4	"		"		"		"
		15						-20 µA		"				"	"	"	"	"	5	"		"		"		"
		16							-20 µA	"				"	"	"	"	"	6	"		"		"		"
		17								"	-20 µA			"	"	"	"	"	7	"		"		"		"
		18								"		-20 µA		"	"	"	"	"	8	"		"		"		"
		19								"			-20 µA	"	"	"	"	"	9	"		"		"		"
V_{OH5}	3006	20	-5.2 mA							"				"	"	"	"	"	0	5.48		5.2		5.48		"
		21		-5.2 mA						"				"	"	"	"	"	1	"		"		"		"
		22			-5.2 mA					"				"	"	"	"	"	2	"		"		"		"
		23				-5.2 mA				"				"	"	"	"	"	3	"		"		"		"
		24					-5.2 mA			"				"	"	"	"	"	4	"		"		"		"
		25						-5.2 mA		"				"	"	"	"	"	5	"		"		"		"
		26							-5.2 mA	"				"	"	"	"	"	6	"		"		"		"
		27								"	-5.2 mA			"	"	"	"	"	7	"		"		"		"
		28								"		-5.2 mA		"	"	"	"	"	8	"		"		"		"
		29								"			-5.2 mA	"	"	"	"	"	9	"		"		"		"
V_{OL3}	3007	30	20 µA							"				1.2 V	1.2 V	1.2 V	1.2 V	"	0		0.05		0.05		0.05	"
		31		20 µA						"				"	"	1.2 V	4.2 V	"	1		"		"		"	"
		32			20 µA					"				"	"	4.2 V	1.2 V	"	2		"		"		"	"
		33				20 µA				"				"	"	4.2 V	4.2 V	"	3		"		"		"	"
		34					20 µA			"				"	4.2 V	1.2 V	1.2 V	"	4		"		"		"	"
		35						20 µA		"				"	"	1.2 V	4.2 V	"	5		"		"		"	"
		36							20 µA	"				"	"	4.2 V	1.2 V	"	6		"		"		"	"
		37								"	20 µA			"	"	4.2 V	4.2 V	"	7		"		"		"	"
		38								"		20 µA		4.2 V	1.2 V	1.2 V	1.2 V	"	8		"		"		"	"
		39								"			20 µA	4.2 V	1.2 V	1.2 V	4.2 V	"	9		"		"		"	"
V_{OL5}	3007	40	5.2 mA							"				1.2 V	1.2 V	1.2 V	1.2 V	"	0		0.26		0.4		0.26	"
		41		5.2 mA						"				"	"	1.2 V	4.2 V	"	1		"		"		"	"
		42			5.2 mA					"				"	"	4.2 V	1.2 V	"	2		"		"		"	"
		43				5.2 mA				"				"	"	4.2 V	4.2 V	"	3		"		"		"	"
		44					5.2 mA			"				"	4.2 V	1.2 V	1.2 V	"	4		"		"		"	"
		45						5.2 mA		"				"	"	1.2 V	4.2 V	"	5		"		"		"	"
		46							5.2 mA	"				"	"	4.2 V	1.2 V	"	6		"		"		"	"
		47								"	5.2 mA			"	"	4.2 V	4.2 V	"	7		"		"		"	"
		48								"		5.2 mA		4.2 V	1.2 V	1.2 V	1.2 V	"	8		"		"		"	"
		49								"			5.2 mA	4.2 V	1.2 V	1.2 V	4.2 V	"	9		"		"		"	"

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TABLE III. Group A inspection for device type 01 - Continued.

Symbol	MIL-STD-883 method	Case	Terminal conditions 1/																Measured terminal	Test limits: Subgroup 1 T_C = +25°C		Subgroup 2 T_C = +125°C		Subgroup 3 T_C = -55°C		Unit
		2	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20								
		E	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16								
		Test no.	0	1	2	3	4	5	6	GND	7	8	9	D	C	B	A	V_{CC}		Min	Max	Min	Max	Min	Max	
I_{OS4}	3011	50	GND							GND				4.0 V	4.0 V	4.0 V	4.0 V	4.0 V	0	-10	-120	-10	-120	-10	-120	mA
		51		GND						"				"	"	"	"	"	1	"	"	"	"	"	"	"
		52			GND					"				"	"	"	"	"	2	"	"	"	"	"	"	"
		53				GND				"				"	"	"	"	"	3	"	"	"	"	"	"	"
		54					GND			"				"	"	"	"	"	4	"	"	"	"	"	"	"
		55						GND		"				"	"	"	"	"	5	"	"	"	"	"	"	"
		56							GND	"				"	"	"	"	"	6	"	"	"	"	"	"	"
		57								"	GND			"	"	"	"	"	7	"	"	"	"	"	"	"
		58								"		GND		"	"	"	"	"	8	"	"	"	"	"	"	"
		59								"			GND	"	"	"	"	"	9	"	"	"	"	"	"	"
I_{IH}	3010	60								"				6.0 V	GND	GND	GND	6.0 V	D		.05		.01			µA
		61								"				GND	6.0 V	GND	"	"	C		"		"			"
		62								"				"	GND	6.0 V	"	"	B		"		"			"
		63								"				"	GND	GND	6.0 V	"	A		"		"			"
I_{IL}	3009	64								"				6.0 V	6.0 V	6.0 V	GND	"	A		-.05		-.01			"
		65								"				"	6.0 V	GND	6.0 V	"	B		"		"			"
		66								"				"	GND	6.0 V	"	"	C		"		"			"
		67								"				GND	6.0 V	6.0 V	"	"	D		"		"			"
																				Subgroup 4 T_C = +25°C						
																				Min	Max					
C_1	3012	68								GND				2/	2/	2/	2/	GND	D		10					pF
		69								"								"	C		"					"
		70								"								"	B		"					"
		71								"								"	A		"					"
																				Subgroup 7 T_C = +25°C		Subgroup 8 T_C = +125°C		Subgroup 8 T_C = -55°C		
																				Min	Max	Min	Max	Min	Max	
Truth table tests 3/	3009	72	L	H	H	H	H	H	H	H	H	GND	H	H	H	H	B	B	B	4.5 V	4/						
		73	H	L	H	H	H	H	H	H	H	"	H	H	H	H	"	B	B	"	"						
		74	"	H	L	H	H	H	H	H	H	"	H	H	H	H	"	A	B	"	"						
		75	"	"	H	L	H	H	H	H	H	"	H	H	H	H	"	B	B	"	"						
		76	"	"	"	H	L	H	H	H	H	"	H	H	H	H	"	A	B	"	"						
		77	"	"	"	"	H	L	H	H	H	"	H	H	H	H	"	B	B	"	"						
		78	"	"	"	"	"	H	L	H	H	"	H	H	H	H	"	A	B	"	"						
		79	"	"	"	"	"	"	H	L	H	"	H	H	H	H	"	B	B	"	"						
		80	"	"	"	"	"	"	"	H	L	"	H	H	H	H	"	A	B	"	"						
		81	"	"	"	"	"	"	"	"	H	"	H	H	H	H	"	B	B	"	"						
		82	"	"	"	"	"	"	"	"	"	"	H	H	H	H	"	A	B	"	"						
		83	"	"	"	"	"	"	"	"	"	"	"	H	H	H	"	B	B	"	"						
		84	"	"	"	"	"	"	"	"	"	"	"	"	H	H	"	A	B	"	"						
		85	"	"	"	"	"	"	"	"	"	"	"	"	"	H	"	B	B	"	"						
		86	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	A	B	"	"						
		87	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	B	B	"	"						

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TABLE III. Group A inspection for device type 01 - Continued.

Symbol	MIL-STD-883 method	Case 2 / E	Terminal conditions 1/ 2 / 1	3 / 2	4 / 3	5 / 4	7 / 5	8 / 6	9 / 7	10 / 8	12 / 9	13 / 10	14 / 11	15 / 12	17 / 13	18 / 14	19 / 15	20 / 16	Measured terminal	Test limits: Subgroup 9 T_C = +25°C 5/ Min	Subgroup 9 Max	Subgroup 10 T_C = +125°C Min	Subgroup 10 Max	Subgroup 11 T_C = -55°C 5/ Min	Subgroup 11 Max	Unit
		Test no.	0	1	2	3	4	5	6	GND	7	8	9	D	C	B	A	V_{CC}								
t_{PHL1}	3003 (Fig. 6)	88	OUT							GND				GND	GND	GND	IN	4.5 V	A to O	4	26	4	35	4	26	ns
		89	"							"				"	GND	IN	GND	"	B to O	"	"	"	"	"	"	"
		90	"							"				IN	IN	GND	"	"	C to O	"	"	"	"	"	"	"
		91								"				IN	GND	"	"	"	D to O	"	"	"	"	"	"	"
		92		OUT						"				GND	"	"	IN	"	A to 1	"	"	"	"	"	"	"
		93		"						"				"	"	IN	4.5 V	"	B to 1	"	"	"	"	"	"	"
		94		"						"				"	IN	GND	"	"	C to 1	"	"	"	"	"	"	"
		95								"				IN	GND	GND	"	"	D to 1	"	"	"	"	"	"	"
		96			OUT					"				GND	"	4.5 V	IN	"	A to 2	"	"	"	"	"	"	"
		97			"					"				"	"	IN	GND	"	B to 2	"	"	"	"	"	"	"
		98			"					"				"	IN	4.5 V	"	"	C to 2	"	"	"	"	"	"	"
		99								"				IN	GND	"	"	"	D to 2	"	"	"	"	"	"	"
		100				OUT				"				GND	"	"	IN	"	A to 3	"	"	"	"	"	"	"
		101				"				"				"	"	IN	4.5 V	"	B to 3	"	"	"	"	"	"	"
		102				"				"				IN	IN	4.5 V	"	"	C to 3	"	"	"	"	"	"	"
		103								"				IN	GND	4.5 V	"	"	D to 3	"	"	"	"	"	"	"
		104					OUT			"				GND	4.5 V	GND	IN	"	A to 4	"	"	"	"	"	"	"
		105					"			"				"	4.5 V	IN	GND	"	B to 4	"	"	"	"	"	"	"
		106					"			"				"	IN	GND	"	"	C to 4	"	"	"	"	"	"	"
		107								"				IN	4.5 V	"	"	"	D to 4	"	"	"	"	"	"	"
		108						OUT		"				GND	"	"	IN	"	A to 5	"	"	"	"	"	"	"
		109						"		"				"	"	IN	4.5 V	"	B to 5	"	"	"	"	"	"	"
		110						"		"				"	IN	GND	"	"	C to 5	"	"	"	"	"	"	"
		111								"				IN	4.5 V	GND	"	"	D to 5	"	"	"	"	"	"	"
		112							OUT	"				GND	"	4.5 V	IN	"	A to 6	"	"	"	"	"	"	"
		113							"	"				"	"	IN	GND	"	B to 6	"	"	"	"	"	"	"
		114							"	"				"	IN	4.5 V	"	"	C to 6	"	"	"	"	"	"	"
		115								"				IN	4.5 V	"	"	"	D to 6	"	"	"	"	"	"	"
		116								"	OUT			GND	"	"	IN	"	A to 7	"	"	"	"	"	"	"
		117								"	"			"	"	IN	4.5 V	"	B to 7	"	"	"	"	"	"	"
		118								"	"			"	IN	4.5 V	"	"	C to 7	"	"	"	"	"	"	"
		119								"	"			IN	4.5 V	4.5 V	"	"	D to 7	"	"	"	"	"	"	"
		120								"	"	OUT		4.5 V	GND	GND	IN	"	A to 8	"	"	"	"	"	"	"
		121								"	"	"		"	GND	IN	GND	"	B to 8	"	"	"	"	"	"	"
		122								"	"	"		"	IN	GND	"	"	C to 8	"	"	"	"	"	"	"
		123								"	"	"		IN	GND	"	"	"	D to 8	"	"	"	"	"	"	"
		124								"	"	"	OUT	4.5 V	"	"	IN	"	A to 9	"	"	"	"	"	"	"
		125								"	"	"	"	"	"	IN	4.5 V	"	B to 9	"	"	"	"	"	"	"
		126								"	"	"	"	"	IN	GND	"	"	C to 9	"	"	"	"	"	"	"
		127								"	"	"	"	IN	GND	"	"	"	D to 9	"	"	"	"	"	"	"
t_{PLH1}	3003 (Fig. 6)	128	OUT							"				GND	"	"	IN	"	A to O	"	"	"	"	"	"	"
		129	"							"				"	"	IN	GND	"	B to O	"	"	"	"	"	"	"
		130	"							"				"	IN	GND	"	"	C to O	"	"	"	"	"	"	"
		131								"				IN	GND	"	"	"	D to O	"	"	"	"	"	"	"
		132		OUT						"				GND	"	"	IN	"	A to 1	"	"	"	"	"	"	"
		133		"						"				"	"	IN	4.5 V	"	B to 1	"	"	"	"	"	"	"
		134		"						"				"	IN	GND	"	"	C to 1	"	"	"	"	"	"	"
		135								"				IN	GND	GND	"	"	D to 1	"	"	"	"	"	"	"
		136			OUT					"				GND	"	4.5 V	IN	"	A to 2	"	"	"	"	"	"	"
		137			"					"				"	"	IN	GND	"	B to 2	"	"	"	"	"	"	"
		138			"					"				"	IN	4.5 V	"	"	C to 2	"	"	"	"	"	"	"
		139								"				IN	GND	"	"	"	D to 2	"	"	"	"	"	"	"
		140				OUT				"				GND	"	"	IN	"	A to 3	"	"	"	"	"	"	"
		141				"				"				"	"	IN	4.5 V	"	B to 3	"	"	"	"	"	"	"
		142				"				"				IN	IN	4.5 V	"	"	C to 3	"	"	"	"	"	"	"
		143								"				IN	GND	4.5 V	"	"	D to 3	"	"	"	"	"	"	"

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TABLE III. Group A inspection for device type 01 - Continued.

Symbol	MIL-STD-883 method	Case	Terminal conditions 1/																Measured terminal	Test limits: Subgroup 9 T_C = +25°C 5/		Subgroup 10 T_C = +125°C		Subgroup 11 T_C = -55°C 5/		Unit
		2	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20								
		E	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16								
		Test no.	0	1	2	3	4	5	6	GND	7	8	9	D	C	B	A	V_{CC}		Min	Max	Min	Max	Min	Max	
t_{PLH1}	3003 (Fig. 6)	144					OUT			GND				GND	4.5 V	GND	IN	4.5 V	A to 4	4	26	4	35	4	26	ns
		145					"			"				"	4.5 V	IN	GND	"	B to 4	"	"	"	"	"	"	"
		146					"			"				IN	IN	GND	"	"	C to 4	"	"	"	"	"	"	"
		147								"				IN	4.5 V	"	"	"	D to 4	"	"	"	"	"	"	"
		148						OUT		"				GND	"	"	IN	"	A to 5	"	"	"	"	"	"	"
		149						"		"				"	"	IN	4.5 V	"	B to 5	"	"	"	"	"	"	"
		150						"		"				IN	IN	GND	"	"	C to 5	"	"	"	"	"	"	"
		151						"		"				IN	4.5 V	GND	"	"	D to 5	"	"	"	"	"	"	"
		152							OUT	"				GND	"	4.5 V	IN	"	A to 6	"	"	"	"	"	"	"
		153							"	"				"	"	IN	GND	"	B to 6	"	"	"	"	"	"	"
		154							"	"				"	IN	4.5 V	"	"	C to 6	"	"	"	"	"	"	"
		155							"	"				IN	4.5 V	"	"	"	D to 6	"	"	"	"	"	"	"
		156								"	OUT			GND	"	"	IN	"	A to 7	"	"	"	"	"	"	"
		157								"	"			"	"	IN	4.5 V	"	B to 7	"	"	"	"	"	"	"
		158								"	"			"	IN	4.5 V	"	"	C to 7	"	"	"	"	"	"	"
		159								"	"			IN	4.5 V	4.5 V	"	"	D to 7	"	"	"	"	"	"	"
		160								"	"	OUT		4.5 V	GND	GND	IN	"	A to 8	"	"	"	"	"	"	"
		161								"	"	"		"	GND	IN	GND	"	B to 8	"	"	"	"	"	"	"
		162								"	"	"		"	IN	GND	"	"	C to 8	"	"	"	"	"	"	"
		163								"	"	"		IN	GND	"	"	"	D to 8	"	"	"	"	"	"	"
		164								"	"	"	OUT	4.5 V	"	"	IN	"	A to 9	"	"	"	"	"	"	"
		165								"	"	"	"	"	"	IN	4.5 V	"	B to 9	"	"	"	"	"	"	"
		166								"	"	"	"	"	IN	GND	"	"	C to 9	"	"	"	"	"	"	"
		167								"	"	"	"	IN	GND	"	"	"	D to 9	"	"	"	"	"	"	"
t_{THL}	3004 (Fig. 6)	168	OUT											GND	"	"	IN	"	0	3	15	3	20	3	15	ns
		169		OUT										"	"	"	"	"	1	"	"	"	"	"	"	"
		170			OUT									"	"	4.5 V	"	"	2	"	"	"	"	"	"	"
		171				OUT								"	"	4.5 V	"	"	3	"	"	"	"	"	"	"
		172					OUT							"	4.5 V	GND	"	"	4	"	"	"	"	"	"	"
		173						OUT						"	"	GND	"	"	5	"	"	"	"	"	"	"
		174							OUT					"	"	4.5 V	"	"	6	"	"	"	"	"	"	"
		175									OUT			"	"	4.5 V	"	"	7	"	"	"	"	"	"	"
		176										OUT		4.5 V	GND	GND	"	"	8	"	"	"	"	"	"	"
		177											OUT	4.5 V	"	"	"	"	9	"	"	"	"	"	"	"
t_{TLH}	3004 (Fig. 6)	178	OUT											GND	"	"	"	"	0	"	"	"	"	"	"	ns
		179		OUT										"	"	"	"	"	1	"	"	"	"	"	"	"
		180			OUT									"	"	4.5 V	"	"	2	"	"	"	"	"	"	"
		181				OUT								"	"	4.5 V	"	"	3	"	"	"	"	"	"	"
		182					OUT							"	4.5 V	GND	"	"	4	"	"	"	"	"	"	"
		183						OUT						"	"	GND	"	"	5	"	"	"	"	"	"	"
		184							OUT					"	"	4.5 V	"	"	6	"	"	"	"	"	"	"
		185									OUT			"	"	4.5 V	"	"	7	"	"	"	"	"	"	"
		186										OUT		4.5 V	GND	GND	"	"	8	"	"	"	"	"	"	"
		187											OUT	4.5 V	GND	GND	"	"	9	"	"	"	"	"	"	"

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TABLE III. Group A inspection for device type 02.

Symbol	MIL-STD-883 method	Case	Terminal conditions 1/																Measured terminal	Test Limits						Unit
																				Subgroup 1 T_C = +25°C		Subgroup 2 T_C = +125°C		Subgroup 3 T_C = -55°C		
		2	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20								
		E	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16								
		Test no.	A	B	C	$\overline{G2A}$	$\overline{G2B}$	G1	Y7	GND	Y6	Y5	Y4	Y3	Y2	Y1	Y0	V_{CC}		Min	Max	Min	Max	Min	Max	
$V_{IC(pos)}$		1	1 mA							1/								GND	A		1.5					V
		2		1 mA						"								"	B		"					"
		3			1 mA					"								"	C		"					"
		4				1 mA				"								"	$\overline{G2A}$		"					"
		5					1 mA			"								"	$\overline{G2B}$		"					"
		6						1 mA		"								"	G1		"					"
$V_{IC(neg)}$		7	-1 mA							GND								1/	A		-1.5					V
		8		-1 mA						"								"	B		"					"
		9			-1 mA					"								"	C		"					"
		10				-1 mA				"								"	$\overline{G2A}$		"					"
		11					-1 mA			"								"	$\overline{G2B}$		"					"
		12						-1 mA		"								"	G1		"					"
I_{CC}	3005	13	6.0 V	6.0 V	6.0 V	6.0 V	6.0 V	6.0 V		"								6.0 V	V_{CC}		0.2		20			µA
V_{OH3}	3006	14	1.2 V	1.2 V	1.2 V	4.2 V	1.2 V	1.2 V		"					-20 µA	-20 µA	-20 µA	"	Y0	5.95		5.95		5.95		V
		15	"	"	"	"	"	"		"					"	"	"	"	Y1	"		"		"		"
		16	"	"	"	"	"	"		"					"	"	"	"	Y2	"		"		"		"
		17	"	"	"	"	"	"		"					"	"	"	"	Y3	"		"		"		"
		18	"	"	"	"	"	"		"			-20 µA		"	"	"	"	Y4	"		"		"		"
		19	"	"	"	"	"	"		"		-20 µA			"	"	"	"	Y5	"		"		"		"
		20	"	"	"	"	"	"		"	-20 µA				"	"	"	"	Y6	"		"		"		"
		21	"	"	"	"	"	"	-20 µA	"				-20 µA	"	"	"	"	Y7	"		"		"		"
V_{OH5}	3006	22	"	"	"	"	"	"		"					-5.2 mA	-5.2 mA	-5.2 mA	"	Y0	5.48		5.2		5.48		"
		23	"	"	"	"	"	"		"					"	"	"	"	Y1	"		"		"		"
		24	"	"	"	"	"	"		"					"	"	"	"	Y2	"		"		"		"
		25	"	"	"	"	"	"		"					"	"	"	"	Y3	"		"		"		"
		26	"	"	"	"	"	"		"			-5.2 mA		"	"	"	"	Y4	"		"		"		"
		27	"	"	"	"	"	"		"		-5.2 mA			"	"	"	"	Y5	"		"		"		"
		28	"	"	"	"	"	"		"	-5.2 mA				"	"	"	"	Y6	"		"		"		"
		29	"	"	"	"	"	"	-5.2 mA	"				-5.2 mA	"	"	"	"	Y7	"		"		"		"
V_{OL3}	3007	30	"	"	"	1.2 V	"	4.2 V		"					20 µA	20 µA	20 µA	"	Y0		0.05		0.05		0.05	"
		31	4.2 V	"	"	"	"	"		"					"	"	"	"	Y1		"		"		"	"
		32	1.2 V	4.2 V	"	"	"	"		"					"	"	"	"	Y2		"		"		"	"
		33	4.2 V	4.2 V	"	"	"	"		"					"	"	"	"	Y3		"		"		"	"
		34	1.2 V	1.2 V	4.2 V	"	"	"		"			20 µA		"	"	"	"	Y4		"		"		"	"
		35	4.2 V	1.2 V	"	"	"	"		"		20 µA			"	"	"	"	Y5		"		"		"	"
		36	1.2 V	4.2 V	"	"	"	"		"	20 µA				"	"	"	"	Y6		"		"		"	"
		37	4.2 V	4.2 V	"	"	"	"	20 µA	"				20 µA	"	"	"	"	Y7		"		"		"	"
V_{OL5}	3007	38	1.2 V	1.2 V	1.2 V	"	"	"		"					5.2 mA	5.2 mA	5.2 mA	"	Y0		0.26		0.4		0.26	"
		39	4.2 V	1.2 V	"	"	"	"		"					"	"	"	"	Y1		"		"		"	"
		40	1.2 V	4.2 V	"	"	"	"		"					"	"	"	"	Y2		"		"		"	"
		41	4.2 V	4.2 V	"	"	"	"		"					"	"	"	"	Y3		"		"		"	"
		42	1.2 V	1.2 V	4.2 V	"	"	"		"			5.2 mA		"	"	"	"	Y4		"		"		"	"
		43	4.2 V	1.2 V	"	"	"	"		"		5.2 mA			"	"	"	"	Y5		"		"		"	"
		44	1.2 V	4.2 V	"	"	"	"		"	5.2 mA				"	"	"	"	Y6		"		"		"	"
		45	4.2 V	4.2 V	"	"	"	"	5.2 mA	"				5.2 mA	"	"	"	"	Y7		"		"		"	"

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TABLE III. Group A inspection for device type 02 - Continued.

Symbol	MIL-STD-883 method	Case 2 / L / Test no.	Terminal conditions 1/ 2 / 1 / A	3 / 2 / B	4 / 3 / C	5 / 4 / $\overline{G2A}$	7 / 5 / $\overline{G2B}$	8 / 6 / G1	9 / 7 / Y7	10 / 8 / GND	12 / 9 / Y6	13 / 10 / Y5	14 / 11 / Y4	15 / 12 / Y3	17 / 13 / Y2	18 / 14 / Y1	19 / 15 / Y0	20 / 16 / V_{CC}	Measured terminal	Test limits: Subgroup 1 T_C = +25°C Min	Subgroup 1 T_C = +25°C Max	Subgroup 2 T_C = +125°C Min	Subgroup 2 T_C = +125°C Max	Subgroup 3 T_C = -55°C Min	Subgroup 3 T_C = -55°C Max	Unit
I_{OS4}	3011	46	GND	GND	GND	4.0 V	4.0 V	GND		GND					GND			4.0 V	Y0	-10	-120	-10	-120	-10	-120	mA
		47	"	"	"	"	"	"		"						GND		"	Y1	"	"	"	"	"	"	"
		48	"	"	"	"	"	"		"							GND	"	Y2	"	"	"	"	"	"	"
		49	"	"	"	"	"	"		"								"	Y3	"	"	"	"	"	"	"
		50	"	"	"	"	"	"		"			GND					"	Y4	"	"	"	"	"	"	"
		51	"	"	"	"	"	"		"		GND						"	Y5	"	"	"	"	"	"	"
		52	"	"	"	"	"	"		"	GND							"	Y6	"	"	"	"	"	"	"
		53	"	"	"	"	"	"	GND	"				GND				"	Y7	"	"	"	"	"	"	"
I_{IH}	3010	54	6.0 V	GND	GND	GND	GND	GND										6.0 V	A		0.05		0.1			uA
		55	GND	6.0 V	GND	"	"	"										"	B		"		"			"
		56	"	GND	6.0 V	6.0 V	"	"										"	C		"		"			"
		57	"	"	GND	GND	6.0 V	"										"	$\overline{G2A}$		"		"			"
		58	"	"	"	6.0 V	GND	"										"	$\overline{G2B}$		"		"			"
		59	"	"	"	GND	GND	6.0 V										"	G1		"		"			"
I_{IL}	3009	60	"	6.0 V	6.0 V	6.0 V	6.0 V	"										"	A		-0.05		-0.1			uA
		61	6.0 V	GND	6.0 V	"	"	"										"	B		"		"			"
		62	"	6.0 V	GND	GND	"	"										"	C		"		"			"
		63	"	"	6.0 V	"	GND	"										"	$\overline{G2A}$		"		"			"
		64	"	"	"	GND	6.0 V	"										"	$\overline{G2B}$		"		"			"
		65	"	"	"	6.0 V	GND	GND										"	G1		"		"			"
																				Subgroup 4 T_C = +25°C Min	Subgroup 4 T_C = +25°C Max					
C_I	3012	66	2/							GND								GND	A		10					pF
		67		2/						"								"	B		"					"
		68			2/					"								"	C		"					"
		69				2/				"								"	$\overline{G2A}$		"					"
		70					2/			"								"	$\overline{G2B}$		"					"
		71						2/		"								"	G1		"					"
																				Subgroup 7 T_C = +25°C Min	Subgroup 7 T_C = +25°C Max	Subgroup 8 T_C = +125°C Min	Subgroup 8 T_C = +125°C Max	Subgroup 8 T_C = -55°C Min	Subgroup 8 T_C = -55°C Max	
Truth table tests 3/	3014	72	A	A	A	A	A	A	H	GND	H	H	H	H	H	H	H	4.5 V	4/							
		73	B	A	"	"	"	"	"	"	"	"	"	"	"	"	"	"									
		74	"	B	B	"	"	"	"	"	"	"	"	"	"	"	"	"									
		75	"	"	"	B	"	"	"	"	"	"	"	"	"	"	"	"									
		76	"	"	"	"	B	"	"	"	"	"	"	"	"	"	"	"									
		77	"	"	"	"	"	B	"	"	"	"	"	"	"	"	"	"									
		78	"	"	"	"	"	"	A	"	"	"	"	"	"	"	"	"									
		79	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"									
		80	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"									
		81	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"									
		82	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"									
		83	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"									
		84	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"									
		85	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"									
		86	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"									
		87	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"									
		88	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"									

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TABLE III. Group A inspection for device type 02 - Continued.

Symbol	MIL-STD-883 method	Case	Terminal conditions 1/																Measured terminal	Test limits						Unit
																				Subgroup 9 T_C = +25°C 5/		Subgroup 10 T_C = +125°C		Subgroup 11 T_C = -55°C 5/		
		2	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20								
		L	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16								
		Test no.	A	B	C	$\overline{G2A}$	$\overline{G2B}$	G1	Y7	GND	Y6	Y5	Y4	Y3	Y2	Y1	Y0	V_{CC}		Min	Max	Min	Max	Min	Max	
t_{PHL1}	3003 (Fig. 6)	89	IN	GND	GND	GND	GND	4.5 V		GND							OUT	4.5 V	A to Y0	5	35	5	47	5	35	ns
t_{PHL1}		90	GND	IN	GND	"	"	"		"							"	"	B to Y0	"	"	"	"	"	"	"
t_{PHL1}		91	"	GND	IN	"	"	"		"							"	"	C to Y0	"	"	"	"	"	"	"
t_{PHL2}		92	"	"	GND	IN	"	"		"							"	"	$\overline{G2A}$ to Y0	4	31	4	41	4	31	"
t_{PHL2}		93	"	"	"	GND	IN	"		"							"	"	$\overline{G2B}$ to Y0	"	"	"	"	"	"	"
t_{PHL2}		94	"	"	"	"	GND	IN		"							"	"	G1 to Y0	"	"	"	"	"	"	"
t_{PHL1}		95	IN	"	"	"	"	4.5 V		"						OUT	"	"	A to Y1	5	35	5	47	5	35	"
t_{PHL1}		96	4.5 V	IN	"	"	"	"		"						"	"	"	B to Y1	"	"	"	"	"	"	"
t_{PHL1}		97	"	GND	IN	"	"	"		"						"	"	"	C to Y1	"	"	"	"	"	"	"
t_{PHL2}		98	"	"	GND	IN	"	"		"						"	"	"	$\overline{G2A}$ to Y1	4	31	4	41	4	31	"
t_{PHL2}		99	"	"	"	GND	IN	"		"						"	"	"	$\overline{G2B}$ to Y1	"	"	"	"	"	"	"
t_{PHL2}		100	"	"	"	"	GND	IN		"						"	"	"	G1 to Y1	"	"	"	"	"	"	"
t_{PHL1}		101	IN	4.5 V	"	"	"	4.5 V		"					OUT	"	"	"	A to Y2	5	35	5	47	5	35	"
t_{PHL1}		102	GND	IN	"	"	"	"		"					"	"	"	"	B to Y2	"	"	"	"	"	"	"
t_{PHL1}		103	"	4.5 V	IN	"	"	"		"					"	"	"	"	C to Y2	"	"	"	"	"	"	"
t_{PHL2}		104	"	"	GND	IN	"	"		"					"	"	"	"	$\overline{G2A}$ to Y2	4	31	4	41	4	31	"
t_{PHL2}		105	"	"	"	GND	IN	"		"					"	"	"	"	$\overline{G2B}$ to Y2	"	"	"	"	"	"	"
t_{PHL2}		106	"	"	"	"	GND	IN		"					"	"	"	"	G1 to Y2	"	"	"	"	"	"	"
t_{PHL1}		107	IN	"	"	"	"	4.5 V		"				OUT	"	"	"	"	A to Y3	5	35	5	47	5	35	"
t_{PHL1}		108	4.5 V	IN	"	"	"	"		"				"	"	"	"	"	B to Y3	"	"	"	"	"	"	"
t_{PHL1}		109	"	4.5 V	IN	"	"	"		"				"	"	"	"	"	C to Y3	"	"	"	"	"	"	"
t_{PHL2}		110	"	"	"	IN	"	"		"				"	"	"	"	"	$\overline{G2A}$ to Y3	4	31	4	41	4	31	"
t_{PHL2}		111	"	"	"	GND	IN	"		"				"	"	"	"	"	$\overline{G2B}$ to Y3	"	"	"	"	"	"	"
t_{PHL2}		112	"	"	"	"	GND	IN		"				"	"	"	"	"	G1 to Y3	"	"	"	"	"	"	"
t_{PHL1}		113	IN	GND	4.5 V	"	"	4.5 V		"			OUT	"	"	"	"	"	A to Y4	5	35	5	47	5	35	"
t_{PHL1}		114	GND	IN	4.5 V	"	"	"		"			"	"	"	"	"	"	B to Y4	"	"	"	"	"	"	"
t_{PHL1}		115	"	GND	IN	"	"	"		"			"	"	"	"	"	"	C to Y4	"	"	"	"	"	"	"
t_{PHL2}		116	"	"	4.5 V	IN	"	"		"			"	"	"	"	"	"	$\overline{G2A}$ to Y4	4	31	4	41	4	31	"
t_{PHL2}		117	"	"	"	GND	IN	"		"			"	"	"	"	"	"	$\overline{G2B}$ to Y4	"	"	"	"	"	"	"
t_{PHL2}		118	"	"	"	"	GND	IN		"			"	"	"	"	"	"	G1 to Y4	"	"	"	"	"	"	"
t_{PHL1}		119	IN	GND	"	"	"	4.5 V		"		OUT	"	"	"	"	"	"	A to Y5	5	35	5	47	5	35	"
t_{PHL1}		120	4.5 V	IN	"	"	"	"		"		"	"	"	"	"	"	"	B to Y5	"	"	"	"	"	"	"
t_{PHL1}		121	"	GND	IN	"	"	"		"		"	"	"	"	"	"	"	C to Y5	"	"	"	"	"	"	"
t_{PHL2}		122	"	"	4.5 V	IN	"	"		"		"	"	"	"	"	"	"	$\overline{G2A}$ to Y5	4	31	4	41	4	31	"
t_{PHL2}		123	"	"	"	GND	IN	"		"		"	"	"	"	"	"	"	$\overline{G2B}$ to Y5	"	"	"	"	"	"	"
t_{PHL2}		124	"	"	"	"	GND	IN		"		"	"	"	"	"	"	"	G1 to Y5	"	"	"	"	"	"	"
t_{PHL1}		125	IN	4.5 V	"	"	"	4.5 V		"	OUT	"	"	"	"	"	"	"	A to Y6	5	35	5	47	5	35	"
t_{PHL1}		126	GND	IN	"	"	"	"		"	"	"	"	"	"	"	"	"	B to Y6	"	"	"	"	"	"	"
t_{PHL1}		127	"	4.5 V	IN	"	"	"		"	"	"	"	"	"	"	"	"	C to Y6	"	"	"	"	"	"	"
t_{PHL2}		128	"	"	4.5 V	IN	"	"		"	"	"	"	"	"	"	"	"	$\overline{G2A}$ to Y6	4	31	4	41	4	31	"
t_{PHL2}		129	"	"	"	GND	IN	"		"	"	"	"	"	"	"	"	"	$\overline{G2B}$ to Y6	"	"	"	"	"	"	"
t_{PHL2}		130	"	"	"	"	GND	IN		"	"	"	"	"	"	"	"	"	G1 to Y6	"	"	"	"	"	"	"
t_{PHL1}		131	IN	"	"	"	"	4.5 V	OUT	"	"	"	"	"	"	"	"	"	A to Y7	5	35	5	47	5	35	"
t_{PHL1}		132	4.5 V	IN	"	"	"	"	"	"	"	"	"	"	"	"	"	"	B to Y7	"	"	"	"	"	"	"
t_{PHL1}		133	"	4.5 V	IN	"	"	"	"	"	"	"	"	"	"	"	"	"	C to Y7	"	"	"	"	"	"	"
t_{PHL2}		134	"	"	4.5 V	IN	"	"	"	"	"	"	"	"	"	"	"	"	$\overline{G2A}$ to Y7	4	31	4	41	4	31	"
t_{PHL2}		135	"	"	"	"	IN	"	"	"	"	"	"	"	"	"	"	"	$\overline{G2B}$ to Y7	"	"	"	"	"	"	"
t_{PHL2}		136	"	"	"	"	IN	IN	"	"	"	"	"	"	"	"	"	"	G1 to Y7	"	"	"	"	"	"	"

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TABLE III. Group A inspection for device type 02 - Continued.

Symbol	MIL-STD-883 method	Case	Terminal conditions 1/																Measured terminal	Test Limits: Subgroup 9 T_C = +25°C 5/		Subgroup 10 T_C = +125°C		Subgroup 11 T_C = -55°C 5/		Unit
		2	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20								
		E	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16								
		Test no.	A	B	C	$\overline{G2A}$	$\overline{G2B}$	G1	Y7	GND	Y6	Y5	Y4	Y3	Y2	Y1	Y0	V_{CC}		Min	Max	Min	Max	Min	Max	
t_{PLH1}	3003 (Fig. 6)	137	IN	GND	GND	GND	GND	4.5 V		GND							OUT	4.5 V	A to Y0	4	35	4	47	4	35	ns
t_{PLH1}		138	GND	IN	GND	"	"	"		"							"	"	B to Y0	"	"	"	"	"	"	
t_{PLH1}		139	"	GND	IN	"	"	"		"							"	"	C to Y0	"	"	"	"	"	"	
t_{PLH2}		140	"	"	GND	IN	"	"		"							"	"	$\overline{G2A}$ to Y0	"	31	"	41	"	31	
t_{PLH2}		141	"	"	"	GND	IN	"		"							"	"	$\overline{G2B}$ to Y0	"	"	"	"	"	"	
t_{PLH2}		142	"	"	"	"	GND	IN		"							"	"	G1 to Y0	"	"	"	"	"	"	
t_{PLH1}		143	IN	"	"	"	"	4.5 V		"						OUT	"	"	A to Y1	"	35	"	47	"	35	
t_{PLH1}		144	4.5 V	IN	"	"	"	"		"						"	"	"	B to Y1	"	"	"	"	"	"	
t_{PLH1}		145	"	GND	IN	"	"	"		"						"	"	"	C to Y1	"	"	"	"	"	"	
t_{PLH2}		146	"	"	GND	IN	"	"		"						"	"	"	$\overline{G2A}$ to Y1	"	31	"	41	"	31	
t_{PLH2}		147	"	"	"	GND	IN	"		"						"	"	"	$\overline{G2B}$ to Y1	"	"	"	"	"	"	
t_{PLH2}		148	"	"	"	"	GND	IN		"						"	"	"	G1 to Y1	"	"	"	"	"	"	
t_{PLH1}		149	IN	4.5 V	"	"	"	4.5 V		"					OUT	"	"	"	A to Y2	"	35	"	47	"	35	
t_{PLH1}		150	GND	IN	"	"	"	"		"					"	"	"	"	B to Y2	"	"	"	"	"	"	
t_{PLH1}		151	"	4.5 V	IN	"	"	"		"					"	"	"	"	C to Y2	"	"	"	"	"	"	
t_{PLH2}		152	"	"	GND	IN	"	"		"					"	"	"	"	$\overline{G2A}$ to Y2	"	31	"	41	"	31	
t_{PLH2}		153	"	"	"	GND	IN	"		"					"	"	"	"	$\overline{G2B}$ to Y2	"	"	"	"	"	"	
t_{PLH2}		154	"	"	"	"	GND	IN		"					"	"	"	"	G1 to Y2	"	"	"	"	"	"	
t_{PLH1}		155	IN	4.5 V	"	"	"	4.5 V		"				OUT	"	"	"	"	A to Y3	"	35	"	47	"	35	
t_{PLH1}		156	4.5 V	IN	"	"	"	"		"				"	"	"	"	"	B to Y3	"	"	"	"	"	"	
t_{PLH1}		157	"	4.5 V	IN	"	"	"		"				"	"	"	"	"	C to Y3	"	"	"	"	"	"	
t_{PLH2}		158	"	"	GND	IN	"	"		"				"	"	"	"	"	$\overline{G2A}$ to Y3	"	31	"	41	"	31	
t_{PLH2}		159	"	"	"	GND	IN	"		"				"	"	"	"	"	$\overline{G2B}$ to Y3	"	"	"	"	"	"	
t_{PLH2}		160	"	"	"	"	GND	IN		"				"	"	"	"	"	G1 to Y3	"	"	"	"	"	"	
t_{PLH1}		161	IN	GND	4.5 V	"	"	4.5 V		"			OUT	"	"	"	"	"	A to Y4	"	35	"	47	"	35	
t_{PLH1}		162	GND	IN	4.5 V	"	"	"		"			"	"	"	"	"	"	B to Y4	"	"	"	"	"	"	
t_{PLH1}		163	"	GND	IN	"	"	"		"			"	"	"	"	"	"	C to Y4	"	"	"	"	"	"	
t_{PLH2}		164	"	"	4.5 V	IN	"	"		"			"	"	"	"	"	"	$\overline{G2A}$ to Y4	"	31	"	41	"	31	
t_{PLH2}		165	"	"	"	GND	IN	"		"			"	"	"	"	"	"	$\overline{G2B}$ to Y4	"	"	"	"	"	"	
t_{PLH2}		166	"	"	"	"	GND	IN		"			"	"	"	"	"	"	G1 to Y4	"	"	"	"	"	"	
t_{PLH1}		167	IN	"	"	"	"	4.5 V		"		OUT	"	"	"	"	"	"	A to Y5	"	35	"	47	"	35	
t_{PLH1}		168	4.5 V	IN	"	"	"	"		"		"	"	"	"	"	"	"	B to Y5	"	"	"	"	"	"	
t_{PLH1}		169	"	GND	IN	"	"	"		"		"	"	"	"	"	"	"	C to Y5	"	"	"	"	"	"	
t_{PLH2}		170	"	"	4.5 V	IN	"	"		"		"	"	"	"	"	"	"	$\overline{G2A}$ to Y5	"	31	"	41	"	31	
t_{PLH2}		171	"	"	"	GND	IN	"		"		"	"	"	"	"	"	"	$\overline{G2B}$ to Y5	"	"	"	"	"	"	
t_{PLH2}		172	"	"	"	"	GND	IN		"		"	"	"	"	"	"	"	G1 to Y5	"	"	"	"	"	"	
t_{PLH1}		173	IN	4.5 V	"	"	"	4.5 V		"	OUT	"	"	"	"	"	"	"	A to Y6	"	35	"	47	"	35	
t_{PLH1}		174	GND	IN	"	"	"	"		"	"	"	"	"	"	"	"	"	B to Y6	"	"	"	"	"	"	
t_{PLH1}		175	"	4.5 V	IN	"	"	"		"	"	"	"	"	"	"	"	"	C to Y6	"	"	"	"	"	"	
t_{PLH2}		176	"	"	4.5 V	IN	"	"		"	"	"	"	"	"	"	"	"	$\overline{G2A}$ to Y6	"	31	"	41	"	31	
t_{PLH2}		177	"	"	"	GND	IN	"		"	"	"	"	"	"	"	"	"	$\overline{G2B}$ to Y6	"	"	"	"	"	"	
t_{PLH2}		178	"	"	"	"	GND	IN		"	"	"	"	"	"	"	"	"	G1 to Y6	"	"	"	"	"	"	
t_{PLH1}		179	IN	"	"	"	"	4.5 V	OUT	"	"	"	"	"	"	"	"	"	A to Y7	"	35	"	47	"	35	
t_{PLH1}		180	4.5 V	IN	"	"	"	"	"	"	"	"	"	"	"	"	"	"	B to Y7	"	"	"	"	"	"	
t_{PLH1}		181	"	4.5 V	IN	"	"	"	"	"	"	"	"	"	"	"	"	"	C to Y7	"	"	"	"	"	"	
t_{PLH2}		182	"	"	4.5 V	IN	"	"	"	"	"	"	"	"	"	"	"	"	$\overline{G2A}$ to Y7	"	31	"	41	"	31	
t_{PLH2}		183	"	"	"	GND	IN	"	"	"	"	"	"	"	"	"	"	"	$\overline{G2B}$ to Y7	"	"	"	"	"	"	
t_{PLH2}		184	"	"	"	GND	GND	IN	"	"	"	"	"	"	"	"	"	"	G1 to Y7	"	"	"	"	"	"	

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TABLE III. Group A inspection for device type 02 - Continued.

Symbol	MIL-STD-883 method	Case	Terminal conditions 1/																Measured terminal	Test limits Subgroup 9 T_C = +25°C 5/		Subgroup 10 T_C = +125°C		Subgroup 11 T_C = -55°C 5/		Unit
		2	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20								
		E	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16								
		Test no.	A	B	C	G2A	G2B	G1	Y7	GND	Y6	Y5	Y4	Y3	Y2	Y1	Y0	V_{CC}		Min	Max	Min	Max	Min	Max	
t_{THL}	3004 (Fig. 6)	185	GND	GND	GND	GND	GND	IN		GND					OUT	OUT	OUT	4.5 V	Y0	3	15	3	20	3	15	ns
		186	4.5 V	GND	"	"	"	"		"								"	Y1	"	"	"	"	"	"	"
		187	GND	4.5 V	"	"	"	"		"								"	Y2	"	"	"	"	"	"	"
		188	4.5 V	4.5 V	"	"	"	"		"								"	Y3	"	"	"	"	"	"	"
		189	GND	GND	4.5 V	"	"	"		"			OUT	OUT				"	Y4	"	"	"	"	"	"	"
		190	4.5 V	GND	"	"	"	"		"		OUT						"	Y5	"	"	"	"	"	"	"
		191	GND	4.5 V	"	"	"	"		"	OUT							"	Y6	"	"	"	"	"	"	"
		192	4.5 V	4.5 V	"	"	"	"	OUT	"								"	Y7	"	"	"	"	"	"	"
t_{TLH}	3004 (Fig. 6)	193	GND	GND	GND	"	"	"		"					OUT	OUT	OUT	"	Y0	"	"	"	"	"	"	"
		194	4.5 V	GND	"	"	"	"		"								"	Y1	"	"	"	"	"	"	"
		195	GND	4.5 V	"	"	"	"		"								"	Y2	"	"	"	"	"	"	"
		196	4.5 V	4.5 V	"	"	"	"		"								"	Y3	"	"	"	"	"	"	"
		197	GND	GND	4.5 V	"	"	"		"			OUT	OUT				"	Y4	"	"	"	"	"	"	"
		198	4.5 V	GND	"	"	"	"		"		OUT						"	Y5	"	"	"	"	"	"	"
		199	GND	4.5 V	"	"	"	"		"	OUT							"	Y6	"	"	"	"	"	"	"
		200	4.5 V	4.5 V	"	"	"	"	OUT	"								"	Y7	"	"	"	"	"	"	"

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TABLE III. Group A inspection for device type 03.

Symbol	MIL-STD-883 method	Case	Terminal conditions 1/																Measured terminal	Test Limits						Unit
																				Subgroup 1 T_C = +25°C		Subgroup 2 T_C = +125°C		Subgroup 3 T_C = -55°C		
		2	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20								
		t	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16		Min	Max	Min	Max	Min	Max	
		Test no.	G1	A1	B1	1Y0	1Y1	1Y2	1Y3	GND	2Y3	2Y2	2Y1	2Y0	B2	A2	G2	V_{CC}								
$V_{IC(pos)}$		1	1 mA							1/								GND	G1		1.5					V
		2		1 mA						"								"	A1		"					"
		3			1 mA					"								"	B1		"					"
		4								"							1 mA	"	G2		"					"
		5								"						1 mA		"	A2		"					"
		6								"					1 mA			"	B2		"					"
$V_{IC(neg)}$		7	-1 mA							GND								1/	G1		-1.5					"
		8		-1 mA						"								"	A1		"					"
		9			-1 mA					"								"	B1		"					"
		10								"							-1 mA	"	G2		"					"
		11								"						-1 mA		"	A2		"					"
		12								"					-1 mA			"	B2		"					"
I_{CC}	3005	13	6.0 V	6.0 V	6.0 V					"					6.0 V	6.0 V	6.0 V	6.0 V	V_{CC}		0.2		20			µA
V_{OH3}	3006	14	4.2 V	GND	GND	-20 µA									GND	GND			1Y0	5.95		5.95		5.95		V
		15	"	"	"		-20 µA								"	"			1Y1	"		"		"		"
		16	"	"	"			-20 µA							"	"			1Y2	"		"		"		"
		17	"	"	"				-20 µA						"	"			1Y3	"		"		"		"
		18		"	"					"					"	"	4.2 V	"	2Y0	"		"		"		"
		19		"	"					"			-20 µA		"	"	"	"	2Y1	"		"		"		"
		20		"	"					"		-20 µA			"	"	"	"	2Y2	"		"		"		"
		21		"	"					"	-20 µA				"	"	"	"	2Y3	"		"		"		"
V_{OH5}	3006	22	4.2 V	"	"	-5.2 mA									"	"			1Y0	5.48		5.2		5.48		"
		23	"	"	"		-5.2 mA								"	"			1Y1	"		"		"		"
		24	"	"	"			-5.2 mA							"	"			1Y2	"		"		"		"
		25		"	"				-5.2 mA						"	"			1Y3	"		"		"		"
		26		"	"					"					"	"	4.2 V	"	2Y0	"		"		"		"
		27		"	"					"			-5.2 mA		"	"	"	"	2Y1	"		"		"		"
		28		"	"					"		-5.2 mA			"	"	"	"	2Y2	"		"		"		"
		29		"	"					"	-5.2 mA				"	"	"	"	2Y3	"		"		"		"
V_{OL3}	3007	30	GND	"	"	20 µA									"	"	GND	"	1Y0		0.05		0.05		0.05	"
		31	"	4.2 V	"		20 µA								"	"	"	"	1Y1		"		"		"	"
		32	"	GND	4.2 V			20 µA							"	"	"	"	1Y2		"		"		"	"
		33	"	4.2 V	4.2 V				20 µA						"	"	"	"	1Y3		"		"		"	"
		34	"	GND	GND					"					"	"	"	"	2Y0		"		"		"	"
		35	"	"	"					"				20 µA	"	4.2 V	"	"	2Y1		"		"		"	"
		36	"	"	"					"		20 µA	20 µA		4.2 V	GND	"	"	2Y2		"		"		"	"
		37	"	"	"					"	20 µA				4.2 V	4.2 V	"	"	2Y3		"		"		"	"
V_{OL5}	3007	38	"	"	"	5.2 mA									GND	GND	"	"	1Y0		0.26		0.4		0.26	"
		39	"	4.2 V	"		5.2 mA								"	"	"	"	1Y1		"		"		"	"
		40	"	GND	4.2 V			5.2 mA							"	"	"	"	1Y2		"		"		"	"
		41	"	4.2 V	4.2 V				5.2 mA						"	"	"	"	1Y3		"		"		"	"
		42	"	GND	GND					"					"	"	"	"	2Y0		"		"		"	"
		43	"	"	"					"			5.2 mA	5.2 mA	"	4.2 V	"	"	2Y1		"		"		"	"
		44	"	"	"					"		5.2 mA			4.2 V	GND	"	"	2Y2		"		"		"	"
		45	"	"	"					"	5.2 mA				4.2 V	4.2 V	"	"	2Y3		"		"		"	"

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TABLE III. Group A inspection for device type 03 - Continued.

Symbol	MIL-STD-883 method	Case	Terminal conditions 1/																Measured terminal	Test limits						Unit
		2	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20		Subgroup 1 T_C = +25°C		Subgroup 2 T_C = +125°C		Subgroup 3 T_C = -55°C		
		E	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16		Min	Max	Min	Max	Min	Max	
		Test no.	G1	A1	B1	1Y0	1Y1	1Y2	1Y3	GND	2Y3	2Y2	2Y1	2Y0	B2	A2	G2	V_{CC}								
I_{OS4}	3011	46	4.0 V	GND	GND	GND	GND	GND	GND	"					GND	GND		4.0 V	1Y0	-10	-120	-10	-120	-10	-120	mA
		47	"	"	"					"					"	"		"	1Y1	"	"	"	"	"	"	"
		48	"	"	"					"					"	"		"	1Y2	"	"	"	"	"	"	"
		49		"	"					"					"	"		"	1Y3	"	"	"	"	"	"	"
		50		"	"					"					"	"	4.0 V	"	2Y0	"	"	"	"	"	"	"
		51		"	"					"					"	"	"	"	2Y1	"	"	"	"	"	"	"
		52		"	"					"					"	"	"	"	2Y2	"	"	"	"	"	"	"
		53		"	"					"	GND				"	"	"	"	2Y3	"	"	"	"	"	"	"
I_{IH}	3010	54	6.0 V	"	"					"					"	"	GND	6.0 V	G1		.05		0.1			µA
		55	GND	6.0 V	"					"					"	"	"	"	A1		"		"			"
		56	"	GND	6.0 V					"					"	"	6.0 V	"	B1		"		"			"
		57	"	"	GND					"					"	"	GND	"	G2		"		"			"
		58	"	"	"					"					"	6.0 V	GND	"	A2		"		"			"
		59	"	"	"					"					6.0 V	GND	GND	"	B2		"		"			"
I_{IL}	3009	60	GND	6.0 V	6.0 V					"					"	6.0 V	6.0 V	6.0 V	G1		-.05		-0.1			µA
		61	6.0 V	GND	6.0 V					"					"	"	"	"	A1		"		"			"
		62	"	6.0 V	GND					"					"	"	GND	"	B1		"		"			"
		63	"	"	6.0 V					"					"	"	6.0 V	"	G2		"		"			"
		64	"	"	"					"					"	GND	6.0 V	"	A2		"		"			"
		65	"	"	"					"					GND	6.0 V	6.0 V	"	B2		"		"			"
																				Subgroup 4 T_C = +25°C						
																				Min	Max					
C_i	3012	66	2/							GND								GND	G1	10						pF
		67		2/						"								"	A1	"						"
		68			2/					"								"	B1	"						"
		69								"								"	G2	"						"
		70								"								"	A2	"						"
		71								"					2/	2/	2/	"	B2	"						"
																				Subgroup 7 T_C = +25°C		Subgroup 8 T_C = +125°C		Subgroup 8 T_C = -55°C		
																				Min	Max	Min	Max	Min	Max	
Truth table tests 3/	3014	72	A	A	A	H	H	H	H	GND	H	H	H	H	A	A	A	4.5 V	4/							
		73	"	B	B	"	"	"	"	"	"	"	"	"	A	B	"	"	"							
		74	"	"	A	"	"	"	"	"	"	"	"	"	B	A	"	"	"							
		75	B	A	B	L	L	"	"	"	"	"	L	"	"	B	B	"	"							
		76	"	B	A	H	H	L	"	"	"	L	H	"	"	A	B	"	"							
		77	"	A	B	"	"	H	L	"	L	H	H	"	"	B	A	"	"							
		78	"	B	B	"	"	"	H	"	H	"	"	L	A	A	A	"	"							
		79	"	A	A	"	"	"	L	"	L	H	H	H	A	B	A	"	"							

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TABLE III. Group A inspection for device type 03 - Continued.

Symbol	MIL-STD-883 method	Case	Terminal conditions 1/																Measured terminal	Test Limits						Unit
																				Subgroup 9 T_C = +25°C 5/		Subgroup 10 T_C = +125°C		Subgroup 11 T_C = -55°C 5/		
		2	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20							
		E	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16							
		Test no.	G1	A1	B1	1Y0	1Y1	1Y2	1Y3	GND	2Y3	2Y2	2Y1	2Y0	B2	A2	G2	V_{CC}		Min	Max	Min	Max	Min	Max	
t_{PHL2}	3003 (Fig. 6)	80	IN	GND	GND	OUT				GND					GND	GND	GND	4.5 V	G1 to 1Y0	6	31	6	41	6	31	ns
t_{PHL1}		81	GND	IN	GND	"				"					"	"	"	"	A1 to 1Y0	"	39	"	51	"	39	
t_{PHL1}		82	GND	GND	IN	"				"					"	"	"	"	B1 to 1Y0	"	39	"	51	"	39	
t_{PHL2}		83	IN	4.5 V	GND		OUT			"					"	"	"	"	G1 to 1Y1	"	31	"	41	"	31	
t_{PHL1}		84	GND	IN	GND		"			"					"	"	"	"	A1 to 1Y1	"	39	"	51	"	39	
t_{PHL1}		85	GND	4.5 V	IN		"			"					"	"	"	"	B1 to 1Y1	"	39	"	51	"	39	
t_{PHL2}		86	IN	GND	4.5 V			OUT		"					"	"	"	"	G1 to 1Y2	"	31	"	41	"	31	
t_{PHL1}		87	GND	IN	4.5 V			"		"					"	"	"	"	A1 to 1Y2	"	39	"	51	"	39	
t_{PHL1}		88	GND	GND	IN			"		"					"	"	"	"	B1 to 1Y2	"	39	"	51	"	39	
t_{PHL2}		89	IN	4.5 V	4.5 V				OUT	"					"	"	"	"	G1 to 1Y3	"	31	"	41	"	31	
t_{PHL1}		90	GND	IN	4.5 V				"	"					"	"	"	"	A1 to 1Y3	8	39	8	51	8	39	
t_{PHL1}		91	"	4.5 V	IN					"					"	"	"	"	B1 to 1Y3	8	39	8	51	8	39	
t_{PHL2}		92	"	GND	GND					"				OUT	"	"	IN	"	G2 to 2Y0	6	31	6	41	6	31	
t_{PHL1}		93	"	"	"					"				"	"	IN	GND	"	A2 to 2Y0	"	39	"	51	"	39	
t_{PHL1}		94	"	"	"					"					IN	GND	GND	"	B2 to 2Y0	"	39	"	51	"	39	
t_{PHL2}		95	"	"	"					"			OUT		GND	4.5 V	IN	"	G2 to 2Y1	"	31	"	41	"	31	
t_{PHL1}		96	"	"	"					"			"		GND	IN	GND	"	A2 to 2Y1	"	39	"	51	"	39	
t_{PHL1}		97	"	"	"					"					IN	4.5 V	GND	"	B2 to 2Y1	"	39	"	51	"	39	
t_{PHL2}		98	"	"	"					"		OUT			4.5 V	GND	IN	"	G2 to 2Y2	"	31	"	41	"	31	
t_{PHL1}		99	"	"	"					"		"			4.5 V	IN	GND	"	A2 to 2Y2	"	39	"	51	"	39	
t_{PHL1}		100	"	"	"					"					IN	GND	GND	"	B2 to 2Y2	"	39	"	51	"	39	
t_{PHL2}		101	"	"	"					"	OUT				4.5 V	4.5 V	IN	"	G2 to 2Y3	"	31	"	41	"	31	
t_{PHL1}		102	"	"	"					"	"				4.5 V	IN	GND	"	A2 to 2Y3	8	39	8	51	8	39	
t_{PHL1}		103	"	"	"					"					IN	4.5 V	"	"	B2 to 2Y3	8	39	8	51	8	39	
t_{PLH2}		104	IN	"	"	OUT				"					GND	GND	"	"	G1 to 1Y0	6	31	6	41	6	31	
t_{PLH1}		105	GND	IN	"	"				"					"	"	"	"	A1 to 1Y0	"	39	"	51	"	39	
t_{PLH1}		106	GND	GND	IN	"				"					"	"	"	"	B1 to 1Y0	"	39	"	51	"	39	
t_{PLH2}		107	IN	4.5 V	GND		OUT			"					"	"	"	"	G1 to 1Y1	"	31	"	41	"	31	
t_{PLH1}		108	GND	IN	GND		"			"					"	"	"	"	A1 to 1Y1	"	39	"	51	"	39	
t_{PLH1}		109	GND	4.5 V	IN		"			"					"	"	"	"	B1 to 1Y1	"	39	"	51	"	39	
t_{PLH2}		110	IN	GND	4.5 V			OUT		"					"	"	"	"	G1 to 1Y2	"	31	"	41	"	31	
t_{PLH1}		111	GND	IN	4.5 V			"		"					"	"	"	"	A1 to 1Y2	"	39	"	51	"	39	
t_{PLH1}		112	GND	GND	IN			"		"					"	"	"	"	B1 to 1Y2	"	39	"	51	"	39	
t_{PLH2}		113	IN	4.5 V	4.5 V				OUT	"					"	"	"	"	G1 to 1Y3	"	31	"	41	"	31	
t_{PLH1}		114	GND	IN	4.5 V				"	"					"	"	"	"	A1 to 1Y3	8	39	8	51	8	39	
t_{PLH1}		115	"	GND	IN					"					"	"	"	"	B1 to 1Y3	8	39	8	51	8	39	
t_{PLH2}		116	"	"	GND					"				OUT	"	"	IN	"	G2 to 2Y0	6	31	6	41	6	31	
t_{PLH1}		117	"	"	"					"				"	"	IN	GND	"	A2 to 2Y0	"	39	"	51	"	39	
t_{PLH1}		118	"	"	"					"					IN	GND	GND	"	B2 to 2Y0	"	39	"	51	"	39	
t_{PLH2}		119	"	"	"					"			OUT		GND	4.5 V	IN	"	G2 to 2Y1	"	31	"	41	"	31	
t_{PLH1}		120	"	"	"					"			"		GND	IN	GND	"	A2 to 2Y1	"	39	"	51	"	39	
t_{PLH1}		121	"	"	"					"					IN	4.5 V	GND	"	B2 to 2Y1	"	39	"	51	"	39	
t_{PLH2}		122	"	"	"					"		OUT			4.5 V	GND	IN	"	G2 to 2Y2	"	31	"	41	"	31	
t_{PLH1}		123	"	"	"					"		"			4.5 V	IN	GND	"	A2 to 2Y2	"	39	"	51	"	39	
t_{PLH1}		124	"	"	"					"					IN	GND	GND	"	B2 to 2Y2	"	39	"	51	"	39	
t_{PLH2}		125	"	"	"					"	OUT				4.5 V	4.5 V	IN	"	G2 to 2Y3	"	31	"	41	"	31	
t_{PLH1}		126	"	"	"					"	"				4.5 V	IN	GND	"	A2 to 2Y3	8	39	8	51	8	39	
t_{PLH1}		127	"	"	"					"					IN	4.5 V	"	"	B2 to 2Y3	8	39	8	51	8	39	
t_{THL}	3003 (Fig. 6)	128	IN	"	"					"					GND	GND	"	"	1Y0	3	15	3	20	3	15	ns
		129	"	4.5 V	"					"					"	"	"	"	1Y1	"	"	"	"	"	"	
		130	"	GND	4.5 V					"					"	"	"	"	1Y2	"	"	"	"	"	"	
		131	"	4.5 V	4.5 V					"					"	"	"	"	1Y3	"	"	"	"	"	"	
		132	GND	GND	GND					"					"	GND	IN	"	2Y0	"	"	"	"	"	"	
		133	"	"	"					"					"	4.5 V	"	"	2Y1	"	"	"	"	"	"	
		134	"	"	"					"					4.5 V	GND	"	"	2Y2	"	"	"	"	"	"	
		135	"	"	"					"					4.5 V	4.5 V	"	"	2Y3	"	"	"	"	"	"	
t_{TLH}	3003 (Fig. 6)	136	IN	"	"					"					GND	GND	GND	"	1Y0	"	"	"	"	"	"	ns
		137	"	4.5 V	"					"					"	"	"	"	1Y1	"	"	"	"	"	"	
		138	"	GND	4.5 V					"					"	"	"	"	1Y2	"	"	"	"	"	"	
		139	"	4.5 V	4.5 V					"					"	"	"	"	1Y3	"	"	"	"	"	"	
		140	GND	GND	GND					"					"	"	IN	"	2Y0	"	"	"	"	"	"	
		141	"	"	"					"					"	4.5 V	"	"	2Y1	"	"	"	"	"	"	
		142	"	"	"					"					4.5 V	GND	"	"	2Y2	"	"	"	"	"	"	
		143	"	"	"					"					4.5 V	4.5 V	"	"	2Y3	"	"	"	"	"	"	

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TABLE III. Group A inspection for device type 04.

Symbol	MIL-STD-883 method	Terminal condition 1/																								Measured terminal	Test limits						Unit
		Case 3	2	3	4	5	6	7	9	10	11	12	13	14	16	17	18	19	20	21	23	24	25	26	27	28		Subgroup 1 T_C = +25°C		Subgroup 2 T_C = +125°C		Subgroup 3 T_C = -55°C		
		L	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24								
		Test no.	Y0	Y1	Y2	Y3	Y4	Y5	Y6	Y7	Y8	Y9	Y10	GND	Y11	Y12	Y13	Y14	Y15	G1	G2	D	C	B	A	V_{CC}		Min	Max	Min	Max	Min	Max	
$V_{IC(pos)}$		1 2 3 4 5 6												1/ " " " " "						1 mA	1 mA	1 mA	1 mA	1 mA	1 mA	GND " " " " "	A B C D G2 G1		1.5 " " " " "					V " " " " "
$V_{IC(neg)}$		7 8 9 10 11 12												GND " " " " "						-1 mA	-1 mA	-1 mA	-1 mA	-1 mA	-1 mA	1/ " " " " "	A B C D G2 G1		-1.5 " " " " "					V " " " " "
I_{CC}	3005	13												"						6.0 V	6.0 V	6.0 V	6.0 V	6.0 V	6.0 V	6.0 V	V_{CC}		0.2		20			uA
V_{OH3}	3006	14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29	-20 uA	-20 uA	-20 uA	-20 uA	-20 uA	-20 uA	-20 uA	-20 uA	-20 uA	-20 uA	-20 uA	" " " " " " " " " " " " " " " "	-20 uA	-20 uA	-20 uA	-20 uA	-20 uA	4.2 V " " " " " " " " " " " " " " "	4.2 V " " " " " " " " " " " " " " "	4.2 V " " " " " " " " " " " " " " "	4.2 V " " " " " " " " " " " " " " "	4.2 V " " " " " " " " " " " " " " "	4.2 V " " " " " " " " " " " " " " "	" " " " " " " " " " " " " " " "	Y0 Y1 Y2 Y3 Y4 Y5 Y6 Y7 Y8 Y9 Y10 Y11 Y12 Y13 Y14 Y15	5.95 " " " " " " " " " " " " " " "		5.95 " " " " " " " " " " " " " " "		5.95 " " " " " " " " " " " " " " "		V " " " " " " " " " " " " " " "
V_{OH5}	3006	30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45	-5.2 mA	-5.2 mA	-5.2 mA	-5.2 mA	-5.2 mA	-5.2 mA	-5.2 mA	-5.2 mA	-5.2 mA	-5.2 mA	-5.2 mA	" " " " " " " " " " " " " " " "	-5.2 mA	-5.2 mA	-5.2 mA	-5.2 mA	-5.2 mA	" " " " " " " " " " " " " " " "	" " " " " " " " " " " " " " " "	" " " " " " " " " " " " " " " "	" " " " " " " " " " " " " " " "	" " " " " " " " " " " " " " " "	" " " " " " " " " " " " " " " "	Y0 Y1 Y2 Y3 Y4 Y5 Y6 Y7 Y8 Y9 Y10 Y11 Y12 Y13 Y14 Y15	5.48 " " " " " " " " " " " " " " "		5.2 " " " " " " " " " " " " " " "		5.48 " " " " " " " " " " " " " " "		V " " " " " " " " " " " " " " "

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TABLE III. Group A inspection for device type 04 - Continued.

Symbol	MIL-STD-883 method	Terminal conditions 1/																								Measured terminal	Test limits						Unit
		Case 3	2	3	4	5	6	7	9	10	11	12	13	14	16	17	18	19	20	21	23	24	25	26	27	28		Subgroup 1 T_C = +25°C		Subgroup 2 T_C = +125°C		Subgroup 3 T_C = -55°C		
		L	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24								
		Test no.	Y0	Y1	Y2	Y3	Y4	Y5	Y6	Y7	Y8	Y9	Y10	GND	Y11	Y12	Y13	Y14	Y15	G1	G2	D	C	B	A	V_{CC}		Min	Max	Min	Max	Min	Max	
V_{OL3}	3007	46	20 µA											GND						GND	GND	GND	GND	GND	GND	6.0 V	Y0		0.05		0.05		0.05	V
		47		20 µA										"						"	"	"	"	GND	4.2 V	"	Y1		"		"		"	"
		48			20 µA									"						"	"	"	"	4.2 V	GND	"	Y2		"		"		"	"
		49				20 µA								"						"	"	"	"	GND	4.2 V	"	Y3		"		"		"	"
		50					20 µA							"						"	"	"	4.2 V	GND	GND	"	Y4		"		"		"	"
		51						20 µA						"						"	"	"	"	GND	4.2 V	"	Y5		"		"		"	"
		52							20 µA					"						"	"	"	"	4.2 V	GND	"	Y6		"		"		"	"
		53								20 µA				"						"	"	"	"	GND	4.2 V	"	Y7		"		"		"	"
		54									20 µA			"						"	"	4.2 V	GND	GND	GND	"	Y8		"		"		"	"
		55										20 µA		"						"	"	"	"	GND	4.2 V	"	Y9		"		"		"	"
		56											20 µA	"						"	"	"	"	4.2 V	GND	"	Y10		"		"		"	"
		57												"						"	"	"	"	GND	4.2 V	"	Y11		"		"		"	"
		58												"						"	"	"	"	GND	GND	"	Y12		"		"		"	"
		59												"						"	"	"	"	GND	4.2 V	"	Y13		"		"		"	"
		60												"						"	"	"	"	4.2 V	GND	"	Y14		"		"		"	"
		61												"	20 µA									4.2 V	4.2 V		Y15							
V_{OL5}	3007	62	5.2 mA											"						"	"	GND	GND	GND	GND	"	Y0		0.26		0.4		0.26	"
		63		5.2 mA										"						"	"	"	"	GND	4.2 V	"	Y1		"		"		"	"
		64			5.2 mA									"						"	"	"	"	4.2 V	GND	"	Y2		"		"		"	"
		65				5.2 mA								"						"	"	"	"	GND	4.2 V	"	Y3		"		"		"	"
		66					5.2 mA							"						"	"	"	4.2 V	GND	GND	"	Y4		"		"		"	"
		67						5.2 mA						"						"	"	"	"	GND	4.2 V	"	Y5		"		"		"	"
		68							5.2 mA					"						"	"	"	"	4.2 V	GND	"	Y6		"		"		"	"
		69								5.2 mA				"						"	"	"	"	GND	4.2 V	"	Y7		"		"		"	"
		70									5.2 mA			"						"	"	4.2 V	GND	GND	GND	"	Y8		"		"		"	"
		71										5.2 mA		"						"	"	"	"	GND	4.2 V	"	Y9		"		"		"	"
		72											5.2 mA	"						"	"	"	"	4.2 V	GND	"	Y10		"		"		"	"
		73												"						"	"	"	"	GND	4.2 V	"	Y11		"		"		"	"
		74												"						"	"	"	"	GND	GND	"	Y12		"		"		"	"
		75												"						"	"	"	"	GND	4.2 V	"	Y13		"		"		"	"
		76												"						"	"	"	"	4.2 V	GND	"	Y14		"		"		"	"
		77												"	5.2 mA									4.2 V	4.2 V		Y15							
I_{OS4}	3011	78	GND											"						4.0 V	4.0 V	GND	GND	GND	GND	4.0 V	Y0	-10	-120	-10	-120	-10	-120	mA
		79		GND										"						"	"	"	"	"	"	"	Y1	"	"	"	"	"	"	"
		80			GND									"						"	"	"	"	"	"	"	Y2	"	"	"	"	"	"	"
		81				GND								"						"	"	"	"	"	"	"	Y3	"	"	"	"	"	"	"
		82					GND							"						"	"	"	"	"	"	"	Y4	"	"	"	"	"	"	"
		83						GND						"						"	"	"	"	"	"	"	Y5	"	"	"	"	"	"	"
		84							GND					"						"	"	"	"	"	"	"	Y6	"	"	"	"	"	"	"
		85								GND				"						"	"	"	"	"	"	"	Y7	"	"	"	"	"	"	"
		86									GND			"						"	"	"	"	"	"	"	Y8	"	"	"	"	"	"	"
		87										GND		"						"	"	"	"	"	"	"	Y9	"	"	"	"	"	"	"
		88											GND	"						"	"	"	"	"	"	"	Y10	"	"	"	"	"	"	"
		89												"						"	"	"	"	"	"	"	Y11	"	"	"	"	"	"	"
		90												"						"	"	"	"	"	"	"	Y12	"	"	"	"	"	"	"
		91												"						"	"	"	"	"	"	"	Y13	"	"	"	"	"	"	"
		92												"						"	"	"	"	"	"	"	Y14	"	"	"	"	"	"	"
		93												"	GND									"	"		Y15							

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TABLE III. Group A inspection for device type 04 - Continued.

Symbol	MIL-STD-883 method	Terminal conditions 1/																								Measured terminal	Test limits						Unit
		Case 3	2	3	4	5	6	7	9	10	11	12	13	14	16	17	18	19	20	21	23	24	25	26	27	28		Subgroup 1 T_C = +25°C		Subgroup 2 T_C = +125°C		Subgroup 3 T_C = -55°C		
		L	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24		Min	Max	Min	Max	Min	Max	
		Test no.	Y0	Y1	Y2	Y3	Y4	Y5	Y6	Y7	Y8	Y9	Y10	GND	Y11	Y12	Y13	Y14	Y15	G1	G2	D	C	B	A	V_{CC}								
I_{IH}	3010	94												GND						GND	GND	GND	GND	GND	6.0 V	6.0 V	A		0.05		0.1			μA
		95												"						"	"	"	6.0 V	6.0 V	GND	"	B		"		"			
		96												"						"	"	"	"	"	"	"	C		"		"			
		97												"						"	"	"	"	"	"	"	D		"		"			
		98												"						"	6.0 V	"	"	"	"	"	G1		"		"			
		99												"						6.0 V	"	"	"	"	"	"	G2		"		"			
I_{IL}	3009	100																			GND	GND					A		-0.5		-0.1			"
		101																			"	"					B		"		"			
		102																			"	"					C		"		"			
		103																			"	"					D		"		"			
		104																			"	"					G1		"		"			
		105																			6.0 V	6.0 V					G2		"		"			
																												Subgroup 4 T_C = +25°C						
																												Min	Max					
C_I	3012	106												GND						2/	2/	2/	2/	2/	2/	GND	A		10					pF
		107												"												"	B		"					
		108												"												"	C		"					
		109												"												"	D		"					
		110												"												"	G1		"					
		111												"												"	G2		"					
																												Subgroup 7 T_C = +25°C		Subgroup 8 T_C = +125°C		Subgroup 8 T_C = -55°C		
																												Min	Max	Min	Max	Min	Max	
Truth table tests 3/	3014	112	L	H	H	H	H	H	H	H	H	H	H	GND	H	H	H	H	H	B	B	B	B	B	6	4.5 V	4/							
		113	"	L	H	H	H	H	H	H	H	H	H	"	H	H	H	H	H	A	A	X	A	A	A	"								
		114	"	"	L	H	H	H	H	H	H	H	H	"	H	H	H	H	H	B	B	B	B	B	B	"								
		115	"	"	"	L	H	H	H	H	H	H	H	"	H	H	H	H	H	A	A	X	A	A	A	"								
		116	"	"	"	"	L	H	H	H	H	H	H	"	H	H	H	H	H	B	B	B	B	B	B	"								
		117	"	"	"	"	"	L	H	H	H	H	H	"	H	H	H	H	H	A	A	X	A	A	A	"								
		118	"	"	"	"	"	"	L	H	H	H	H	"	H	H	H	H	H	B	B	B	B	B	B	"								
		119	"	"	"	"	"	"	"	L	H	H	H	"	H	H	H	H	H	A	A	X	A	A	A	"								
		120	"	"	"	"	"	"	"	"	L	H	H	"	H	H	H	H	H	B	B	B	B	B	B	"								
		121	"	"	"	"	"	"	"	"	"	L	H	"	H	H	H	H	H	A	A	X	A	A	A	"								
		122	"	"	"	"	"	"	"	"	"	"	L	"	H	H	H	H	H	B	B	B	B	B	B	"								
		123	"	"	"	"	"	"	"	"	"	"	"	"	L	H	H	H	H	A	A	X	A	A	A	"								
		124	"	"	"	"	"	"	"	"	"	"	"	"	"	L	H	H	H	B	B	B	B	B	B	"								
		125	"	"	"	"	"	"	"	"	"	"	"	"	"	"	L	H	H	A	A	X	A	A	A	"								
		126	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	L	H	B	B	B	B	B	B	"								
		127	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	L	A	A	X	A	A	A	"								
		128	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	B	B	B	B	B	B	"								
		129	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	A	A	X	A	A	A	"								
		130	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	B	B	B	B	B	B	"								

See footnotes at end of table.

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TABLE III. Group A inspection for device type 04 - Continued.

Symbol	MIL-STD-883 method	Terminal conditions 1/																									Measured terminal	Test limits						Unit
		Case 3	2	3	4	5	6	7	9	10	11	12	13	14	16	17	18	19	20	21	23	24	25	26	27	28		Subgroup 9 T_C = +25°C 5/		Subgroup 10 T_C = +125°C		Subgroup 11 T_C = -55°C 5/		
		L	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24								
		Test no.	Y0	Y1	Y2	Y3	Y4	Y5	Y6	Y7	Y8	Y9	Y10	GND	Y11	Y12	Y13	Y14	Y15	G1	G2	D	C	B	A	V_{CC}		Min	Max	Min	Max	Min	Max	
t_{PHL1}	3003 (Fig. 6)	131	OUT											GND						GND	GND	GND	GND	GND	IN	4.5 V	Y0	6	33	6	44	6	33	ns
		132		OUT										"						"	"	"	GND	IN	4.5 V	"	Y1	"	"	"	"	"	"	"
		133			OUT									"						"	"	IN	IN	4.5 V	GND	"	Y2	"	"	"	"	"	"	"
		134				OUT								"						"	"	GND	GND	4.5 V	4.5 V	"	Y3	"	"	"	"	"	"	"
		135					OUT							"						"	"	"	4.5 V	GND	IN	"	Y4	"	"	"	"	"	"	"
		136						OUT						"						"	"	IN	"	IN	4.5 V	"	Y5	"	"	"	"	"	"	"
		137							OUT					"						"	"	4.5 V	"	4.5 V	IN	"	Y6	"	"	"	"	"	"	"
		138								OUT				"						"	"	"	GND	GND	4.5 V	"	Y7	"	"	"	"	"	"	"
		139									OUT			"						"	"	"	GND	IN	IN	"	Y8	"	"	"	"	"	"	"
		140										OUT		"						"	"	"	IN	4.5 V	GND	"	Y9	"	"	"	"	"	"	"
		141											OUT	"						"	"	"	IN	GND	4.5 V	"	Y10	"	"	"	"	"	"	"
		142												"						"	"	"	4.5 V	4.5 V	IN	"	Y11	"	"	"	"	"	"	"
		143												"						"	"	"	"	GND	IN	"	Y12	"	"	"	"	"	"	"
		144												"						"	"	"	"	IN	4.5 V	"	Y13	"	"	"	"	"	"	"
		145												"						"	"	"	"	"	IN	"	Y14	"	"	"	"	"	"	"
t_{PHL2}	3003 (Fig. 6)	146	OUT											"					OUT	IN	"	GND	GND	GND	GND	"	Y0	"	32	"	42	"	32	"
		147												"						GND	IN	4.5 V	4.5 V	4.5 V	4.5 V	"	Y15	"	32	"	42	"	32	"
t_{PLH1}	3003 (Fig. 6)	148		OUT										"						"	GND	GND	GND	GND	IN	"	Y1	"	33	"	44	"	33	"
		149			OUT									"						"	"	"	"	IN	GND	"	Y2	"	"	"	"	"	"	"
		150				OUT								"						"	"	"	IN	IN	4.5 V	"	Y3	"	"	"	"	"	"	"
		151					OUT							"						"	"	IN	"	GND	GND	"	Y4	"	"	"	"	"	"	"
		152						OUT						"						"	"	"	"	GND	4.5 V	"	Y5	"	"	"	"	"	"	"
		153							OUT					"						"	"	"	4.5 V	4.5 V	IN	"	Y6	"	"	"	"	"	"	"
		154								OUT				"						"	"	"	"	GND	GND	"	Y7	"	"	"	"	"	"	"
		155									OUT			"						"	"	IN	GND	"	GND	"	Y8	"	"	"	"	"	"	"
		156										OUT		"						"	"	"	"	GND	4.5 V	"	Y9	"	"	"	"	"	"	"
		157											OUT	"						"	"	"	"	4.5 V	GND	"	Y10	"	"	"	"	"	"	"
		158												"						"	"	"	"	4.5 V	4.5 V	"	Y11	"	"	"	"	"	"	"
		159												"						"	"	"	4.5 V	GND	GND	"	Y12	"	"	"	"	"	"	"
		160												"						"	"	4.5 V	4.5 V	GND	IN	"	Y13	"	"	"	"	"	"	"
		161												"						"	"	4.5 V	IN	4.5 V	GND	"	Y14	"	"	"	"	"	"	"
		162												"						"	"	IN	"	4.5 V	4.5 V	"	Y15	"	"	"	"	"	"	"
t_{PLH2}	3003 (Fig. 6)	163	OUT											"					OUT	IN	"	GND	GND	GND	GND	"	Y0	"	32	"	42	"	32	"
		164												"						GND	IN	4.5 V	4.5 V	4.5 V	4.5 V	"	Y15	"	32	"	42	"	32	"
t_{THL}	3004 (Fig. 6)	165	OUT											"						IN	GND	GND	GND	GND	GND	"	Y0	3	15	3	20	3	15	"
		166		OUT										"						"	"	"	"	GND	4.5 V	"	Y1	"	"	"	"	"	"	"
		167			OUT									"						"	"	"	"	4.5 V	GND	"	Y2	"	"	"	"	"	"	"
		168				OUT								"						"	"	"	"	4.5 V	4.5 V	"	Y3	"	"	"	"	"	"	"
		169					OUT							"						"	"	"	4.5 V	GND	GND	"	Y4	"	"	"	"	"	"	"
		170						OUT						"						"	"	"	"	GND	4.5 V	"	Y5	"	"	"	"	"	"	"
		171							OUT					"						"	"	"	"	4.5 V	GND	"	Y6	"	"	"	"	"	"	"
		172								OUT				"						"	"	"	"	4.5 V	4.5 V	"	Y7	"	"	"	"	"	"	"
		173									OUT			"						"	"	4.5 V	GND	GND	GND	"	Y8	"	"	"	"	"	"	"
		174										OUT		"						"	"	"	"	GND	4.5 V	"	Y9	"	"	"	"	"	"	"
		175											OUT	"						"	"	"	"	4.5 V	GND	"	Y10	"	"	"	"	"	"	"
		176												"						"	"	"	"	4.5 V	4.5 V	"	Y11	"	"	"	"	"	"	"
		177												"						"	"	"	"	GND	GND	"	Y12	"	"	"	"	"	"	"
		178												"						"	"	"	"	GND	4.5 V	"	Y13	"	"	"	"	"	"	"
		179												"						"	"	"	"	4.5 V	GND	"	Y14	"	"	"	"	"	"	"
		180												"						"	"	"	"	4.5 V	4.5 V	"	Y15	"	"	"	"	"	"	"

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TABLE III. Group A inspection for device type 04 - Continued.

Symbol	MIL-STD-883 method	Terminal conditions 1/																								Measured terminal	Test limits						Unit
		Case	2	3	4	5	6	7	9	10	11	12	13	14	16	17	18	19	20	21	23	24	25	26	27	28		Subgroup 9 T_C = +25°C 5/		Subgroup 10 T_C = +125°C		Subgroup 11 T_C = -55°C 5/		
		L	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24								
		Test no.	Y0	Y1	Y2	Y3	Y4	Y5	Y6	Y7	Y8	Y9	Y10	GND	Y11	Y12	Y13	Y14	Y15	G1	G2	0	C	B	A	V_{CC}		Min	Max	Min	Max	Min	Max	
t_{TLH}	3004 (Fig. 6)	181	OUT											GND						IN	GND	GND	GND	GND	GND	4.5 V	Y0	3	15	3	20	3	15	ns
		192		OUT										"						"	"	"	"	GND	4.5 V	"	Y1	"	"	"	"	"	"	"
		133			OUT									"						"	"	"	"	4.5 V	GND	"	Y2	"	"	"	"	"	"	"
		194				OUT								"						"	"	"	"	4.5 V	4.5 V	"	Y3	"	"	"	"	"	"	"
		185					OUT							"						"	"	"	4.5 V	GND	GND	"	Y4	"	"	"	"	"	"	"
		196						OUT						"						"	"	"	"	GND	4.5 V	"	Y5	"	"	"	"	"	"	"
		187							OUT					"						"	"	"	"	4.5 V	GND	"	Y6	"	"	"	"	"	"	"
		188								OUT				"						"	"	"	"	4.5 V	4.5 V	"	Y7	"	"	"	"	"	"	"
		189									OUT			"						"	"	4.5 V	GND	GND	GND	"	Y8	"	"	"	"	"	"	"
		190										OUT		"						"	"	"	"	GND	4.5 V	"	Y9	"	"	"	"	"	"	"
		191											OUT	"						"	"	"	"	4.5 V	GND	"	Y10	"	"	"	"	"	"	"
		192												"	OUT					"	"	"	"	4.5 V	4.5 V	"	Y11	"	"	"	"	"	"	"
		193												"		OUT				"	"	"	4.5 V	GND	GND	"	Y12	"	"	"	"	"	"	"
		194												"			OUT			"	"	"	"	GND	4.5 V	"	Y13	"	"	"	"	"	"	"
		195												"				OUT		"	"	"	"	4.5 V	GND	"	Y14	"	"	"	"	"	"	"
		196												"					OUT	"	"	"	"	4.5 V	4.5 V	"	Y15	"	"	"	"	"	"	"

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MIL-M-38510/656

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TABLE III. Group A inspection for device type 52.

Symbol	MIL-STD-883 method	Case 2 / E	Terminal conditions 1/																Measured terminal	Test Limits: Subgroup 1 T_C = +25°C		Subgroup 2 T_C = +125°C		Subgroup 3 T_C = -55°C		Unit
			2 / 1	3 / 2	4 / 3	5 / 4	7 / 5	8 / 6	9 / 7	10 / 8	12 / 9	13 / 10	14 / 11	15 / 12	17 / 13	18 / 14	19 / 15	20 / 16		Min	Max	Min	Max	Min	Max	
		Test no.	A	B	C	$\overline{G2A}$	$\overline{G2B}$	G1	Y7	GND	Y6	Y5	Y4	Y3	Y2	Y1	Y0	V_{CC}								
$V_{IC(pos)}$		1	1 mA							1/								GND	A		1.5					V
		2		1 mA						"								"	B		"					"
		3			1 mA					"								"	C		"					"
		4				1 mA				"								"	$\overline{G2A}$		"					"
		5					1 mA			"								"	$\overline{G2B}$		"					"
		6						1 mA		"								"	G1		"					"
$V_{IC(neg)}$		7	-1 mA							GND								1/	A		-1.5					"
		8		-1 mA						"								"	B		"					"
		9			-1 mA					"								"	C		"					"
		10				-1 mA				"								"	$\overline{G2A}$		"					"
		11					-1 mA			"								"	$\overline{G2B}$		"					"
		12						-1 mA		"								"	G1		"					"
I_{CC}	3005	13	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V		"								5.5 V	V_{CC}		0.2		20			µA
$I_{CC\Delta}$		14	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	2.4 V		"								"	"		.710		1.5			mA
		15	2.4 V	0.8 V	0.8 V	0.8 V	"	5.5 V		"								"	"		.345		"			"
		16				2.4 V		0.8 V		"								"	"		.125					
V_{OH6}	3006	17	0.8 V	0.8 V	0.8 V	2.0 V	0.8 V	0.8 V		"					-20 µA	-20 µA	-20 µA	4.5 V	Y0	4.40		4.40		4.40		V
		18	"	"	"	"	"	"		"					"	"	"	"	Y1	"		"		"		"
		19	"	"	"	"	"	"		"					"	"	"	"	Y2	"		"		"		"
		20	"	"	"	"	"	"		"			-20 µA		"	"	"	"	Y3	"		"		"		"
		21	"	"	"	"	"	"		"		-20 µA	"	-20 µA	"	"	"	"	Y4	"		"		"		"
		22	"	"	"	"	"	"		"	-20 µA	"	"	"	"	"	"	"	Y5	"		"		"		"
		23	"	"	"	"	"	"		"	"	"	"	"	"	"	"	"	Y6	"		"		"		"
		24	"	"	"	"	"	"	-20 µA	"	"	"	"	"	"	"	"	"	Y7	"		"		"		"
V_{OH7}	3006	25	"	"	"	"	"	"		"					-4.0 mA	-4.0 mA	-4.0 mA	5.5 V	Y0	3.98		3.70		3.98		"
		26	"	"	"	"	"	"		"					"	"	"	"	Y1	"		"		"		"
		27	"	"	"	"	"	"		"					"	"	"	"	Y2	"		"		"		"
		28	"	"	"	"	"	"		"			-4.0 mA		"	"	"	"	Y3	"		"		"		"
		29	"	"	"	"	"	"		"		-4.0 mA	"	-4.0 mA	"	"	"	"	Y4	"		"		"		"
		30	"	"	"	"	"	"		"	-4.0 mA	"	"	"	"	"	"	"	Y5	"		"		"		"
		31	"	"	"	"	"	"		"	"	"	"	"	"	"	"	"	Y6	"		"		"		"
		32	"	"	"	"	"	"	-4.0 mA	"	"	"	"	"	"	"	"	"	Y7	"		"		"		"
V_{OL6}	3007	33	"	"	"	0.8 V	"	2.0 V		"					20 µA	20 µA	20 µA	4.5 V	Y0		0.05		0.05		0.05	"
		34	2.0 V	"	"	"	"	"		"					"	"	"	"	Y1		"		"		"	"
		35	0.8 V	2.0 V	"	"	"	"		"					"	"	"	"	Y2		"		"		"	"
		36	2.0 V	2.0 V	"	"	"	"		"			20 µA		"	"	"	"	Y3		"		"		"	"
		37	0.8 V	0.8 V	2.0 V	"	"	"		"		20 µA	"	20 µA	"	"	"	"	Y4		"		"		"	"
		38	2.0 V	0.8 V	"	"	"	"		"	20 µA	"	"	"	"	"	"	"	Y5		"		"		"	"
		39	0.8 V	2.0 V	"	"	"	"		"	"	"	"	"	"	"	"	"	Y6		"		"		"	"
		40	2.0 V	2.0 V	"	"	"	"	20 µA	"	"	"	"	"	"	"	"	"	Y7		"		"		"	"
V_{OL7}	3007	41	0.8 V	0.8 V	0.8 V	"	"	"		"					4.0 mA	4.0 mA	4.0 mA	5.5 V	Y0		0.26		0.4		0.26	"
		42	2.0 V	0.8 V	"	"	"	"		"					"	"	"	"	Y1		"		"		"	"
		43	0.8 V	2.0 V	"	"	"	"		"					"	"	"	"	Y2		"		"		"	"
		44	2.0 V	2.0 V	"	"	"	"		"			4.0 mA		"	"	"	"	Y3		"		"		"	"
		45	0.8 V	0.8 V	2.0 V	"	"	"		"		4.0 mA	"	4.0 mA	"	"	"	"	Y4		"		"		"	"
		46	2.0 V	0.8 V	"	"	"	"		"	4.0 mA	"	"	"	"	"	"	"	Y5		"		"		"	"
		47	0.8 V	2.0 V	"	"	"	"		"	"	"	"	"	"	"	"	"	Y6		"		"		"	"
		48	2.0 V	2.0 V	"	"	"	"	4.0 mA	"	"	"	"	"	"	"	"	"	Y7		"		"		"	"

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TABLE III. Group A inspection for device type 52 - Continued.

Symbol	MIL-STD-883 method	Case 2 / E	Terminal conditions 1/ 2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20	Measured terminal	Test limits: Subgroup 1 T_C = +25°C Min	Subgroup 1 T_C = +25°C Max	Subgroup 2 T_C = +125°C Min	Subgroup 2 T_C = +125°C Max	Subgroup 3 T_C = -55°C Min	Subgroup 3 T_C = -55°C Max	Unit
		Test no. / 1	A	B	C	$\overline{G2A}$	$\overline{G2B}$	G1	Y7	GND	Y6	Y5	Y4	Y3	Y2	Y1	YO	V_{CC}								
I_{OS4}	3011	49	GND	GND	GND	4.0 V	4.0 V	GND		GND					GND	GND	GND	4.0 V	YO	-10	-120	-10	-120	-10	-120	mA
		50	"	"	"	"	"	"		"								"	Y1	"	"	"	"	"	"	"
		51	"	"	"	"	"	"		"								"	Y2	"	"	"	"	"	"	"
		52	"	"	"	"	"	"		"								"	Y3	"	"	"	"	"	"	"
		53	"	"	"	"	"	"		"			GND					"	Y4	"	"	"	"	"	"	"
		54	"	"	"	"	"	"		"		GND						"	Y5	"	"	"	"	"	"	"
		55	"	"	"	"	"	"		"	GND							"	Y6	"	"	"	"	"	"	"
		56	"	"	"	"	"	"	GND	"				GND				"	Y7	"	"	"	"	"	"	"
I_{IH}	3010	57	5.5 V	"	"	GND	GND	"		"								5.5 V	A		0.5		0.1			µA
		58	GND	5.5 V	"	"	"	"		"								"	B		"		"			"
		59	"	GND	5.5 V	"	"	"		"								"	C		"		"			"
		60	"	"	GND	5.5 V	"	"		"								"	$\overline{G2A}$		"		"			"
		61	"	"	"	GND	5.5 V	"		"								"	$\overline{G2B}$		"		"			"
		62	"	"	"	GND	GND	5.5 V		"								"	G1		"		"			"
I_{IL}	3009	63	"	5.5 V	5.5 V	5.5 V	5.5 V	"		"								"	A		-0.5		-0.1			"
		64	5.5 V	GND	5.5 V	"	"	"		"								"	B		"		"			"
		65	"	5.5 V	GND	GND	"	"		"								"	C		"		"			"
		66	"	"	5.5 V	"	GND	"		"								"	$\overline{G2A}$		"		"			"
		67	"	"	"	5.5 V	5.5 V	"		"								"	$\overline{G2B}$		"		"			"
		68	"	"	"	5.5 V	"	GND		"								"	G1		"		"			"
																				Subgroup 4 T_C = +25°C Min	Subgroup 4 T_C = +25°C Max					
C_I	3012	69	2/	2/	2/	2/	2/	2/		GND								GND	A		10					pF
		70								"								"	B		"					"
		71								"								"	C		"					"
		72								"								"	$\overline{G2A}$		"					"
		73								"								"	$\overline{G2B}$		"					"
		74								"								"	G1		"					"
																				Subgroup 7 T_C = +25°C Min	Subgroup 7 T_C = +25°C Max	Subgroup 8 T_C = +125°C Min	Subgroup 8 T_C = +125°C Max	Subgroup 8 T_C = -55°C Min	Subgroup 8 T_C = -55°C Max	
Truth table tests 3/	3014	75	A	A	A	A	A	A	H	GND	H	H	H	H	H	H	H	4.5 V	4/							
		76	B	A	"	"	"	"	"	"	"	"	"	"	"	"	"	"									
		77	"	B	"	"	"	"	"	"	"	"	"	"	"	"	"	"									
		78	"	"	B	"	"	"	"	"	"	"	"	"	"	"	"	"									
		79	"	"	"	B	A	"	"	"	"	"	"	"	"	"	"	"									
		80	"	"	"	A	B	"	"	"	"	"	"	"	"	"	"	"									
		81	"	"	"	B	A	"	"	"	"	"	"	"	"	"	"	"									
		82	"	"	"	A	B	"	"	"	"	"	"	"	"	"	"	"									
		83	"	"	"	"	"	A	"	"	"	"	"	"	"	"	"	"									
		84	"	"	"	"	"	B	"	"	"	"	"	"	"	"	"	"									
		85	"	"	"	"	"	A	"	"	"	"	"	"	"	"	"	"									
		86	"	"	"	"	"	B	"	"	"	"	"	"	"	"	"	"									
		87	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"									
		88	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"									
		89	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"									
		90	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"									
		91	"	"	"	"	"	"	L	"	L	L	L	L	L	L	L	"									

See footnotes at end of table.

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TABLE III. Group A inspection for device type 52 - Continued.

Symbol	MIL-STD-883 method	Case 2 / E	Terminal conditions 1/																Measured terminal	Test Limits: Subgroup 9 T_C = +25°C 5/		Subgroup 10 T_C = +125°C		Subgroup 11 T_C = -55°C 5/		Unit
			2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20								
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16								
		Test no.	A	B	C	$\overline{G2A}$	$\overline{G2B}$	G1	Y7	GND	Y6	Y5	Y4	Y3	Y2	Y1	Y0	V_{CC}		Min	Max	Min	Max	Min	Max	
t_{PHL1}	3003 (Fig. 6)	92	IN	GND	GND	GND	GND	4.5 V		GND							OUT	4.5 V	A to Y0	6	35	6	47	6	35	ns
t_{PHL1}		93	GND	IN	GND	"	"	"		"							"	"	B to Y0	"	"	"	"	"	"	
t_{PHL1}		94	"	GND	IN	"	"	"		"							"	"	C to Y0	"	"	"	"	"	"	
t_{PHL2}		95	"	"	GND	IN	"	"		"							"	"	$\overline{G2A}$ to Y0	4	"	4	"	4	"	
t_{PHL2}		96	"	"	"	GND	IN	"		"							"	"	$\overline{G2B}$ to Y0	"	"	"	"	"	"	
t_{PHL2}		97	"	"	"	"	GND	IN		"							"	"	G1 to Y0	"	"	"	"	"	"	
t_{PHL1}		98	IN	"	"	"	"	4.5 V		"						OUT	"	"	A to Y1	6	"	6	"	6	"	
t_{PHL1}		99	4.5 V	IN	"	"	"	"		"						"	"	"	B to Y1	"	"	"	"	"	"	
t_{PHL1}		100	"	GND	IN	"	"	"		"						"	"	"	C to Y1	"	"	"	"	"	"	
t_{PHL2}		101	"	"	GND	IN	"	"		"						"	"	"	$\overline{G2A}$ to Y1	4	"	4	"	4	"	
t_{PHL2}		102	"	"	"	GND	IN	"		"						"	"	"	$\overline{G2B}$ to Y1	"	"	"	"	"	"	
t_{PHL2}		103	"	"	"	"	GND	IN		"						"	"	"	G1 to Y1	"	"	"	"	"	"	
t_{PHL1}		104	IN	4.5 V	"	"	"	4.5 V		"					OUT	"	"	"	A to Y2	6	"	6	"	6	"	
t_{PHL1}		105	GND	IN	"	"	"	"		"					"	"	"	"	B to Y2	"	"	"	"	"	"	
t_{PHL1}		106	"	4.5 V	IN	"	"	"		"					"	"	"	"	C to Y2	"	"	"	"	"	"	
t_{PHL2}		107	"	"	GND	IN	"	"		"					"	"	"	"	$\overline{G2A}$ to Y2	4	"	4	"	4	"	
t_{PHL2}		108	"	"	"	GND	IN	"		"					"	"	"	"	$\overline{G2B}$ to Y2	"	"	"	"	"	"	
t_{PHL2}		109	"	"	"	"	GND	IN		"					"	"	"	"	G1 to Y2	"	"	"	"	"	"	
t_{PHL1}		110	IN	"	"	"	"	4.5 V		"					"	"	"	"	A to Y3	6	"	6	"	6	"	
t_{PHL1}		111	4.5 V	IN	"	"	"	"		"					"	"	"	"	B to Y3	"	"	"	"	"	"	
t_{PHL1}		112	"	4.5 V	IN	"	"	"		"					"	"	"	"	C to Y3	"	"	"	"	"	"	
t_{PHL2}		113	"	"	"	IN	"	"		"					"	"	"	"	$\overline{G2A}$ to Y3	4	"	4	"	4	"	
t_{PHL2}		114	"	"	"	GND	IN	"		"					"	"	"	"	$\overline{G2B}$ to Y3	"	"	"	"	"	"	
t_{PHL2}		115	"	"	"	"	GND	IN		"					"	"	"	"	G1 to Y3	"	"	"	"	"	"	
t_{PHL1}		116	IN	GND	4.5 V	"	"	4.5 V		"			OUT		"	"	"	"	A to Y4	6	"	6	"	6	"	
t_{PHL1}		117	GND	IN	4.5 V	"	"	"		"			"		"	"	"	"	B to Y4	"	"	"	"	"	"	
t_{PHL1}		118	"	GND	IN	"	"	"		"			"		"	"	"	"	C to Y4	"	"	"	"	"	"	
t_{PHL2}		119	"	"	4.5 V	IN	"	"		"			"		"	"	"	"	$\overline{G2A}$ to Y4	4	"	4	"	4	"	
t_{PHL2}		120	"	"	"	GND	IN	"		"			"		"	"	"	"	$\overline{G2B}$ to Y4	"	"	"	"	"	"	
t_{PHL2}		121	"	"	"	"	GND	IN		"			"		"	"	"	"	G1 to Y4	"	"	"	"	"	"	
t_{PHL1}		122	IN	GND	"	"	"	4.5 V		"		OUT	"		"	"	"	"	A to Y5	6	"	6	"	6	"	
t_{PHL1}		123	4.5 V	IN	"	"	"	"		"		"	"		"	"	"	"	B to Y5	"	"	"	"	"	"	
t_{PHL1}		124	"	GND	IN	"	"	"		"		"	"		"	"	"	"	C to Y5	"	"	"	"	"	"	
t_{PHL2}		125	"	"	4.5 V	IN	"	"		"		"	"		"	"	"	"	$\overline{G2A}$ to Y5	4	"	4	"	4	"	
t_{PHL2}		126	"	"	"	GND	IN	"		"		"	"		"	"	"	"	$\overline{G2B}$ to Y5	"	"	"	"	"	"	
t_{PHL2}		127	"	"	"	"	GND	IN		"		"	"		"	"	"	"	G1 to Y5	"	"	"	"	"	"	
t_{PHL1}		128	IN	4.5 V	"	"	"	4.5 V		"	OUT	"	"		"	"	"	"	A to Y6	6	"	6	"	6	"	
t_{PHL1}		129	GND	IN	"	"	"	"		"	"	"	"		"	"	"	"	B to Y6	"	"	"	"	"	"	
t_{PHL1}		130	"	4.5 V	IN	"	"	"		"	"	"	"		"	"	"	"	C to Y6	"	"	"	"	"	"	
t_{PHL2}		131	"	"	4.5 V	IN	"	"		"	"	"	"		"	"	"	"	$\overline{G2A}$ to Y6	4	"	4	"	4	"	
t_{PHL2}		132	"	"	"	GND	IN	"		"	"	"	"		"	"	"	"	$\overline{G2B}$ to Y6	"	"	"	"	"	"	
t_{PHL2}		133	"	"	"	"	GND	IN		"	"	"	"		"	"	"	"	G1 to Y6	"	"	"	"	"	"	
t_{PHL1}		134	IN	"	"	"	"	4.5 V	OUT	"	"	"	"		"	"	"	"	A to Y7	6	"	6	"	6	"	
t_{PHL1}		135	4.5 V	IN	"	"	"	"	"	"	"	"	"		"	"	"	"	B to Y7	"	"	"	"	"	"	
t_{PHL1}		136	"	4.5 V	IN	"	"	"	"	"	"	"	"		"	"	"	"	C to Y7	"	"	"	"	"	"	
t_{PHL2}		137	"	"	4.5 V	IN	"	"	"	"	"	"	"		"	"	"	"	$\overline{G2A}$ to Y7	4	"	4	"	4	"	
t_{PHL2}		138	"	"	"	"	IN	"	"	"	"	"	"		"	"	"	"	$\overline{G2B}$ to Y7	"	"	"	"	"	"	
t_{PHL2}		139	"	"	"	"	IN	IN	"	"	"	"	"		"	"	"	"	G1 to Y7	"	"	"	"	"	"	
t_{PLH1}	3003 (Fig. 6)	140	IN	GND	GND	GND	GND	4.5 V									OUT		A to Y0	5		5		5		
t_{PLH1}		141	GND	IN	GND	"	"	"									"		B to Y0	"		"		"		
t_{PLH1}		142	"	GND	IN	"	"	"									"		C to Y0	"		"		"		
t_{PLH2}		143	"	"	GND	IN	"	"									"		$\overline{G2A}$ to Y0	4		4		4		
t_{PLH2}		144	"	"	"	GND	IN	"									"		$\overline{G2B}$ to Y0	"		"		"		
t_{PLH2}		145	"	"	"	"	GND	IN									"		G1 to Y0	"		"		"		
t_{PLH1}		146	IN	"	"	"	"	4.5 V								OUT	"		A to Y1	5		5		5		
t_{PLH1}		147	4.5 V	IN	"	"	"	"								"	"		B to Y1	"		"		"		
t_{PLH1}		148	"	GND	IN	"	"	"								"	"		C to Y1	"		"		"		
t_{PLH2}		149	"	"	GND	IN	"	"								"	"		$\overline{G2A}$ to Y1	4		4		4		
t_{PLH2}		150	"	"	"	GND	IN	"								"	"		$\overline{G2B}$ to Y1	"		"		"		
t_{PLH2}		151	"	"	"	"	GND	IN								"	"		G1 to Y1	"		"		"		
t_{PLH1}		152	IN	4.5 V	"	"	"	4.5 V							OUT	"	"		A to Y2	5		5		5		
t_{PLH1}		153	GND	IN	"	"	"	"							"	"	"		B to Y2	"		"		"		
t_{PLH1}		154	"	4.5 V	IN	"	"	"							"	"	"		C to Y2	"		"		"		

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TABLE III. Group A inspection for device type 52 - Continued.

Symbol	MIL-STD-883 method	Case	Terminal conditions 1/																Measured terminal	Test Limits						Unit
																				Subgroup 9 T_C = +25°C 5/		Subgroup 10 T_C = +125°C		Subgroup 11 T_C = -55°C 5/		
		2	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20								
		E	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16								
		Test no.	A	B	C	$\overline{G2A}$	$\overline{G2B}$	G1	Y7	GND	Y6	Y5	Y4	Y3	Y2	Y1	Y0	V_{CC}		Min	Max	Min	Max	Min	Max	
t_{PLH2}	3003 (Fig. 6)	155	GND	4.5 V	GND	IN	GND	4.5 V		GND					OUT			4.5 V	$\overline{G2A}$ to Y2	4	35	4	47	4	35	ns
t_{PLH2}		156	"	"	"	GND	IN	4.5 V		"					"			"	$\overline{G2B}$ to Y2	"	"	"	"	"	"	"
t_{PLH2}		157	"	"	"	"	GND	IN		"					"			"	G1 to Y2	"	"	"	"	"	"	"
t_{PLH1}		158	IN	"	"	"	"	4.5 V		"				OUT				"	A to Y3	5	"	5	"	5	"	"
t_{PLH1}		159	4.5 V	IN	"	"	"	"		"				"				"	B to Y3	"	"	"	"	"	"	"
t_{PLH1}		160	"	4.5 V	IN	"	"	"		"				"				"	C to Y3	"	"	"	"	"	"	"
t_{PLH2}		161	"	"	GND	IN	"	"		"				"				"	$\overline{G2A}$ to Y3	4	"	4	"	4	"	"
t_{PLH2}		162	"	"	"	GND	IN	"		"				"				"	$\overline{G2B}$ to Y3	"	"	"	"	"	"	"
t_{PLH2}		163	"	"	"	"	GND	IN		"				"				"	G1 to Y3	"	"	"	"	"	"	"
t_{PLH1}		164	IN	GND	4.5 V	"	"	4.5 V		"			OUT					"	A to Y4	5	"	5	"	5	"	"
t_{PLH1}		165	GND	IN	4.5 V	"	"	"		"			"					"	B to Y4	"	"	"	"	"	"	"
t_{PLH1}		166	"	GND	IN	"	"	"		"			"					"	C to Y4	"	"	"	"	"	"	"
t_{PLH2}		167	"	"	4.5 V	IN	"	"		"			"					"	$\overline{G2A}$ to Y4	4	"	4	"	4	"	"
t_{PLH2}		168	"	"	"	GND	IN	"		"			"					"	$\overline{G2B}$ to Y4	"	"	"	"	"	"	"
t_{PLH2}		169	"	"	"	"	GND	IN		"			"					"	G1 to Y4	"	"	"	"	"	"	"
t_{PLH1}		170	IN	"	"	"	"	4.5 V		"		OUT						"	A to Y5	5	"	5	"	5	"	"
t_{PLH1}		171	4.5 V	IN	"	"	"	"		"		"						"	B to Y5	"	"	"	"	"	"	"
t_{PLH1}		172	"	GND	IN	"	"	"		"		"						"	C to Y5	"	"	"	"	"	"	"
t_{PLH2}		173	"	"	4.5 V	IN	"	"		"		"						"	$\overline{G2A}$ to Y5	4	"	4	"	4	"	"
t_{PLH2}		174	"	"	"	GND	IN	"		"		"						"	$\overline{G2B}$ to Y5	"	"	"	"	"	"	"
t_{PLH2}		175	"	"	"	"	GND	IN		"		"						"	G1 to Y5	"	"	"	"	"	"	"
t_{PLH1}		176	IN	4.5 V	"	"	"	4.5 V		"	OUT							"	A to Y6	5	"	5	"	5	"	"
t_{PLH1}		177	GND	IN	"	"	"	"		"	"							"	B to Y6	"	"	"	"	"	"	"
t_{PLH1}		178	"	4.5 V	IN	"	"	"		"	"							"	C to Y6	"	"	"	"	"	"	"
t_{PLH2}		179	"	"	4.5 V	IN	"	"		"	"							"	$\overline{G2A}$ to Y6	4	"	4	"	4	"	"
t_{PLH2}		180	"	"	"	GND	IN	"		"	"							"	$\overline{G2B}$ to Y6	"	"	"	"	"	"	"
t_{PLH2}		181	"	"	"	"	GND	IN		"	"							"	G1 to Y6	"	"	"	"	"	"	"
t_{PLH1}		182	IN	"	"	"	"	4.5 V	OUT	"	"							"	A to Y7	5	"	5	"	5	"	"
t_{PLH1}		183	4.5 V	IN	"	"	"	"	"	"	"							"	B to Y7	"	"	"	"	"	"	"
t_{PLH1}		184	"	4.5 V	IN	"	"	"	"	"	"							"	C to Y7	"	"	"	"	"	"	"
t_{PLH2}		185	"	"	4.5 V	IN	"	"	"	"	"							"	$\overline{G2A}$ to Y7	4	"	4	"	4	"	"
t_{PLH2}		186	"	"	"	GND	IN	"	"	"	"							"	$\overline{G2B}$ to Y7	"	"	"	"	"	"	"
t_{PLH2}		187	"	"	"	"	GND	IN	"	"	"							"	G1 to Y7	"	"	"	"	"	"	"
t_{THL}	3004 (Fig. 6)	188	GND	GND	GND	"	"	"		"						OUT	OUT	"	Y0	3	15	3	20	3	15	"
		189	4.5 V	GND	"	"	"	"		"					OUT	"	"	"	Y1	"	"	"	"	"	"	"
		190	GND	4.5 V	"	"	"	"		"					"	"	"	"	Y2	"	"	"	"	"	"	"
		191	4.5 V	4.5 V	"	"	"	"		"				OUT	"	"	"	"	Y3	"	"	"	"	"	"	"
		192	GND	GND	4.5 V	"	"	"		"			OUT	"	"	"	"	"	Y4	"	"	"	"	"	"	"
		193	4.5 V	GND	"	"	"	"		"		OUT	"	"	"	"	"	"	Y5	"	"	"	"	"	"	"
		194	GND	4.5 V	"	"	"	"		"	OUT	"	"	"	"	"	"	"	Y6	"	"	"	"	"	"	"
		195	4.5 V	4.5 V	"	"	"	"	OUT	"	"	"	"	"	"	"	"	"	Y7	"	"	"	"	"	"	"
t_{TLH}	3004 (Fig. 6)	196	GND	GND	GND	"	"	"		"						OUT	OUT	"	Y0	"	"	"	"	"	"	"
		197	4.5 V	GND	"	"	"	"		"					OUT	"	"	"	Y1	"	"	"	"	"	"	"
		198	GND	4.5 V	"	"	"	"		"					"	"	"	"	Y2	"	"	"	"	"	"	"
		199	4.5 V	4.5 V	"	"	"	"		"				OUT	"	"	"	"	Y3	"	"	"	"	"	"	"
		200	GND	GND	4.5 V	"	"	"		"			OUT	"	"	"	"	"	Y4	"	"	"	"	"	"	"
		201	4.5 V	GND	"	"	"	"		"		OUT	"	"	"	"	"	"	Y5	"	"	"	"	"	"	"
		202	GND	4.5 V	"	"	"	"		"	OUT	"	"	"	"	"	"	"	Y6	"	"	"	"	"	"	"
		203	4.5 V	4.5 V	"	"	"	"	OUT	"	"	"	"	"	"	"	"	"	Y7	"	"	"	"	"	"	"

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TABLE III. Group A inspection for device type 53.

Symbol	MIL-STD-883 method	Case	Terminal conditions 1/																	Measured terminal	Test limits						Unit
																					Subgroup 1 T_C = +25°C		Subgroup 2 T_C = +125°C		Subgroup 3 T_C = -55°C		
		2	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20			Min	Max	Min	Max	Min	Max	
		E	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16									
		Test no.	G1	A1	B1	1Y0	1Y1	1Y2	1Y3	GND	2Y3	2Y2	2Y1	2Y0	B2	A2	G2	V_{CC}									
$V_{IC(pos)}$		1	1 mA							1/								GND	G1		1.5						V
		2		1 mA						"								"	A1		"						"
		3			1 mA					"								"	B1		"						"
		4								"							1 mA	"	G2		"						"
		5								"						1 mA		"	A2		"						"
		6								"					1 mA			"	B2		"						"
$V_{IC(neg)}$		7	-1 mA							GND								1/	G1		-1.5						V
		8		-1 mA						"								"	A1		"						"
		9			-1 mA					"								"	B1		"						"
		10								"							-1 mA	"	G2		"						"
		11								"						-1 mA		"	A2		"						"
		12								"					-1 mA			"	B2		"						"
I_{CC}	3005	13	5.5 V	5.5 V	5.5 V					"					5.5 V	5.5 V	5.5 V	5.5 V	V_{CC}			0.2		20			µA
$I_{CC\Delta}$		14	2.4 V	0.8 V						"					0.8 V	0.8 V	5.5 V	"	V_{CC}			.345		1.5			mA
		15	0.8 V	0.8 V						"					0.8 V	2.4 V	GND	"	V_{CC}			.345		1.5			mA
V_{OH6}	3006	16	2.0 V	0.8 V	0.8 V	-20 µA				"					GND	GND		4.5 V	1Y0	4.40		4.40		4.40			V
		17	"	"	"		-20 µA			"					"	"		"	1Y1	"		"		"			"
		18	"	"	"			-20 µA		"					"	"		"	1Y2	"		"		"			"
		19	"	"	"				-20 µA	"					"	"		"	1Y3	"		"		"			"
		20	"	"	"					"			-20 µA		"	"	2.0 V	"	2Y0	"		"		"			"
		21	"	"	"					"		-20 µA			"	"	"	"	2Y1	"		"		"			"
		22	"	"	"					"	-20 µA				"	"	"	"	2Y2	"		"		"			"
		23	"	"	"					"				-20 µA	"	"	"	"	2Y3	"		"		"			"
V_{OH7}	3006	24	2.0 V	"	"	-4.0 mA				"					"	"		5.5 V	1Y0	3.98		3.70		3.98			V
		25	"	"	"		-4.0 mA			"					"	"		"	1Y1	"		"		"			"
		26	"	"	"			-4.0 mA		"					"	"		"	1Y2	"		"		"			"
		27		"	"				-4.0 mA	"					"	"		"	1Y3	"		"		"			"
		28		"	"					"			-4.0 mA		"	"	2.0 V	"	2Y0	"		"		"			"
		29		"	"					"		-4.0 mA			"	"	"	"	2Y1	"		"		"			"
		30		"	"					"	-4.0 mA				"	"	"	"	2Y2	"		"		"			"
		31		"	"					"				-4.0 mA	"	"	"	"	2Y3	"		"		"			"
V_{OL6}	3007	32	0.8 V	"	"	20 µA				"					0.8 V	0.8 V	0.8 V	4.5 V	1Y0		0.05		0.05		0.05		V
		33	"	2.0 V	"		20 µA			"					"	"	"	"	1Y1		"		"		"		"
		34	"	0.8 V	2.0 V			20 µA		"					"	"	"	"	1Y2		"		"		"		"
		35	"	2.0 V	2.0 V				20 µA	"					"	"	"	"	1Y3		"		"		"		"
		36	"	0.8 V	0.8 V					"					"	"	"	"	2Y0		"		"		"		"
		37	"	"	"					"			20 µA		"	2.0 V	"	"	2Y1		"		"		"		"
		38	"	"	"					"		20 µA			"	0.8 V	"	"	2Y2		"		"		"		"
		39	"	"	"					"	20 µA			20 µA	2.0 V	2.0 V	"	"	2Y3		"		"		"		"
V_{OL7}	3007	40	"	"	"	4.0 mA				"					0.8 V	0.8 V	"	5.5 V	1Y0		0.26		0.4		0.26		V
		41	"	2.0 V	"		4.0 mA			"					"	"	"	"	1Y1		"		"		"		"
		42	"	0.8 V	2.0 V			4.0 mA		"					"	"	"	"	1Y2		"		"		"		"
		43	"	2.0 V	2.0 V				4.0 mA	"					"	"	"	"	1Y3		"		"		"		"
		44	"	0.8 V	0.8 V					"					"	"	"	"	2Y0		"		"		"		"
		45	"	"	"					"			4.0 mA		"	2.0 V	"	"	2Y1		"		"		"		"
		46	"	"	"					"		4.0 mA			"	0.8 V	"	"	2Y2		"		"		"		"
		47	"	"	"					"	4.0 mA			4.0 mA	2.0 V	2.0 V	"	"	2Y3		"		"		"		"

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TABLE III. Group A inspection for device type 53 - Continued.

Symbol	MIL-STD-883 method	Case	Terminal conditions 1/																	Measured terminal	Test Limits						Unit
																					Subgroup 1 T_C = +25°C		Subgroup 2 T_C = +125°C		Subgroup 3 T_C = -55°C		
		2	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20								
		E	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16								
		Test no.	G1	A1	B1	1Y0	1Y1	1Y2	1Y3	GND	2Y3	2Y2	2Y1	2Y0	B2	A2	G2	V_{CC}		Min	Max	Min	Max	Min	Max	
I_{OS4}	3011	48	4.0 V	GND	GND	GND	GND	GND	GND	GND					GND	GND		4.0 V	1Y0	-10	-120	-10	-120	-10	-120	mA
		49	"	"	"					"					"	"		"	1Y1	"	"	"	"	"	"	"
		50	"	"	"					"					"	"		"	1Y2	"	"	"	"	"	"	"
		51		"	"					"					"	"		"	1Y3	"	"	"	"	"	"	"
		52		"	"					"					"	"	4.0 V	"	2Y0	"	"	"	"	"	"	"
		53		"	"					"					"	"	"	"	2Y1	"	"	"	"	"	"	"
		54		"	"					"					"	"	"	"	2Y2	"	"	"	"	"	"	"
		55		"	"					"	GND				"	"	"	"	2Y3	"	"	"	"	"	"	"
I_{IH}	3010	56	5.5 V	"	"					"					"	"	GND	5.5 V	G1		0.05		0.1			µA
		57	GND	5.5 V	"					"					"	"	"	"	A1		"		"			"
		58	"	GND	5.5 V					"					"	"	5.5 V	"	B1		"		"			"
		59	"	"	GND					"					"	"	GND	"	G2		"		"			"
		60	"	"	"					"					"	5.5 V	GND	"	A2		"		"			"
		61	"	"	"					"					5.5 V	GND	GND	"	B2		"		"			"
I_{IL}	3009	62	GND	5.5 V	5.5 V					"					"	5.5 V	5.5 V	"	G1		-0.05		-0.1			µA
		63	5.5 V	GND	5.5 V					"					"	"	"	"	A1		"		"			"
		64	"	5.5 V	GND					"					"	"	GND	"	B1		"		"			"
		65	"	"	5.5 V					"					"	GND	5.5 V	"	G2		"		"			"
		66	"	"	"					"					"	5.5 V	5.5 V	"	A2		"		"			"
		67	"	"	"					"					GND	5.5 V	5.5 V	"	B2		"		"			"
																				Subgroup 4 T_C = +25°C						
																				Min	Max					
C_I	3012	68	2/		2/					GND								GND	G1	10						pF
		69								"								"	A1	"						"
		70			2/					"								"	B1	"						"
		71								"								"	G2	"						"
		72								"								"	A2	"						"
		73								"					2/	2/	2/	"	B2	"						"
																				Subgroup 7 T_C = +25°C		Subgroup 8 T_C = +125°C		Subgroup 8 T_C = -55°C		
																				Min	Max	Min	Max	Min	Max	
Truth table tests 3/	3014	74	A	A	A	H	H	H	H	GND	H	H	H	H	A	A	A	4.5 V	4/							
		75	"	B	A	"	"	"	"	"	"	"	"	"	A	B	"	"	"							
		76	"	A	B	"	"	"	"	"	"	"	"	"	B	A	"	"	"							
		77	"	B	B	"	"	"	"	"	"	"	"	"	B	B	"	"	"							
		78	B	B	A	L	L	L	L	"	L	L	L	L	B	A	B	"	"							
		79	"	A	B	H	H	H	H	"	L	H	H	H	A	B	"	"	"							
		80	"	B	A	"	"	"	"	"	"	"	"	"	A	A	"	"	"							
		81	"	A	A	"	"	"	"	"	"	"	"	"	A	A	"	"	"							

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TABLE III. Group A inspection for device type 53 - Continued.

Symbol	MIL-STD-883 method	Case	Terminal conditions 1/																Measured terminal	Test limits						Unit
																				Subgroup 9 T_C = +25°C 5/		Subgroup 10 T_C = +125°C		Subgroup 11 T_C = -55°C 5/		
		2	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20							
		E	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16		Min	Max	Min	Max	Min	Max	
		Test no.	G1	A1	B1	1Y0	1Y1	1Y2	1Y3	GND	2Y3	2Y2	2Y1	2Y0	B2	A2	G2	V_{CC}								
t_{PHL2}	3003 (Fig. 6)	82	IN	GND	GND	OUT				GND					GND	GND	GND	4.5 V	G1 to 1Y0	5	35	5	47	5	35	ns
t_{PLH1}		83	GND	IN	GND	"				"					"	"	"	"	A1 to 1Y0	"	31	"	41	"	31	"
t_{PLH1}		84	GND	GND	IN	"				"					"	"	"	"	B1 to 1Y0	"	31	"	41	"	31	"
t_{PLH2}		85	IN	4.5 V	GND		OUT			"					"	"	"	"	G1 to 1Y1	"	35	"	47	"	35	"
t_{PLH1}		86	GND	IN	GND		"			"					"	"	"	"	A1 to 1Y1	"	31	"	41	"	31	"
t_{PLH1}		87	GND	4.5 V	IN					"					"	"	"	"	B1 to 1Y1	"	31	"	41	"	31	"
t_{PLH2}		88	IN	GND	4.5 V			OUT		"					"	"	"	"	G1 to 1Y2	"	35	"	47	"	35	"
t_{PLH1}		89	GND	IN	4.5 V			"		"					"	"	"	"	A1 to 1Y2	"	31	"	41	"	31	"
t_{PLH1}		90	GND	GND	IN					"					"	"	"	"	B1 to 1Y2	"	31	"	41	"	31	"
t_{PLH2}		91	IN	4.5 V	4.5 V				OUT	"					"	"	"	"	G1 to 1Y3	"	35	"	47	"	35	"
t_{PLH1}		92	GND	IN	4.5 V				"	"					"	"	"	"	A1 to 1Y3	7	31	7	41	7	31	"
t_{PLH1}		93	"	4.5 V	IN					"					"	"	"	"	B1 to 1Y3	7	31	7	41	7	31	"
t_{PLH2}		94	"	GND	GND					"				OUT	"	"	IN	"	G2 to 2Y0	5	35	5	47	5	35	"
t_{PLH1}		95	"	"	"					"				"	"	IN	GND	"	A2 to 2Y0	"	31	"	41	"	31	"
t_{PLH1}		96	"	"	"					"					IN	GND	GND	"	B2 to 2Y0	"	31	"	41	"	31	"
t_{PLH2}		97	"	"	"					"			OUT		GND	4.5 V	IN	"	G2 to 2Y1	"	35	"	47	"	35	"
t_{PLH1}		98	"	"	"					"			"		GND	IN	GND	"	A2 to 2Y1	"	31	"	41	"	31	"
t_{PLH1}		99	"	"	"					"					IN	4.5 V	GND	"	B2 to 2Y1	"	31	"	41	"	31	"
t_{PLH2}		100	"	"	"					"		OUT			4.5 V	GND	IN	"	G2 to 2Y2	"	35	"	47	"	35	"
t_{PLH1}		101	"	"	"					"		"			4.5 V	IN	GND	"	A2 to 2Y2	"	31	"	41	"	31	"
t_{PLH1}		102	"	"	"					"					IN	GND	GND	"	B2 to 2Y2	"	31	"	41	"	31	"
t_{PLH2}		103	"	"	"					"	OUT				4.5 V	4.5 V	IN	"	G2 to 2Y3	"	35	"	47	"	35	"
t_{PLH1}		104	"	"	"					"	"				4.5 V	IN	GND	"	A2 to 2Y3	7	31	7	41	7	31	"
t_{PLH1}		105	"	"	"					"					IN	4.5 V	"	"	B2 to 2Y3	7	31	7	41	7	31	"
t_{PLH2}		106	IN	"	"	OUT				"					GND	GND	"	"	G1 to 1Y0	5	35	5	47	5	35	"
t_{PLH1}		107	GND	IN	"	"				"					"	"	"	"	A1 to 1Y0	"	31	"	41	"	31	"
t_{PLH1}		108	GND	GND	IN	"				"					"	"	"	"	B1 to 1Y0	"	31	"	41	"	31	"
t_{PLH2}		109	IN	4.5 V	GND		OUT			"					"	"	"	"	G1 to 1Y1	"	35	"	47	"	35	"
t_{PLH1}		110	GND	IN	GND		"			"					"	"	"	"	A1 to 1Y1	"	31	"	41	"	31	"
t_{PLH1}		111	GND	4.5 V	IN					"					"	"	"	"	B1 to 1Y1	"	31	"	41	"	31	"
t_{PLH2}		112	IN	GND	4.5 V			OUT		"					"	"	"	"	G1 to 1Y2	"	35	"	47	"	35	"
t_{PLH1}		113	GND	IN	4.5 V			"		"					"	"	"	"	A1 to 1Y2	"	31	"	41	"	31	"
t_{PLH1}		114	GND	GND	IN					"					"	"	"	"	B1 to 1Y2	"	31	"	41	"	31	"
t_{PLH2}		115	IN	4.5 V	4.5 V				OUT	"					"	"	"	"	G1 to 1Y3	"	35	"	47	"	35	"
t_{PLH1}		116	GND	IN	4.5 V				"	"					"	"	"	"	A1 to 1Y3	7	31	7	41	7	31	"
t_{PLH1}		117	"	GND	IN					"					"	"	"	"	B1 to 1Y3	7	31	7	41	7	31	"
t_{PLH2}		118	"	"	GND					"				OUT	"	"	IN	"	G2 to 2Y0	5	35	5	47	5	35	"
t_{PLH1}		119	"	"	"					"				"	"	IN	GND	"	A2 to 2Y0	"	31	"	41	"	31	"
t_{PLH1}		120	"	"	"					"					IN	GND	GND	"	B2 to 2Y0	"	31	"	41	"	31	"
t_{PLH2}		121	"	"	"					"			OUT		GND	4.5 V	IN	"	G2 to 2Y1	"	35	"	47	"	35	"
t_{PLH1}		122	"	"	"					"			"		GND	IN	GND	"	A2 to 2Y1	"	31	"	41	"	31	"
t_{PLH1}		123	"	"	"					"					IN	4.5 V	GND	"	B2 to 2Y1	"	31	"	41	"	31	"
t_{PLH2}		124	"	"	"					"		OUT			4.5 V	GND	IN	"	G2 to 2Y2	"	35	"	47	"	35	"
t_{PLH1}		125	"	"	"					"		"			4.5 V	IN	GND	"	A2 to 2Y2	"	31	"	41	"	31	"
t_{PLH1}		126	"	"	"					"					IN	GND	GND	"	B2 to 2Y2	"	31	"	41	"	31	"
t_{PLH2}		127	"	"	"					"	OUT				4.5 V	4.5 V	IN	"	G2 to 2Y3	"	35	"	47	"	35	"
t_{PLH1}		128	"	"	"					"	"				4.5 V	IN	GND	"	A2 to 2Y3	7	31	7	41	7	31	"
t_{PLH1}		129	"	"	"					"					IN	4.5 V	GND	"	B2 to 2Y3	7	31	7	41	7	31	"

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TABLE III. Group A inspection for device type 53 - Continued.

Symbol	MIL-STD-883 method	Case	Terminal conditions 1/																	Measured terminal	Test limits						Unit
																					Subgroup 9 T_C = +25°C 5/		Subgroup 10 T_C = +125°C		Subgroup 11 T_C = -55°C 5/		
		2	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20								
		E	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16								
		Test no.	G1	A1	B1	1Y0	1Y1	1Y2	1Y3	GND	2Y3	2Y2	2Y1	2Y0	B2	A2	G2	V_{CC}		Min	Max	Min	Max	Min	Max	
t_{THL}	3003 (Fig. 6)	130	IN	GND	GND					GND					GND	GND	GND	4.5 V	1Y0	3	15	3	20	3	15	ns
		131	"	4.5 V	GND					"					"	"	"	"	1Y1	"	"	"	"	"	"	"
		132	"	GND	4.5 V					"					"	"	"	"	1Y2	"	"	"	"	"	"	"
		133	"	4.5 V	4.5 V					"					"	"	"	"	1Y3	"	"	"	"	"	"	"
		134	GND	GND	GND					"					"	"	IN	"	2Y0	"	"	"	"	"	"	"
		135	"	"	"					"					"	4.5 V	"	"	2Y1	"	"	"	"	"	"	"
		136	"	"	"					"					4.5 V	GND	"	"	2Y2	"	"	"	"	"	"	"
		137	"	"	"					"					4.5 V	4.5 V	"	"	2Y3	"	"	"	"	"	"	"
t_{TLH}	3003 (Fig. 6)	138	IN	"	"					"					GND	GND	GND	"	1Y0	"	"	"	"	"	"	"
		139	"	4.5 V	"					"					"	"	"	"	1Y1	"	"	"	"	"	"	"
		140	"	GND	4.5 V					"					"	"	"	"	1Y2	"	"	"	"	"	"	"
		141	"	4.5 V	4.5 V					"					"	"	"	"	1Y3	"	"	"	"	"	"	"
		142	GND	GND	GND					"					"	"	IN	"	2Y0	"	"	"	"	"	"	"
		143	"	"	"					"					"	4.5 V	"	"	2Y1	"	"	"	"	"	"	"
		144	"	"	"					"					4.5 V	GND	"	"	2Y2	"	"	"	"	"	"	"
		145	"	"	"					"					4.5 V	4.5 V	"	"	2Y3	"	"	"	"	"	"	"

1/ Output pins not designated may be high level logic, low level logic, or open. Input pins not designated may be high level logic or low level logic. Exceptions are as follows:

a. $V_{IC(pos)}$ tests, the GND terminal shall be open. A minimum limit of 0.4 V applies to tests being performed on equipment not capable of opening the GND pin during test.

b. $V_{IC(neg)}$ tests, the V_{CC} terminal shall be open.

c. I_{CC} tests, the output terminal shall be open.

2/ See 4.4.1c.

3/ Tests shall be performed in sequence. A = 2.4 V; B = 0.4 V for device types 52 and 53; A = 3.7 V; B = 0.4 V for device types 01 through 04.

4/ X = indeterminate output voltage. Output shall be H > 2.5 V; L < 2.5 V.

5/ See 4.4.1d.

6/ The input signal for HCT device types shall be 0 to 3 volts, however the 50 percent V_{CC} measuring point is 1.3 V for input and output signals. The 10 percent and 90 percent V_{CC} points are 0.3 V and 2.7 V respectively for HCT devices.

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4.4.2 Group B inspection. Group B inspection shall be in accordance with table II of method 5005 of MIL-STD-883 and as follows:

a. Class S steady-state life (accelerated) test circuits shall be submitted to the qualifying activity for approval. When the alternate steady-state life test is used, the circuit on figure 5, or equivalent, shall be used.

b. Electrostatic Discharge Sensitivity (ESDS) testing shall be performed in accordance with MIL-STD-883, method 3015. The option to categorize devices as ESD sensitive without performing the test is not allowed. Device types categorized as ESD sensitive shall be further tested using method 3015 modified as follows:

(1) Test method 3015, table I pin combinations 4(V +(A) to common (B)) and 5(V +(B) to common (A)) shall be deleted.

(2) The test sequence specified in 3.b of method 3015 shall be repeated an additional four times instead of two.

(3) The category A limits specified in figure 3015-3, ESD sensitivity category, shall be 1,000 V to 2,000 V.

c. End-point electrical parameters shall be as specified in table II herein. Delta limits shall apply only to subgroup 5 of group B inspection and shall consist of tests specified in table IV herein.

4.4.3 Group C inspection. Group C inspection shall be in accordance with table III of method 5005 of MIL-STD-883 and as follows:

a. End-point electrical parameters shall be as specified in table II herein. Delta limits shall apply only to subgroup 1 of group C inspection and shall consist of tests specified in table IV herein.

b. Steady-state life test (method 1005 of MIL-STD-883) conditions:

(1) Test condition D and as specified in 4.5.2 herein and as shown on figure 5 (note 3), or equivalent.

(2) T_A = +125°C minimum.

(3) Test duration, 1,000 hours, except as permitted by method 1005 of MIL-STD-883.

4.4.4 Group D inspection. Group D inspection shall be in accordance with table IV of method 5005 of MIL-STD-883. End-point electrical parameters shall be as specified in table II herein.

4.4.5 Group E inspection. Group E inspection is required only for device types intended to be marked as radiation hardened (see 3.6.1). When group E testing is performed it shall be in accordance with table V of method 5005 of MIL-STD-883 and 4.5.4 herein.

4.5 Methods of inspection. Methods of inspection shall be specified as follows.

4.5.1 Voltage and current. Unless otherwise specified, all voltages given are referenced to the microcircuit GND terminal. Currents given are conventional current and positive when flowing into the referenced terminal.

4.5.2 Burn-in and life test cool down procedures. When the burn-in and life tests are completed and prior to removal of bias voltages, the devices under test (DUT) shall be cooled to within 10°C of their power stable conditions at room temperature; then, electrical parameter end-point measurements shall be performed.

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TABLE IV. Delta limits at +25°C.

Parameter 1/	All device types
I_{CC}	±30 nA

1/ The above parameters shall be recorded before and after the required burn-in and life tests to determine deltas (Δ).

4.5.3 Quiescent supply current (I_{CC} test). When performing quiescent supply current measurements (I_{CC}), the meter shall be placed so that all currents flow through the meter.

4.5.4 Radiation hardness assurance (RHA) testing. The RHA testing shall be performed in accordance with test procedures and sampling specified in table V of method 5005 of MIL-STD-883 and herein:

a. Before irradiation, selected samples shall be assembled in qualified packages and pass the governing electrical parameters (group A subgroup 1 at +25°C) and also be subjected to the threshold-voltage test in table V in order to calculate the delta threshold (ΔV_T) after irradiation.

b. The devices shall be subjected to a total radiation dose as specified in MIL-M-38510 for the radiation hardness assurance (RHA) level being tested, and meet the end-point electrical parameters as defined in table VI at +25°C, after exposure. The start and completion of the end-point electrical parameter measurements shall not exceed 2 hours following irradiation.

c. Threshold-voltage test circuit conditions shall be as specified in table V and figure 7. In situ and remote testing, the tests shall be performed with the devices biased in accordance with table VII and bias may be interrupted for up to 1 minute to remove devices to the remote bias fixture.

d. After irradiation, the devices shall pass the truth table test as specified in subgroup 7 in table III or if subgroup 7 is not required, then an equivalent truth table test shall be performed.

TABLE V. Threshold-voltage test circuit conditions.

Device type	GND	5 V	V_{TN} measured at -10 μA supply	GND	-5.0 V	V_{TP} measured at 10 μA supply
01	13	12, 16	8, 14, 53	13	8, 14, 15	12, 16
02, 52	2	1, 16	3, 4, 5, 6, 8	2	3, 4, 5, 6, 8	1, 16
03, 53	2	3, 16	1, 8, 13, 14, 15	2	1, 8, 13, 14, 15	3, 16
04	18	19, 24	12, 20, 21, 22, 23	18	12, 20, 21, 22, 23	19, 24

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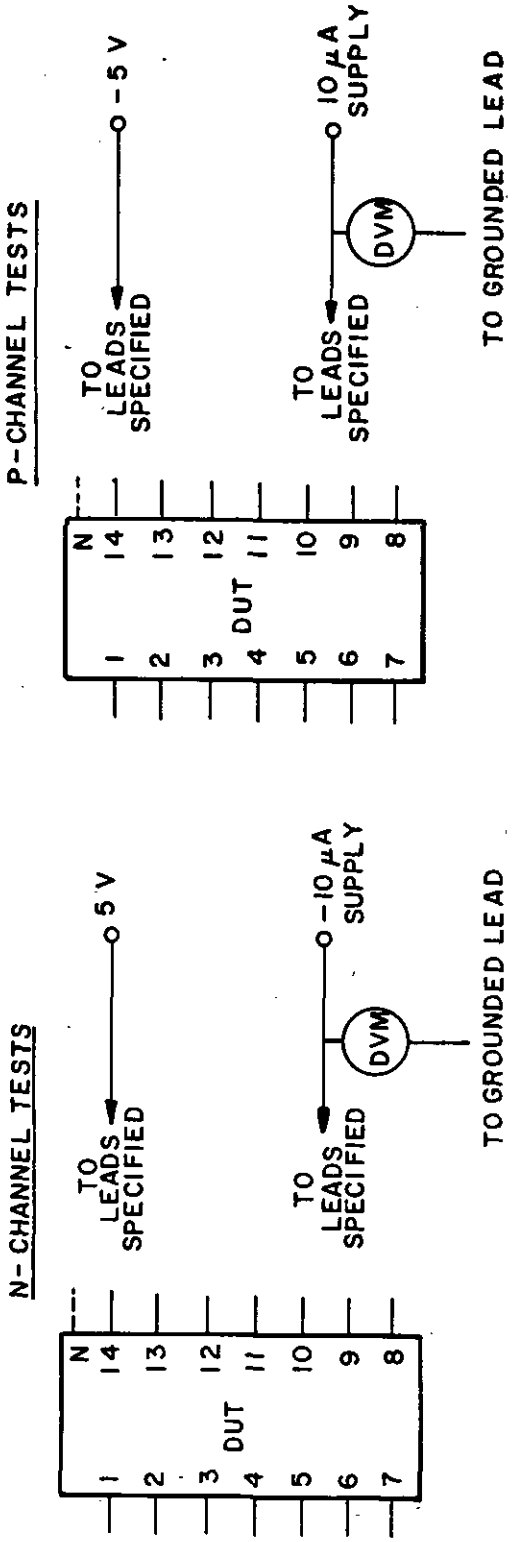


FIGURE 7. Threshold-voltage test circuit.

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TABLE VI. Radiation hardening end-point electrical parameters at +25°C.

Parameter	All device types	V_{CC}	
		Device types	
		01,02,03,04	52, 53
V_{TN}	-0.3 V min	5 V	5 V
V_{TP}	2.8 V max	5 V	5 V
ΔV_T	1.0 V max	5 V	5 V
I_{CC}	100 x max limit	6 V	5.5 V
t_{PLH}	1.35 x max limit	4.5 V	4.5 V
t_{PHL}	1.35 x max limit	4.5 V	4.5 V

TABLE VII. Bias during exposure to radiation.

Device type	Pin connections 1/		
	V_{CC} = 4.5 V dc (through a 30- to 60-kilohm resistor)	GND	V_{CC} = 4.5 V dc
01	12, 13, 14, 15	8	16
02, 52	1, 2, 3, 4, 5, 6	8	16
03, 53	1, 2, 3, 13, 14, 15	8	16
04	18, 19, 20, 21, 22, 23	12	24

1/ Pins not designated are open or connected to 4.5 V dc through a 30- to 60-kilohm resistor.

4.6 Data reporting. When specified in the purchase order or contract, a copy of the following data, as applicable, shall be supplied:

a. Attributes data for all screening tests (see 4.2) and variables data for all static burn-in, dynamic burn-in, and steady-state life tests (see 3.5).

b. A copy of each radiograph.

c. The quality conformance inspection data (see 4.4).

d. Parameter distribution data on parameters evaluated during burn-in (see 3.5).

e. Final electrical parameters data (see 4.2c).

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-38510.

6. NOTES

6.1 Intended use. Microcircuits conforming to this specification are intended for original equipment design application and logistic support of existing equipment.

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6.2 Ordering data. The acquisition document should specify the following:

a. Complete part number (see 1.2).

b. Requirements for delivery of one copy of the quality conformance inspection data pertinent to the device inspection lot to be supplied with each shipment by the device manufacturer, if applicable.

c. Requirements for certificate of compliance, if applicable.

d. Requirements for notification of change of product or process to the contracting activity in addition to notification to the qualifying activity, if applicable.

e. Requirements for failure analysis (including required test condition of method 5003 of MIL-STD-883), corrective action and reporting of results, if applicable.

f. Requirements for product assurance options.

g. Requirements for special carriers, lead lengths, or lead forming, if applicable. These requirements shall not affect the part number. Unless otherwise specified, these requirements shall not apply to direct purchase by, or direct shipment to the Government.

h. Requirements for "JAN" marking.

i. Requirements for total dose radiation testing (see 3.6.1 and 4.5.4), if applicable.

6.3 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-M-38510, MIL-STD-1331, and as follows:

C_I	Input terminal-to-GND capacitance.
GND	Ground zero voltage potential.
I_{CC}	Quiescent supply current.
T_C	Case temperature.
V_{CC}	Positive supply voltage.

6.4 Logistic support. Lead materials and finishes (see 3.3) are interchangeable. Unless otherwise specified, microcircuits acquired for Government logistic support will be acquired to device class S for National Aeronautics and Space Administration or class B for Department of Defense (see 1.2.2), lead finish C (see 3.3). Longer length leads and lead forming shall not affect the part number.

6.5 Substitutability. The cross-reference information below is presented for the convenience of users. Microcircuits covered by this specification will functionally replace the listed generic-industry type. Generic-industry microcircuit types may not have equivalent operational performance characteristics across military temperature ranges or reliability factors equivalent to MIL-M-38510 device types and may have slight physical variations in relation to case size. The presence of this information shall not be deemed as permitting substitution of generic-industry types for MIL-M-38510 types or as a waiver of any of the provisions of MIL-M-38510.

Military device type	Generic-industry type
01	54HC42
02	54HC138
03	54HC139
04	54HC154
52	54HCT138
53	54HCT139

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6.6 Handling. MOS devices must be handled with certain precautions to avoid damage due to accumulation of static charge. Input protective devices have been designed in the chip to minimize the effect of this static buildup. However, the following handling practices are recommended:

a. Devices should be handled on benches with conductive and grounded surface.

b. Ground test equipment and tools.

c. Do not handle devices by the leads.

d. Store devices in conductive foam or carriers.

e. Avoid use of plastic, rubber, or silk in MOS areas.

f. Maintain relative humidity above 50 percent, if practical.

Custodians:
Army - ER
Navy - EC
Air Force - 17
NASA - NA

Review activities:
Army - AR, MI
Air Force - 11, 19, 85, 99
DLA - ES

User activities:
Army - SM
Navy - AS, CG, MC, OS, SH

Preparing activity:
Air Force - 17

Agent:
DLA - ES

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3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
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